SwissBorg

CRYPTO 2024
"HOCUS POCUS"

SwissBorg

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CRYPTO 2024: “HOCUS POCUS”

INTRO

Cloaked in an air of mystery, the magician wore a black cloak with a wide-brimmed hat casting shadows, his smile hinting at intrigue. With a theatrical flourish of his wand, the atmosphere crackled with anticipation. The spotlight honed in on the black hat perched on the table, intensifying the suspense.

Chanting an incantation, the audience collectively held their breath. Though a moment seen repeatedly, the enchantment of the trick never dulled. The magician tapped the hat three times with the wand, each tap resonating through the auditorium.

“Hocus Pocus,” he declared, and like clockwork, a rabbit emerged from the black hat. The classic trick unfolded once more, captivating the audience. Despite their familiarity with the outcome, the magic retained its allure, leaving the spectators entranced and unable to unravel its secrets.

Markets, regardless of their classification – stocks, bonds, cars, cryptocurrencies, and so on – are by nature identical. Like a grand illusion, they appear predictable yet remain enigmatic. Their cyclical nature seems familiar, yet each time it unfolds, a wave of suspense washes over us, a thrilling emotional rollercoaster. When this classic trick plays out again, its timeless appeal captivates both the first-time investor and the seasoned veteran alike. Despite knowing the inevitable outcome, we are drawn in, unable to resist the allure of the illusion.

Welcome to our “Hocus Pocus” Crypto 2024 Edition - We hope you will enjoy the Show.

I. FOREWORD

2023 Retrospective

It's not without a hint of emotions that we kick off our journey by taking a stroll down memory lane, diving into the moments of 2023, the year that ultimately marks the end of the toughest bear market we've ever experienced in crypto assets.

The world is full of contradictions, and human behaviour is no exception. We often make things more complicated than they need to be, especially in the world of investing and trading.

For those like us who have been glued to the market in 2023, it's been a captivating year marked by a mixed bag of emotions. It's an understatement to say that maintaining clarity of mind has been difficult most of the time. Yet, Bitcoin has emerged again as the undisputed champion among 'mature' assets in 2023.

Performance of Major Assets and Asset Classes

2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023 (YTD)
Bitcoin 1,319.09%	Bitcoin 217.26%	Bitcoin 5,537.40%	U.S. REITs 30.38%	Bitcoin 33.74%	Bitcoin 123.31%	Bitcoin 1,414.10%	U.S. Bonds 0.01%	Bitcoin 93.95%	Bitcoin 308.17%	Bitcoin 57.25%	Commodity 21.23%	Bitcoin 62.57%
Gold 10.10%	EM Equity 18.63%	U.S. Equity 32.39%	U.S. Equity 13.69%	U.S. REITs 2.52%	Commodity 19.53%	EM Equity 37.75%	Gold (1.56%)	U.S. Equity 31.49%	Gold 25.12%	U.S. REITs 43.06%	Gold (0.28%)	U.S. Equity 13.07%
U.S. REITs 8.69%	DM Equity 17.90%	DM Equity 23.29%	U.S. Bonds 5.97%	U.S. Equity 1.38%	U.S. Equity 11.96%	DM Equity 25.62%	U.S. Equity (4.38%)	U.S. REITs 25.84%	EM Equity 18.69%	Commodity 42.60%	U.S. Bonds (13.01%)	DM Equity 7.59%
U.S. Bonds 7.84%	U.S. REITs 17.77%	U.S. REITs 2.47%	Gold (1.44%)	U.S. Bonds 0.55%	EM Equity 11.60%	U.S. Equity 21.83%	U.S. REITs (4.57%)	DM Equity 22.66%	U.S. Equity 18.40%	U.S. Equity 28.71%	DM Equity (14.01%)	EM Equity 2.16
U.S. Equity 2.11%	U.S. Equity 16.00%	U.S. Bonds (2.02%)	EM Equity (1.82%)	DM Equity (0.39%)	U.S. REITs 8.60%	Gold 13.53%	Commodity (11.18%)	EM Equity 18.88%	DM Equity 8.28%	DM Equity 11.78%	U.S. Equity (18.11%)	Commodity 1.88%
Commodity (2.39%)	Gold 7.06%	EM Equity (2.27%)	DM Equity (4.48%)	Gold (10.41%)	Gold 8.14%	Commodity 6.16%	DM Equity (13.36%)	Gold 18.31%	U.S. Bonds 7.51%	U.S. Bonds (1.54%)	EM Equity (19.74%)	Gold 1.35%
DM Equity (11.73%)	U.S. Bonds 4.22%	Commodity (6.55%)	Commodity (26.43%)	EM Equity (14.60%)	U.S. Bonds 2.65%	U.S. REITs 5.07%	EM Equity (14.24%)	Commodity 12.94%	Commodity (7.53%)	EM Equity (2.22%)	U.S. REITs (24.51%)	U.S. Bonds (1.21%)
EM Equity (18.17%)	Commodity 4.16%	Gold (28.28%)	Bitcoin (57.74%)	Commodity (26.68%)	DM Equity 1.51%	U.S. Bonds 3.54%	Bitcoin (74.67%)	U.S. Bonds 8.72%	U.S. REITs (7.57%)	Gold (3.64%)	Bitcoin (63.90%)	U.S. REITs (1.95%)

Now, with a clearer perspective and diminished doubt, we find ourselves stepping away from the drama. The Bitcoin chart showcases a performance that will make traditional hedge funds blush.

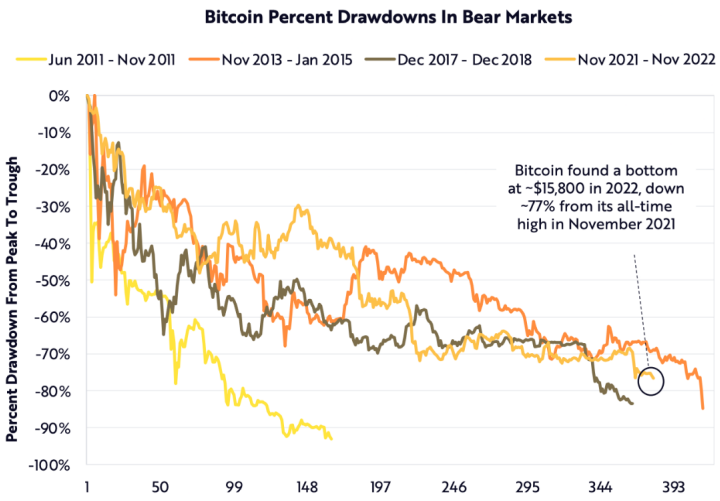
In hindsight, Bitcoin held firm, giving up less on us than we did on it. The price chart, with its relentless upward trajectory, serves as the perfect illustration of this.

Now, with a clearer perspective and diminished doubt, we find ourselves stepping away from the drama. The Bitcoin chart showcases a performance that will make traditional hedge funds blush. In hindsight, Bitcoin held firm, giving up less on us than we did on it. The price chart, with its relentless upward trajectory, serves as
the perfect illustration of this.

Did we truly lose faith in Bitcoin and Crypto this year, or did we just allow the perpetual bears to dominate the narrative for most of it? Let's rewind the clock and replay the sequence of events that defined our journey throughout the year.

#BearMarkets are getting increasingly challenging, which is logical considering the higher the supply of opium, the longer the recovery takes. However, is this truly the case?
When we focus on Bitcoin alone, it's not immediately evident why this market has been more painful than before, given that the price dynamics align with previous cycles.

The Drawdown From Bitcoin's All-Time High Was The Fifth Largest And Second Longest In History



Bitcoin's Bear Market Drawdowns

Peak		Trough		
Date	Price	Date	Price	Drawdown
June 2011	$31.91	November 2011	$2.00	-93.7%
November 2013	$1,242	January 2015	$152	-87.7%
December 2017	$19,891	December 2018	$3,128	-84.3%
November 2021	$69,000	November 2022[1]	$15,797	-76.7%
Peak-to-Peak Average CAGR	157.9%	Trough-to-Trough Average CAGR	153.6%	

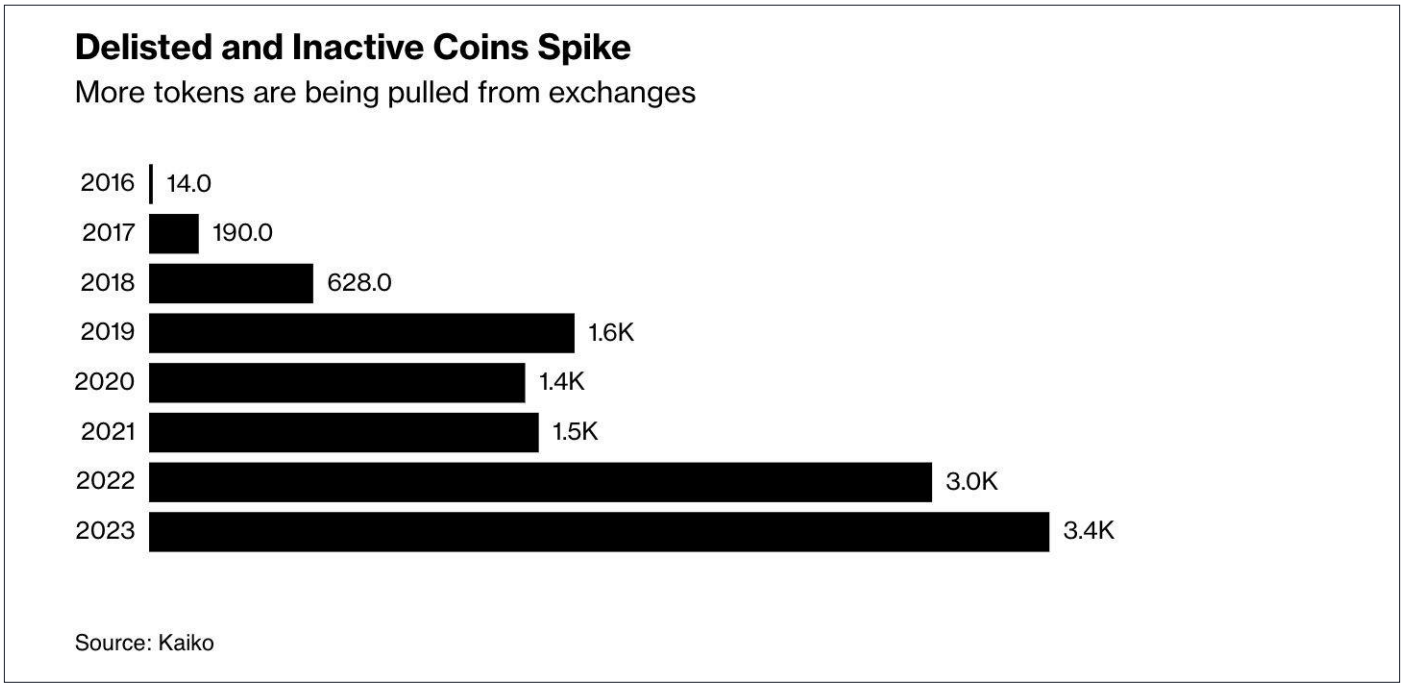
https://research.ark-invest.com/hubfs/1_Download_Files_ARK-Invest/Big_Ideas/ARK%20Invest_Presentation_Big%20Ideas%202023_FINAL_V2.pdf

Judging by the drawdown magnitude, Bitcoin consistently demonstrates an increasing resilience cycle after cycle. Remarkably, the symmetry in the length of completing a potential bottom is striking, averaging around 450 days from peak to trough in each cycle.

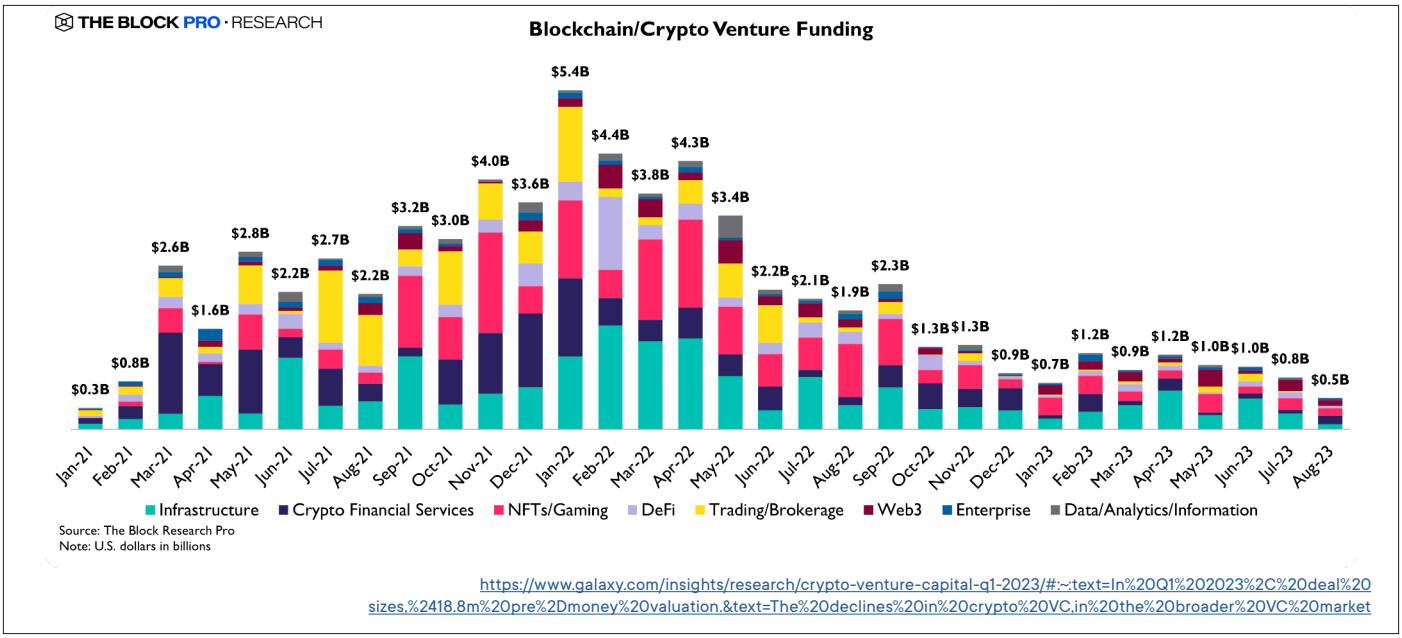
https://twitter.com/thecryptocactus/status/1718963839323005183?s=20

#BRB If Bitcoin's price action was challenging but not dramatic, shifting our focus to the Altcoin market paints a distinctly different picture. We bore the brunt of the excess token launches during 2021 and the early part of 2022. A direct testament to this reality is the significant number of tokens delisted from exchanges.

While exchanges reaped substantial profits, users were left with tokens lacking liquidity. In contrast, SwissBorg only delisted FTX and UST, maintaining a more selective approach



#DryofDryPowder: Venture capital flows have evaporated, mirroring the scarcity of dry powder. The crypto space, still in its infancy, practically guarantees facing phases of exponential growth and prosperity. Navigating the troughs becomes the ultimate challenge for any project, demanding meticulous planning to secure a runway of at least 24 months. Even within the crypto market, maintaining a robust cash position is essential for being able to surf every wave as funds flow in and out swiftly—a game that has caught more founders off guard than anticipated.



https://www.galaxy.com/insights/research/crypto-venture-capital-q1-2023/#:~:text=In%20Q1%202023%2C%20deal%20sizes,%2418.8m%20pre%2Dmoney%20valuation.&text=The%20declines%20in%20crypto%20VC,in%20the%20broader%20VC%20market

#RulesofTheGame have shifted - welcome the big players and their big guns. This Bear Market was unlike any other. On the surface, it seemed like the world was, once more, sceptical about Bitcoin as an asset class. However, behind the scenes, smart money, or forward-thinking investors, saw Bitcoin's price as a generational investment opportunity. This recognition marks a significant paradigm shift. We move from 'if' to 'when'— Bitcoin is becoming the go-to store of value for the digital age.

https://x.com/jamie1coutts/status/1730097855925559508?s=46&t=crsanLFVks0Fs4RUVNQqAA

#HistoryRhymes It is highly probable that 2023 is the post-bull market year, marking the onset of the renowned '4 years' cycle (i.e. the year preceding BTC halving)."

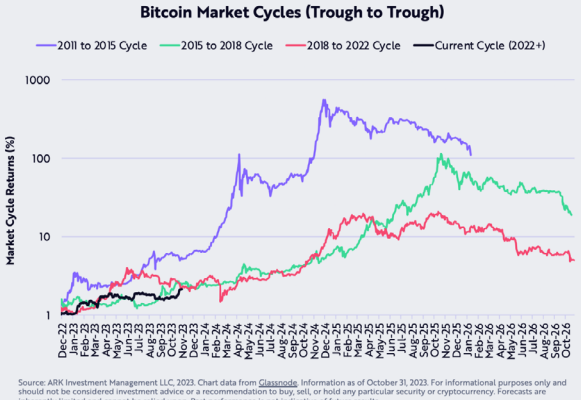
Bitcoin: Market Performance and Four-Years Cycles

"The Bitcoin Cycle"		*"The Ethereum Cycle"*		*"The Early Application Cycle"*		*"The Mainstream Cycle"*	
Year	BTC	Year	BTC	Year	BTC	Year	BTC
2011	1,319%	2015	34%	2019	94%	2023 (YTD)	63%
2012	217%	2016	123%	2020	308%	*2024*	*?*
2013	5,537%	2017	1,414%	2021	57%	*2025*	*?*
2014	(58%)	2018	(75%)	2022	(64%)	*2026*	*?*

Source: Bitwise Asset Management. Data from December 31, 2010 to September 29, 2023

14 Technical And On-Chain Resistance Cleared

Bitcoin's Primary Trend Seems Be At An Early Stage

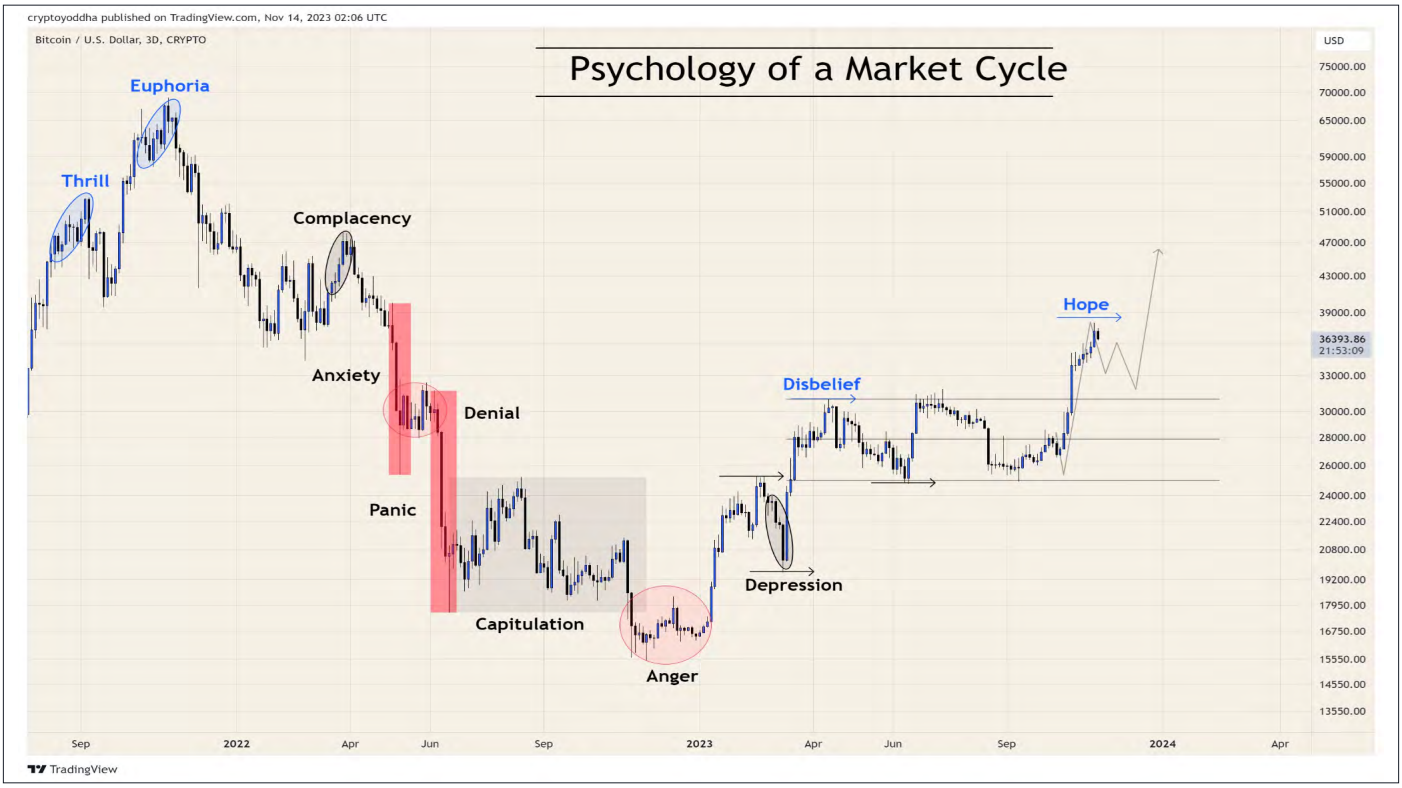


Source: ARK Investment Management LLC, 2023. Chart data from Glassnode. Information as of October 31, 2023. For informational purposes only and should not be considered investment advice or a recommendation to buy, sell, or hold any particular security or cryptocurrency. Forecasts are inherently limited and cannot be relied upon. Past performance is not indicative of future results.

ARK'S VIEW: BULLISH

- This chart showcases every major bitcoin cycle since 2011, fitting every cycle low to the current cycle low in late 2022.
- Depicted in the black line, the current cycle began on November 21, 2022. Compared to previous cycles, the current bull market seems to be at an early stage.
- In previous cycles, the bitcoin price topped 2-3 years after its cyclical low.

#TestYourGut: Like looking at a magician pulling tricks from a hat, we've been down this road many times, yet the element of surprise persists. We find ourselves ensnared in the short-term noise, feeling every twist and turn of the cycle as if it were our first rodeo. 2023 taught us the enduring lesson that discipline is paramount, having a master plan is essential, and the market will forever be there to put our resolve to the test.



#SimpleIsBeautiful: We like complicated theories when it comes to predicting BTC price; every time, it does the trick. It's just astonishing to see how simple rules work the best.

#CenestquunaurevoirCZ, the CEO of Binance, is leaving office. What does it mean for the industry?
A lot of what happened echoes with what occurred with Arthur Hayes and BitMex in 2020. After founding the company in 2016, BitMEX CEO Arthur Hayes left his role following charges by U.S. authorities over allegedly illegal conduct. The parent group of BitMEX announced an executive shakeup in response to the charges.

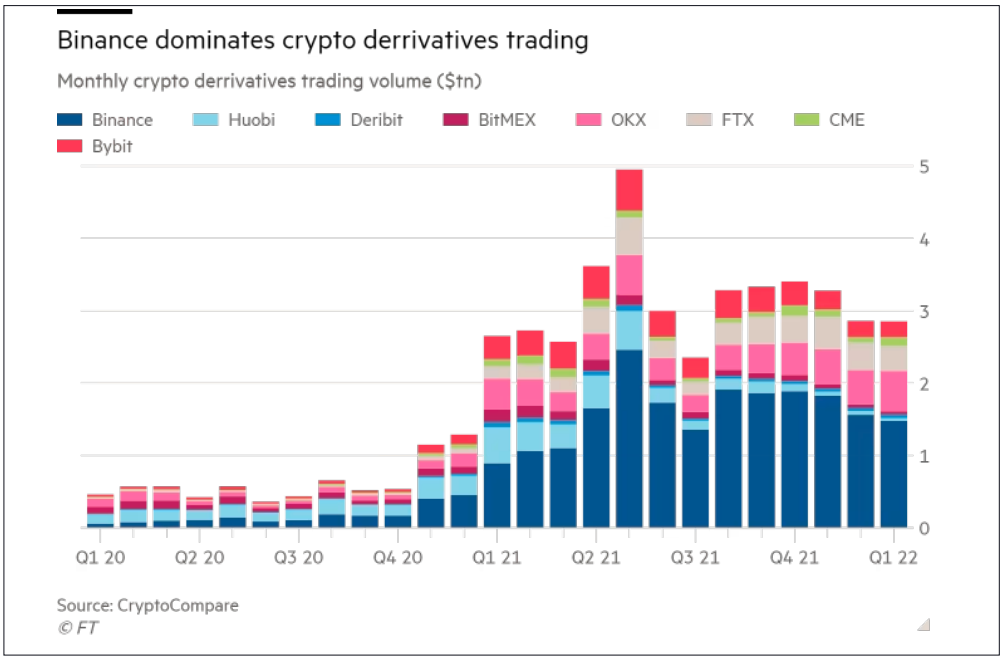
[Link to the article: https://blog.bitmex.com/five-years-ago-the-perpetual-swap-was-born-everything-changed/]

BitMex, once a reference for the crypto derivatives market, entered a downward spiral, allowing Binance to emerge as the biggest winner.

Here is a table of BitMEX's market share on BTC perpetual volume from 2017 to 2023:

Year	Market Share
2017	99%
2018	84%
2019	68%
2020	53%
2021	38%
2022	23%
2023	12%

Source: Google Bard



Will Binance follow the same destiny? The future will tell.

#TheReturnOfProphets: As the market descends, the tourists exit the circus, and the central court in the heart of the village grows eerily quiet. It's winter, and only the true OGs endure. In 2023, I had the pleasure of witnessing the resurgence of our prophets, with Arthur Hayes leading the charge—a voice I strongly recommend delving into from time to time.

https://substack.com/@cryptohayes

#The StrikeBack of Regulation
In 2023, significant regulatory changes in the cryptocurrency sector were implemented in the United Kingdom and announced in the European Union, reflecting a growing emphasis on oversight and consumer protection in this rapidly evolving market. Effective October 8, 2023, the UK introduced stringent crypto regulations requiring all entities dealing with cryptocurrencies and intending to market to UK residents to adhere to new standards. These regulations encompass a wide array of advertising mediums, including online platforms, social media, and mobile applications. To legally advertise, companies must either receive approval from the UK's Financial Conduct Authority (FCA) or satisfy specific criteria set forth by the authority. Non-compliance with these regulations exposes companies to severe repercussions, including substantial fines and legal actions. In support of these regulations, the FCA has also provided comprehensive guidelines to facilitate companies' understanding and adherence.
Simultaneously, the European Union's Markets in Crypto-Assets (MiCA) regulations, initiated in mid-2023 and set to be fully operational by the end of 2024, establish a uniform regulatory framework for cryptocurrency operations across EU member states. MiCA's mandate extends to non-EU entities seeking to operate within the EU, enforcing a comprehensive set of rules encompassing responsible cryptocurrency management and promotion, customer agreement protocols, and environmental considerations associated with digital assets.
Moreover, the EU anticipates future enhancements to these regulations, particularly in areas such as decentralised finance (DeFi) and unique digital assets like non-fungible tokens (NFTs), signalling an ongoing commitment to adapting regulatory measures in line with emerging digital finance trends.
These regulatory developments in both the United Kingdom and the European Union represent a positive and essential step towards the mass adoption of cryptocurrencies. By establishing clear and consistent frameworks, these regulations not only enhance consumer protection but also contribute to the legitimacy and stability of the cryptocurrency market. The UK's focus on advertising standards, the FCA's oversight, and the EU's comprehensive MiCA regulations collectively foster a more trustworthy and transparent environment for consumers and investors. Ultimately, these regulatory measures are pivotal in paving the way for broader acceptance and mainstream adoption of cryptocurrencies.

#SBFTrial - On July 26, 2023, Sam Bankman-Fried was found guilty for his involvement in the collapse of the crypto exchange FTX. The repercussions extend beyond individuals, impacting the entire space, especially centralised apps and exchanges like SwissBorg. We sincerely hope that we collectively learn and exercise greater caution in choosing and supporting products and ecosystems. While it may be a controversial opinion, we believe that the failure of FTX is not solely the result of one person's actions. Many participants, including venture capitalists, might need to be shouldering more responsibility for the events that unfolded. The lesson here is clear: besides appearances, no one is immune to greed and FOMO, even within the supposedly 'reputable' firms.

#BitcoinETF - The catalyst triggering a new wave of optimism and speculation in the space. As our CFO would say, the magnitude of this event can be enormous, akin to when Asia finally opened borders, allowing international capital to invest in its onshore companies. It marked the beginning of China's Stock Gold Rush for TradeFi investors. Before the possibility of a BTC ETF, it was so challenging for TradeFi investors to invest in BTC that they were cut off from this opportunity. The ETF is opening this frontier for them. The final outcome regarding price action is hard to predict, but don't worry—we've done the exercise for you. In case you missed it, here is the link to our article on the topic.

https://academy.swissborg.com/en/learn/bitcoin-etf-potential-impact-on-price

#OPOSonSolana - We often say count the wealth of a project by the number of its community members. What better illustration of that motto than the rebirth of the Phoenix Solana? We deftly see Solana becoming the 'consumer app' backbone in Web 3.0. This is extremely strategic for the "adoption era", where the protocol that creates the maximum value is the one that supports widely adopted apps.

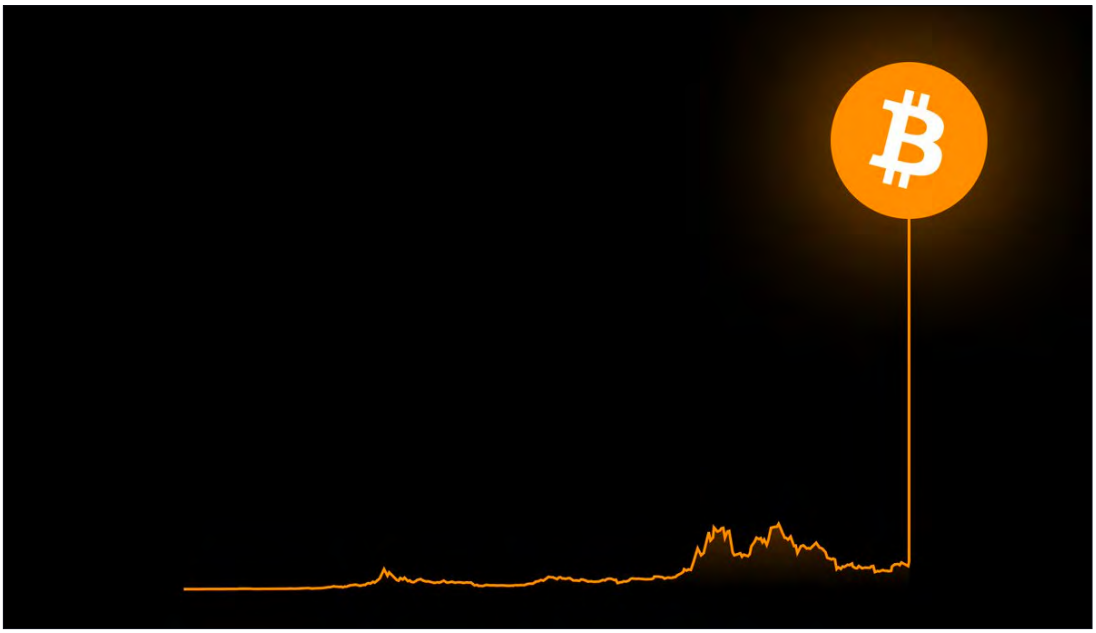
https://x.com/MessariCrypto/status/1663968723773046784?s=20

Project to follow on Sol

- Teleport. xyz & rideshare protocol (https://www.trip.dev/)

#TheMerge - Welcome to the era of ultrasound money. 2023 marks a pivotal year in the history of ETH. The most difficult challenge, 'The completion of the Shanghai milestones,' was achieved this year, allowing Proof of Stake (POS) to become fully operational by enabling withdrawals. On April 12, 2023, Ethereum successfully executed the planned Shanghai update, also known as Shapella. The upgrade allows validators the flexibility to unstake their ETH and retrieve their accumulated rewards.
Additionally, it offers the option to withdraw the staked ether, providing greater control and liquidity to the validators.

#BitSignal - A marketing master class or an ambitious bet? Whatever the reasoning behind the story, this headline marked a pivotal event in 2023, delineating the divide among very smart people. On one side are those who still believe BTC will eventually fade and the design of Fiat will triumph. On the other, the new school is ready to proclaim to the world what the new crypto cycle is all about.

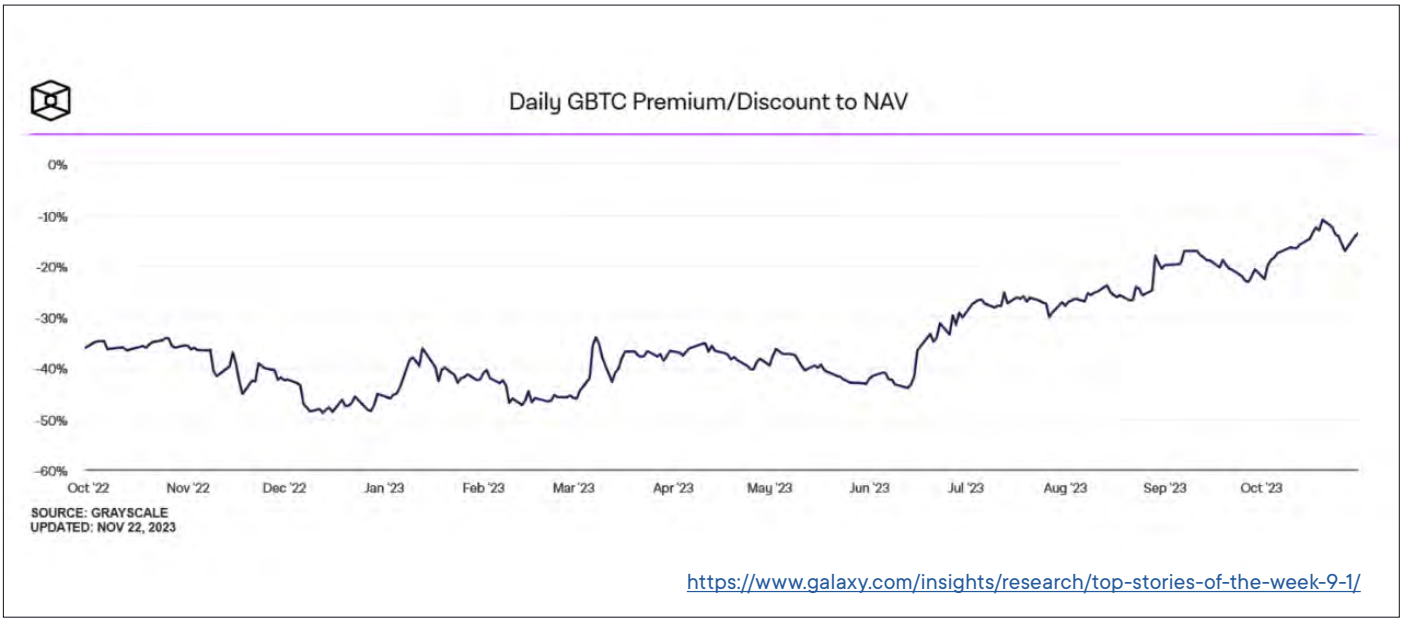


https://twitter.com/balajis/status/1636429773865095168?lang=en

#XRPVsSEC - Another positive outcome for the industry, not only for XRP. If it's ruled that XRP is not a security, many tokens similarly classified as securities may see relief. As of today, July 13, 2023, US District Court Judge Analisa Torres has ruled that Ripple did not violate federal securities law by selling XRP tokens. This is a major victory for Ripple and the cryptocurrency industry as a whole. Can we call it a win? Too soon to tell, but it's solid progress in positive territory. An episode will be closely followed in 2024.

https://cointelegraph.com/news/ripple-wins-case-against-sec-as-judge-rules-xrp-is-not-a-security

#GBTC - The best trade of the year! Grayscale Bitcoin Trust Discount to BTC is closing the gap, with Grayscale winning their battle over the SEC. Beyond being one of the best trades of the year for TradeFi investors (capitalising on exposure to BTC while profiting from a significant discount due to liquidity conditions ns ahand are beginning of this year), this event was unquestionably one of the top Black Swan events that could have potentially devastated the Bitcoin Price Chart. If Grayscale had failed to win its battle, it would have been forced to liquidate its BTC position to reimburse its investors, consequently crashing the price of BTC to the ground. For the record, GBTC is the second-largest BTC holder in the world with over 600K BTC, just after Mr. Satoshi Nakamoto himself with over 1.1M BTC.



https://www.galaxy.com/insights/research/top-stories-of-the-week-9-1/

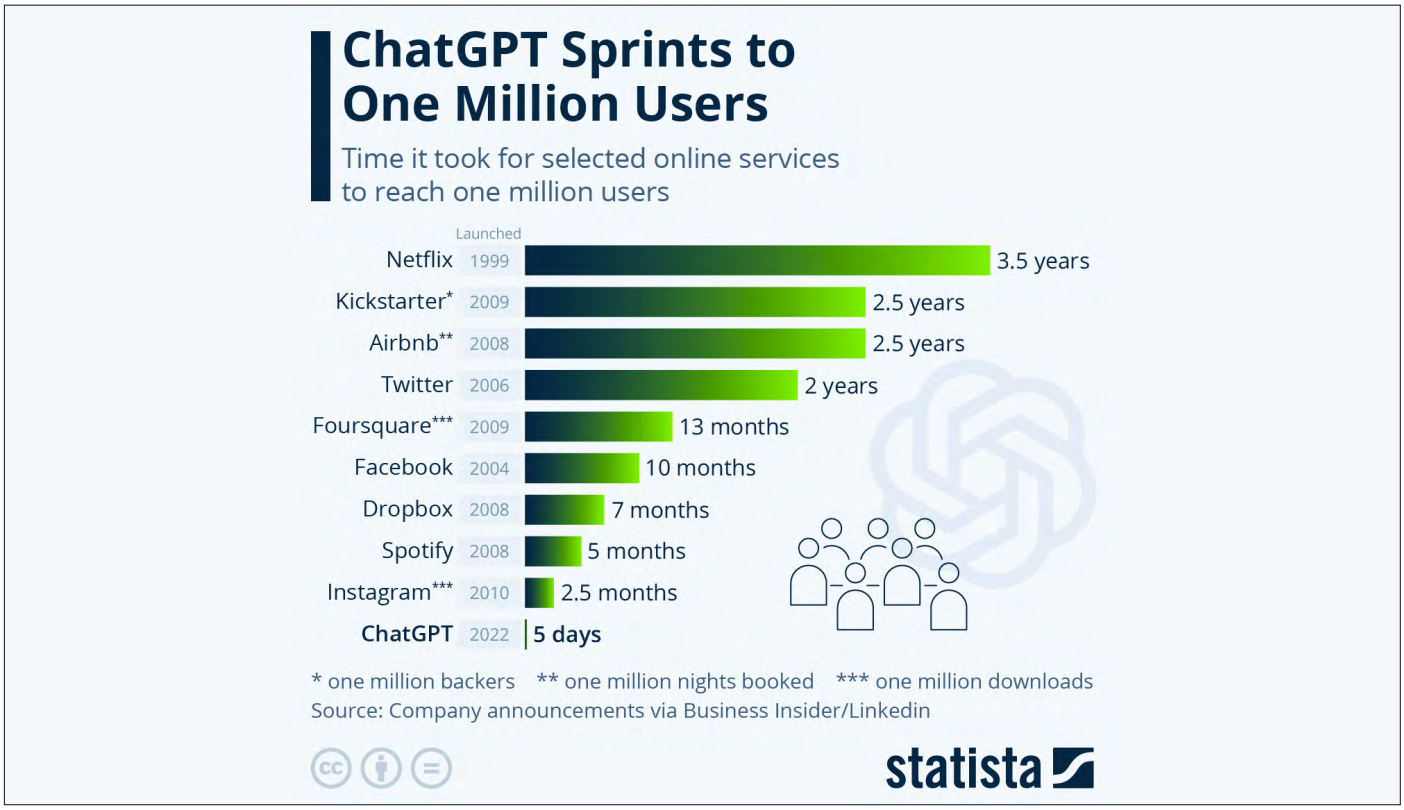
#PEPE - One doesn't make the rules! Nevertheless, what's systematic in every cycle is the creation of a meme coin in the phase preceding the bull run—Dodge for the 2017-18 Bull Market, Shiba for the 2021-22 Bull market, and Pepe for the 2023-2024 Bull Market? Only time will tell. We've undoubtedly witnessed the impressive run of $pepe, demonstrating that meme coins are here to stay and serve as a powerful avenue for straightforward adoption in the space.

Pepe Coin Reached
$1 Billion Market Cap In 19 Days

Source: Spiel Times

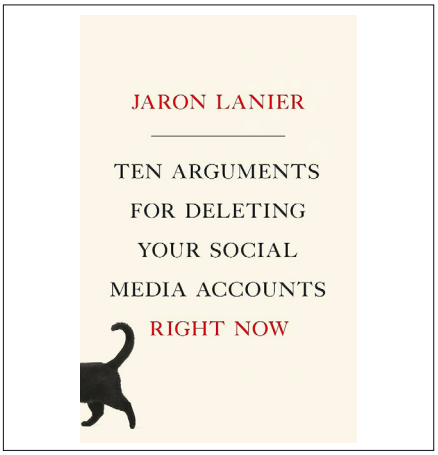
https://cryptorank.io/news/feed/3a353-crypto-hacks-drop-70-in-q1-2023-research

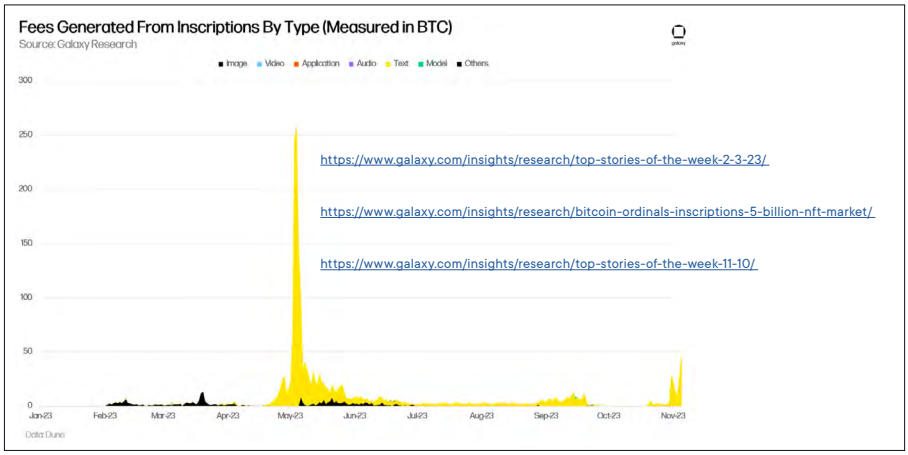
It took Bitcoin 1,579 days post-launch to reach a market cap of $1 billion in 2019. Ethereum took 287 days to climb above the $1 billion market cap. In addition, Pepe's competitor, Dogecoin, needed 2,585 days to reach the $1 billion market cap numbers.
On the other hand, Shiba Inu took only 279 days to reach the same target.



#theBirthofSocialFi More critical than traditional fiat currencies, it's time to disrupt the system governing the modern age's most valuable currency: attention. Understanding why this is fundamental goes beyond your portfolio; it's crucial for humanity. Leinster & Farcaster are paving the way for the SociaFi narrative.

#Ordinals: NFTs on Bitcoin! If Bitcoin is the default digital store of value, it's not necessarily the best payment infrastructure. Therefore, the Bitcoin chain must reinvent itself around utilities beyond a simple transactional backbone. A very controversial topic, Ordinals are here to stay and are an overall bullish initiative for BTC.





#BASE When the most Cefi project is launching its own chain, there is something to notice here :)

#AiFi, the most mysterious theme in crypto, probably pumped the most with the least fundamentals. At this stage, it feels like almost all these tokens are memes, but you definitely can't ignore it—the performances are impressive."

	Performance 2023
Render	905%
TAO	405%
SingularityNet	570%
Nosana	2580%
OLAS	126%
FET	610%

#BuildersTime - Last but not least, What we appreciate most about Bear Markets is the noise-clearing effect, allowing only projects with clear and solid intent to stand out.

> "Bear market create builders,
> builders create bull market,
> bull market create scammers,
> scammers create bear market."

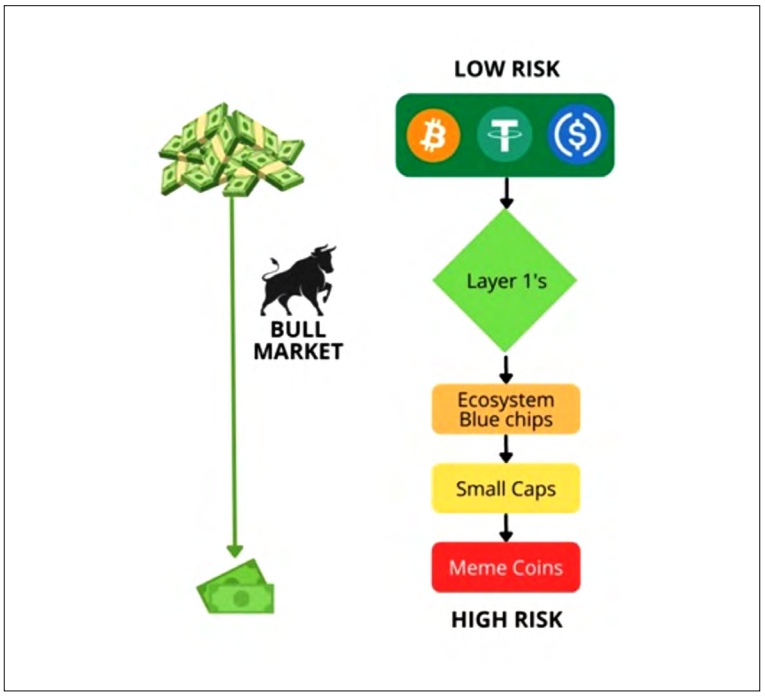
What to expect in 2024

Anthony Lesoismier Twitter: @AnthoLGSB

Florent Koenig Twitter: @tradetheflow_

One does not make the rules; let's start at the end! We hope 2024 to be the continuation of the bull market that has already started by the end of 2023.

The market logic will remain the same as in the last cycle, beginning with a strong momentum in Bitcoin that progressively transitions toward Altcoins and distributes itself across different ecosystems.



However, even when, in retrospect, it would all seem to make sense, living through an actual bull market is far from easy: it is fast, powerful, and riddled with uncertainty along the way. This report is intended to prepare you for what's coming.

Before we get there, let's start with the fundamentals of why we believe that the timing is right for a bull market to start and, more fundamentally, why this is a lifetime opportunity.

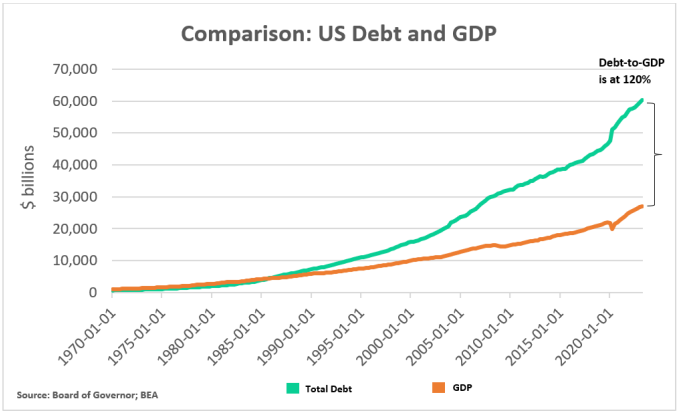
While those ideas will be discussed at length further down our report, we think that repetition is not a bad thing. Much like for a magician pulling a trick, repetition is, in fact, the key to mastery. So let's briefly recap our main ideas here.

Different secular trends are pushing crypto

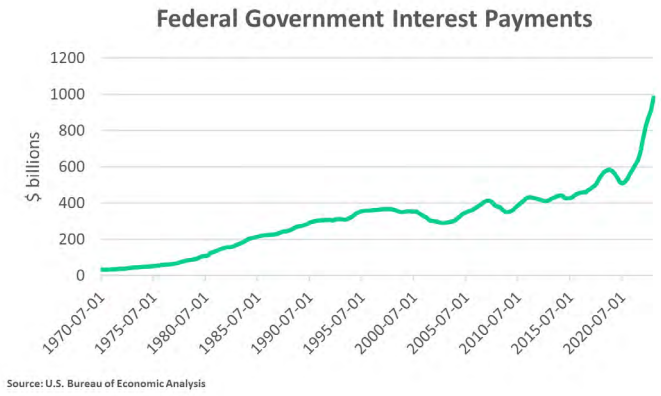
Crypto is at the intersection of two secular trends happening simultaneously: central banks being forced to inject liquidity to save the current fiat system and a technological boom. This makes crypto one of the most asymmetrical bets of our lifetime.

Liquidity injection from central banks

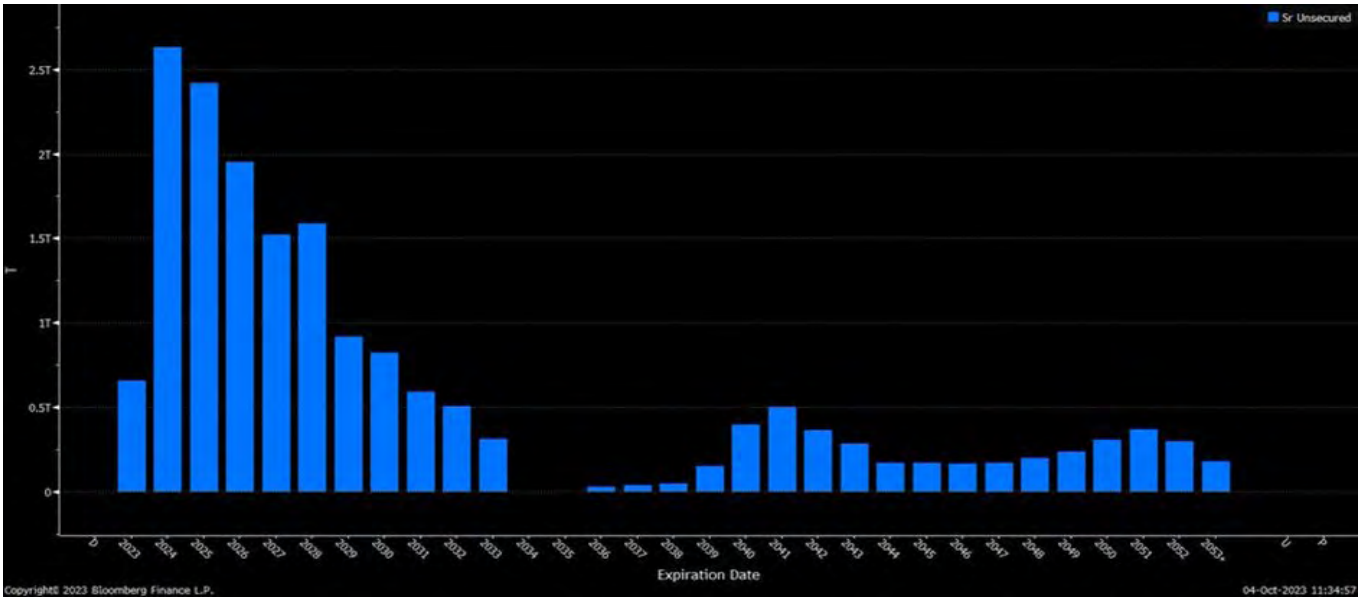
Different elements make us think that central banks, especially the Fed, simply have no choice but to inject more liquidity into the system. One of the main reasons behind this is the increasing debt in major economies. Using the US as a proxy we see that debt has grown at a faster rate than GDP over the last 20 years hitting a debt-to-GDP ratio of 120%.



As a result, interest expenditures have also risen considerably and are taking a greater proportion of GDP each year.

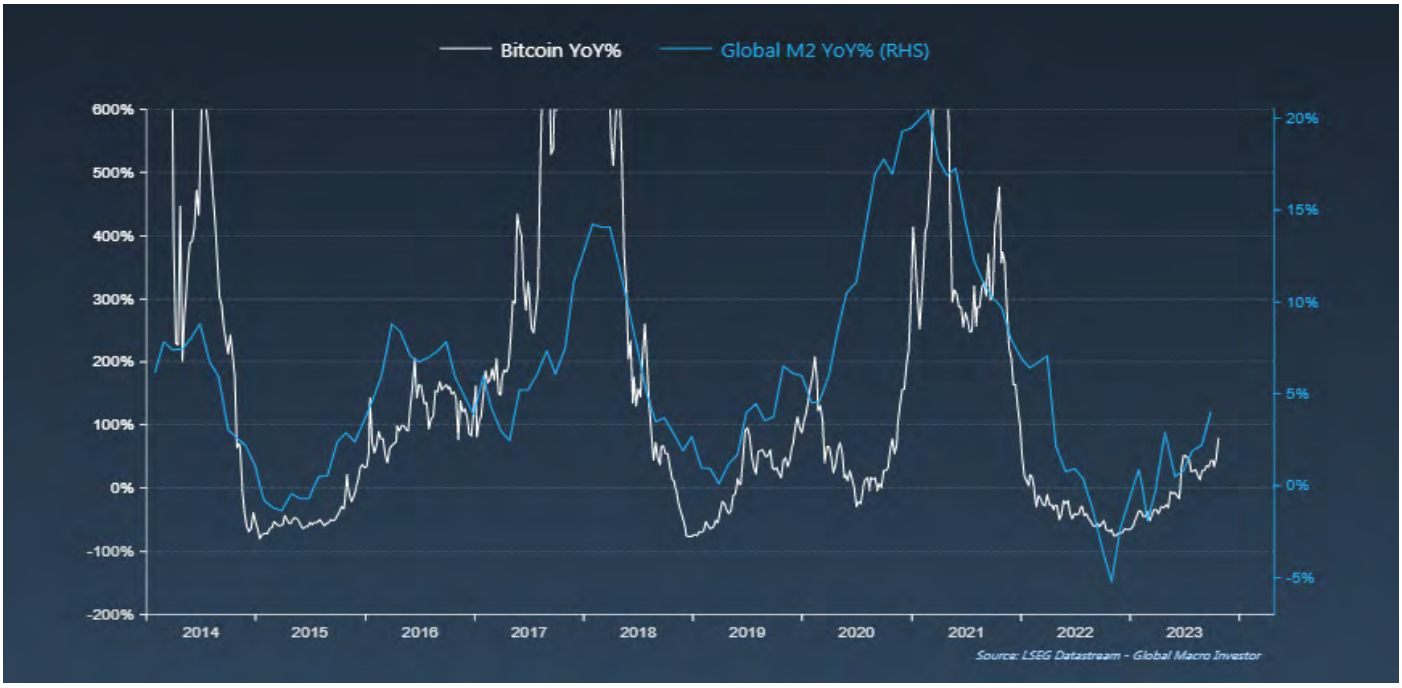


Further, the principal from these debts needs to be repaid at some point in the future. It is estimated that more than ⅓ of all outstanding debt needs to be repaid before 2026.



The US has no financial capacity to repay them, so they are constrained to issue more debt. This creates a difficult situation for the government. If they decide to roll over the debt at the current yield demanded by natural buyers, interest expenditures will rise drastically, leading to a sovereign debt crisis. Therefore, the current yield must decrease so the government can roll over the debt. If they don't, the Fed will be forced to step in and monetise the debt through yield curve control. In both cases, this means further monetary debasement and liquidity injection.

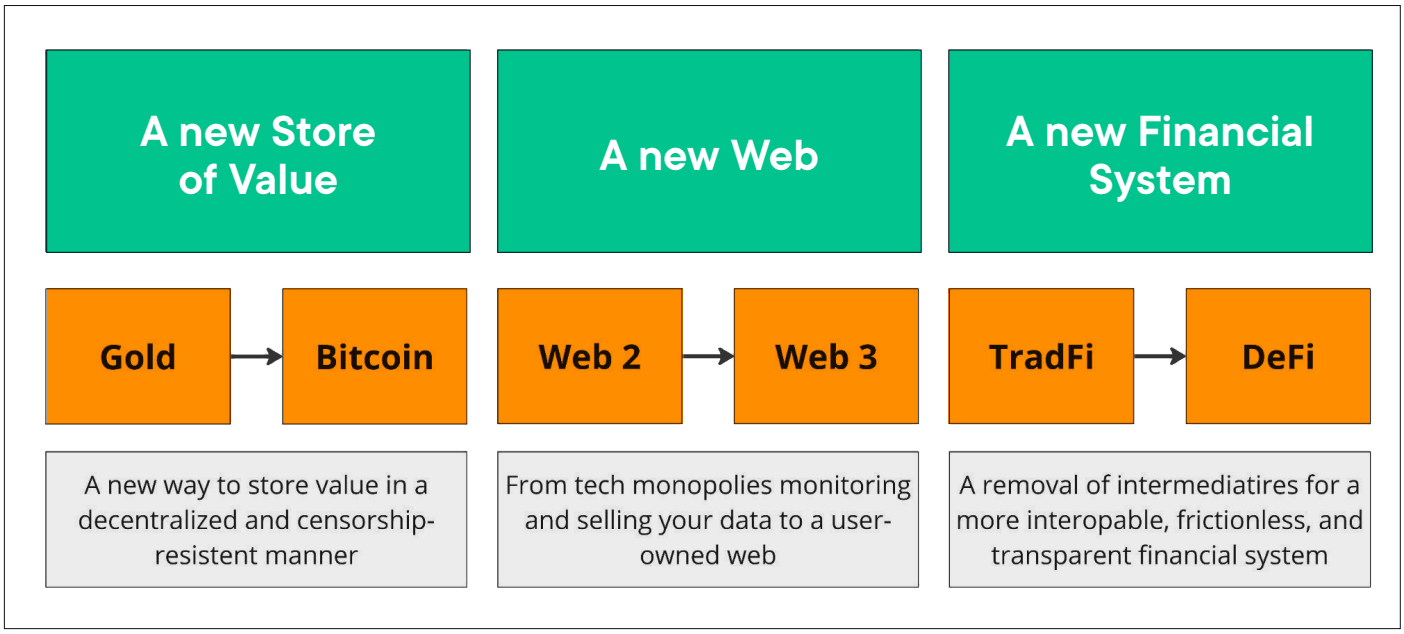
We note that this trend has already started. The Global M2 YoY change bottomed in November of last year, together with Bitcoin. Since then, it has been on an upward trajectory.



As crypto is the most sensitive asset to liquidity and as we are at the forefront of a new liquidity cycle, it seems that being exposed to crypto from now till the end of the liquidity cycle should be a wise investment decision regarding risk-to-rewards ratio.

A technological megatrend:

Crypto users are growing at a pace similar to that of the internet in the 1990s. At this rate, we are expected to reach the 1 billion users milestone by the end of 2024.
Further, crypto infrastructures are at the heart of three major societal shifts.



Unless something changes and we all go back to the stone ages, and nobody wants to use technology again, or if central banks find a magic trick to save the fiat system, crypto is currently one of the most promising asset classes.

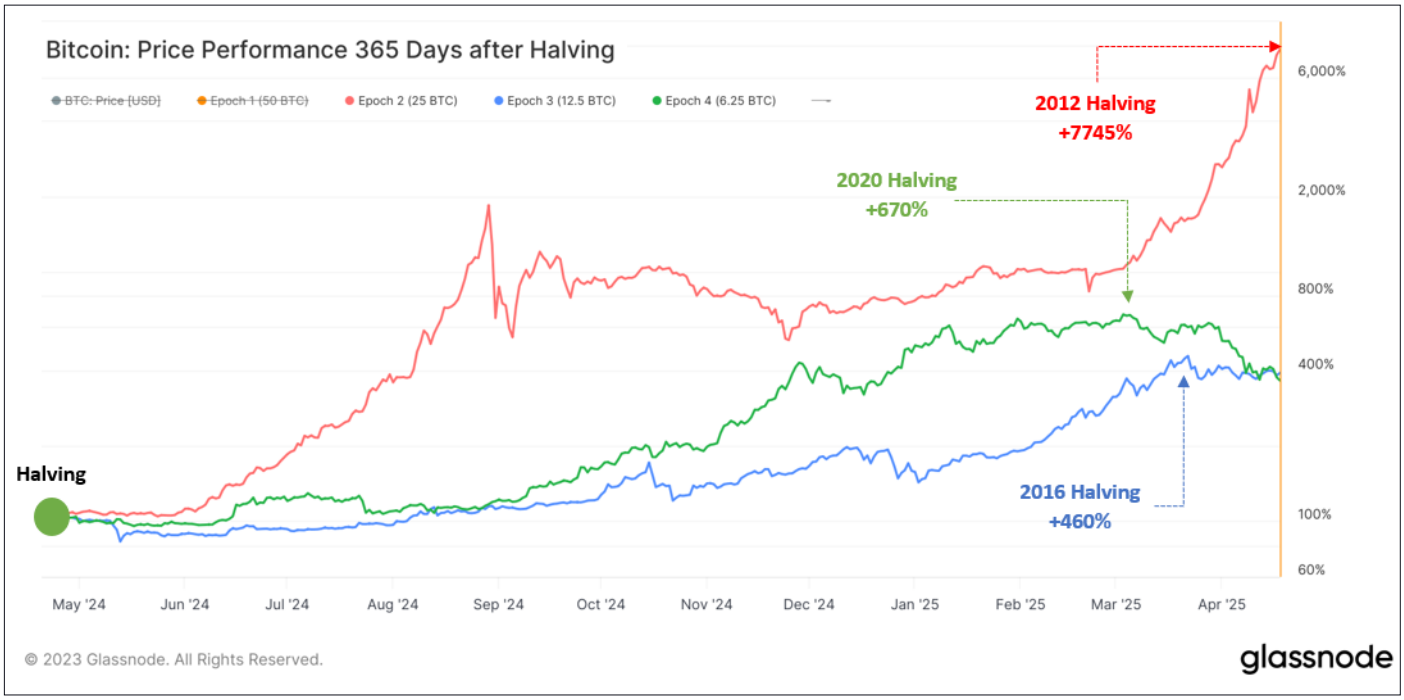
The current business cycle nexus confirms this view.

In addition to liquidity and technological adoption, other elements are pointing towards a continuation of a bull market.

The halving is fast approaching

Historically, Bitcoin has followed a 4 year price cycle where the beginning of each cycle coincided with the halving. So far, Bitcoin seems to be following precisely the same path as during the preceding pre-halving period.

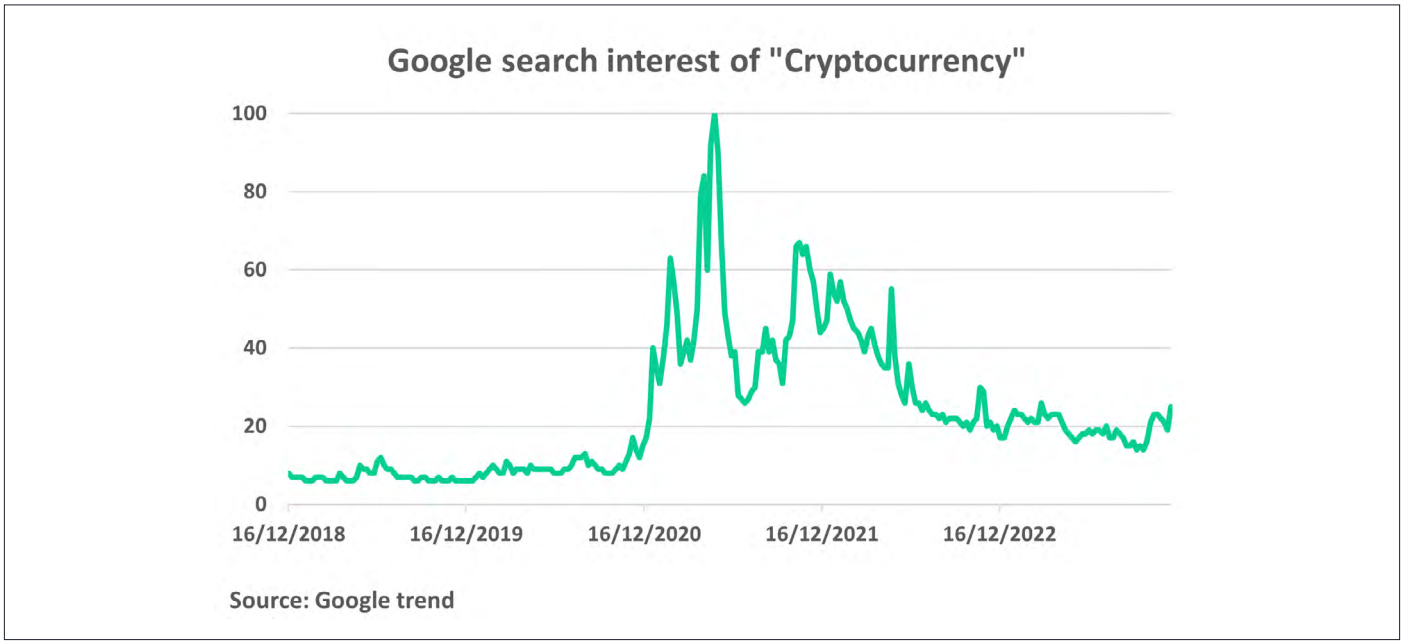
The next halving is expected to occur in April 2024, and we are confident that this will be one of many factors contributing to increased buying pressure on Bitcoin.



Retails are not in yet

Crypto is a game of attention, and similar to a magic trick, success lies in timing and perception. We want to act like magicians in the market: entering the market when attention is low, knowing that magic happens when no one is watching. Conversely, we want to exit when attention peaks, just as the illusion loses its charm when watched too closely.

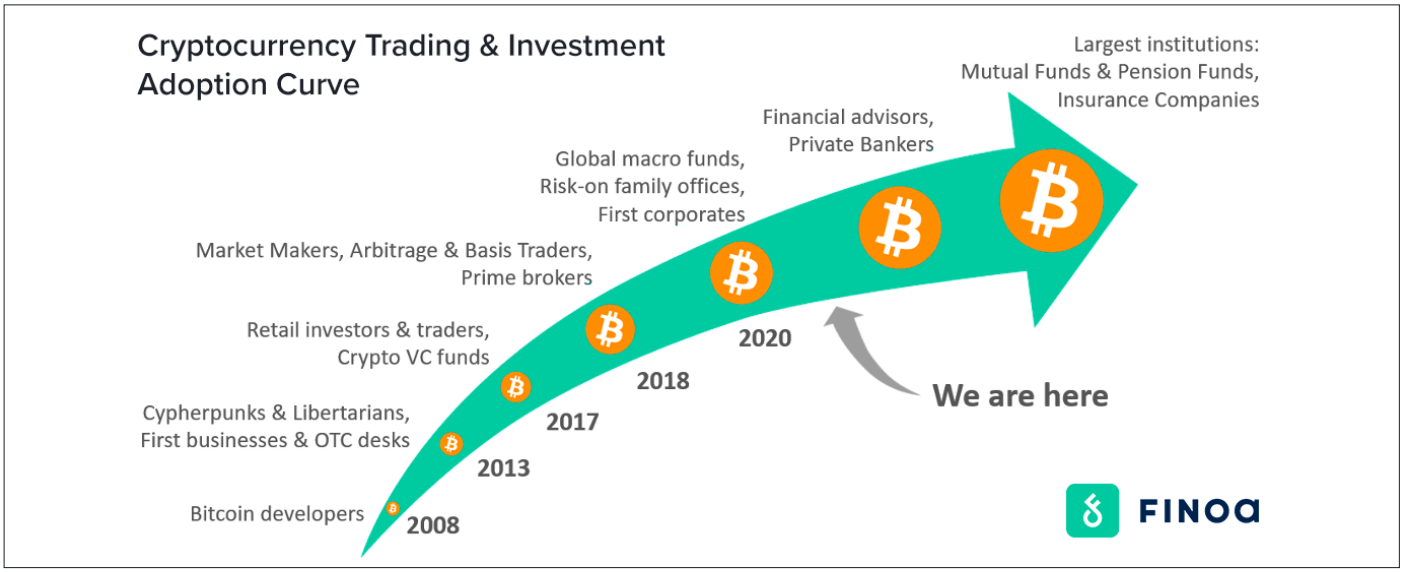
In crypto, the best attention indicator is search interest over time. Currently, we see that it is still low and nowhere near the last bull market peak. This is bullish and signals that we are indeed still early in the cycle.



Coinbase App ranking in the App Store -
https://twitter.com/mhonkasalo/status/1731799814222864815/photo/1

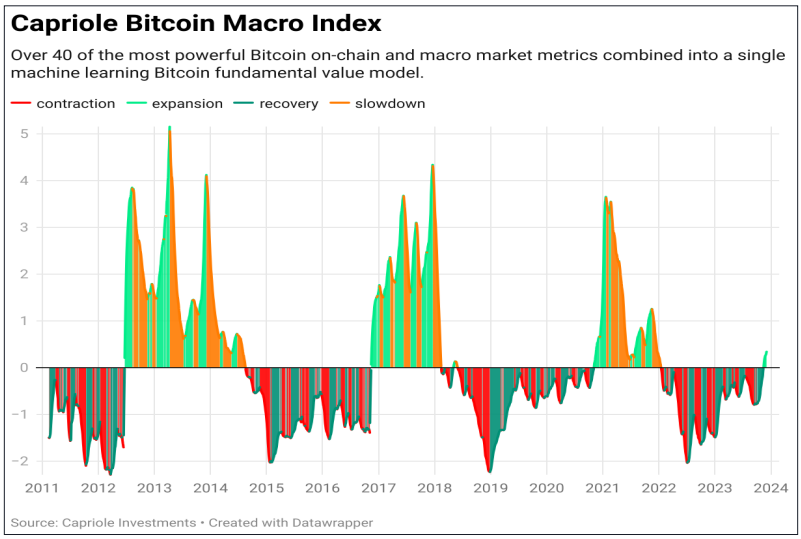
Institutional adoption

We witnessed some key moves from the world of traditional finance in 2023. In June, BlackRock, the largest financial institution in the world, filed for a Spot Bitcoin ETF. A few days later, we saw many other big names from "TradFi" follow. Now, all they are waiting for is the green light from the SEC. We believe this is another factor reinforcing the bullish case for crypto in 2024.



Important Indicators are Flashing Bullish Signals

Looking at the 2-week Bitcoin RSI; we observe that it crossed the 70 mark for the fourth time since 2012. Each past signal flashed during the start of a new bull market.



Capriole Bitcoin Macro Index, a combination of on-chain fundamental and macro market metrics, also flashed an “expansion” signal for the fourth time. Again, all preceding signals marked the beginning of a bull market.

Why is it so important to capture it well?

As further monetary debasement seems inevitable, adoption is growing fast, institutional investors are entering the market, and a combination of important indicators are flashing bullish signals. The message seems clear: crypto is one of the most asymmetric bets for 2024.

Satoshi was a magician, and his biggest trick, Bitcoin, created the spark of a new financial and social revolution. The end game could well be that it transforms the way human beings perceive, store, and exchange value over time and space, and this is our last chance to get in before institutions.

II. Macro Outlook 2024

Florent Koenig Twitter: @tradetheflow_

"The root problem with conventional currency is all the trust that's required to make it work. The central bank must be trusted not to debase the currency, but the history of fiat currencies is full of breaches of that trust." Satoshi Nakamoto

The world economy is in a situation that we've never witnessed in our lifetime and seems on the verge of a big paradigm shift. The causes are well-known: increasing debt problems in major economies, a major demographic shift, a shaky US Treasury market, and geopolitical tensions.

But before trying to understand the implications of these factors on financial markets and, more generally, crypto, it is crucial to briefly take a few steps back and understand how we ended up here.

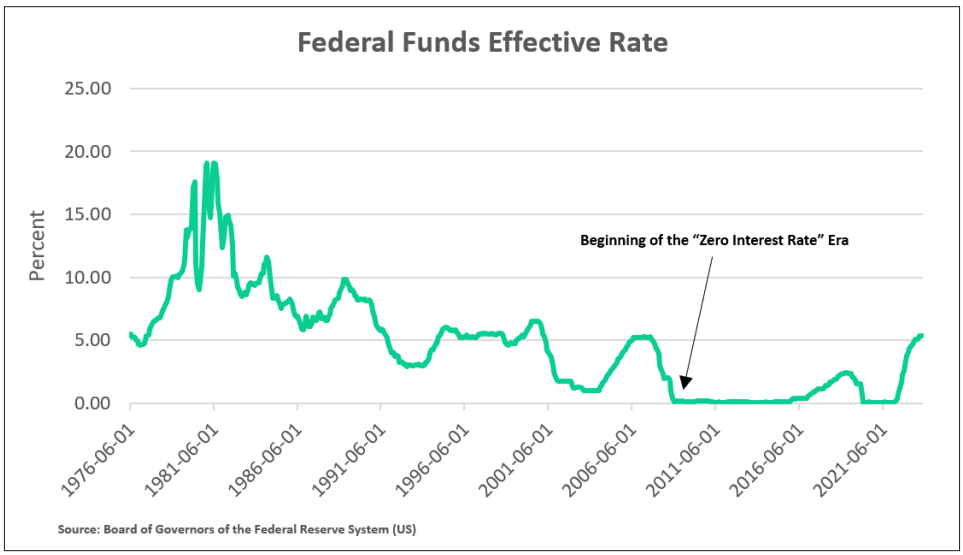
The Beginning of the End

It all started around 50 years ago when the Bretton Woods agreement ended, cutting the link between the US dollar and gold. This marked the beginning of a new chapter in the history of money: **The Fiat Era.**

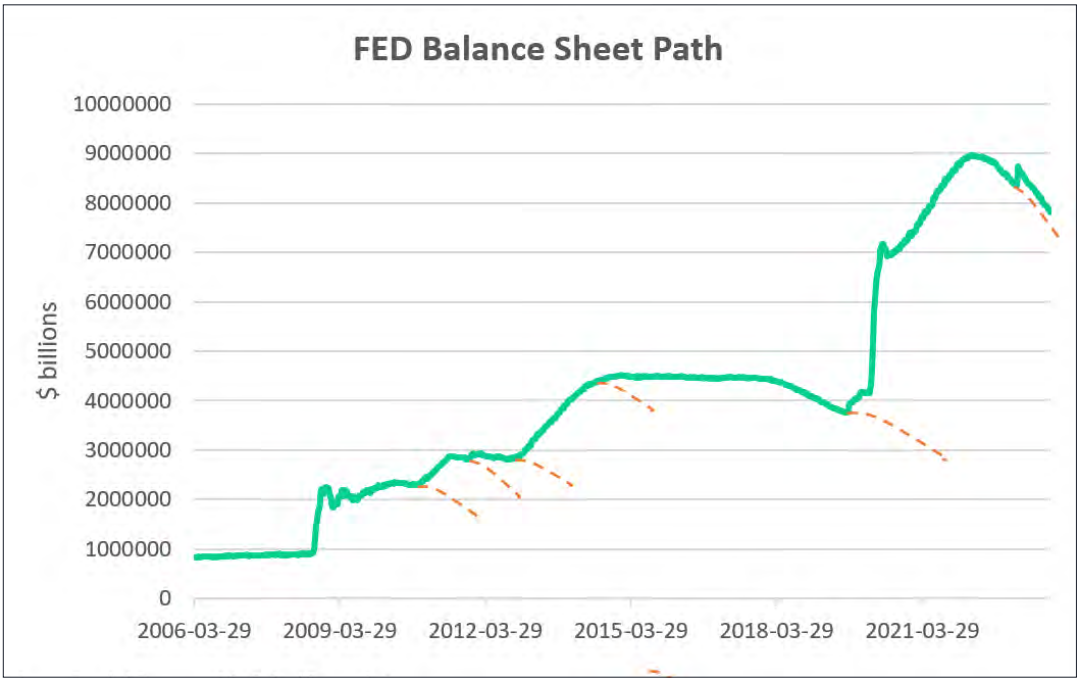
In this paradigm, the money supply (MS) is no longer dependent on gold reserves. Instead, it is entirely controlled by central banks, who, motivated by their mandate of price stability and maximum sustainable employment, are free to use different monetary policies to manipulate the quantity of money in the system.

Before 2008, central banks were more or less conservative in how they used their different tools to influence the economy and only manipulated short-term interest rates. From 1970 to 2008, their Balance sheet, a proxy measure for money supply, grew at a steady rate of around 4%. During that time, market participants monitored monetary policies, but it was far from the most important variable affecting financial asset prices.

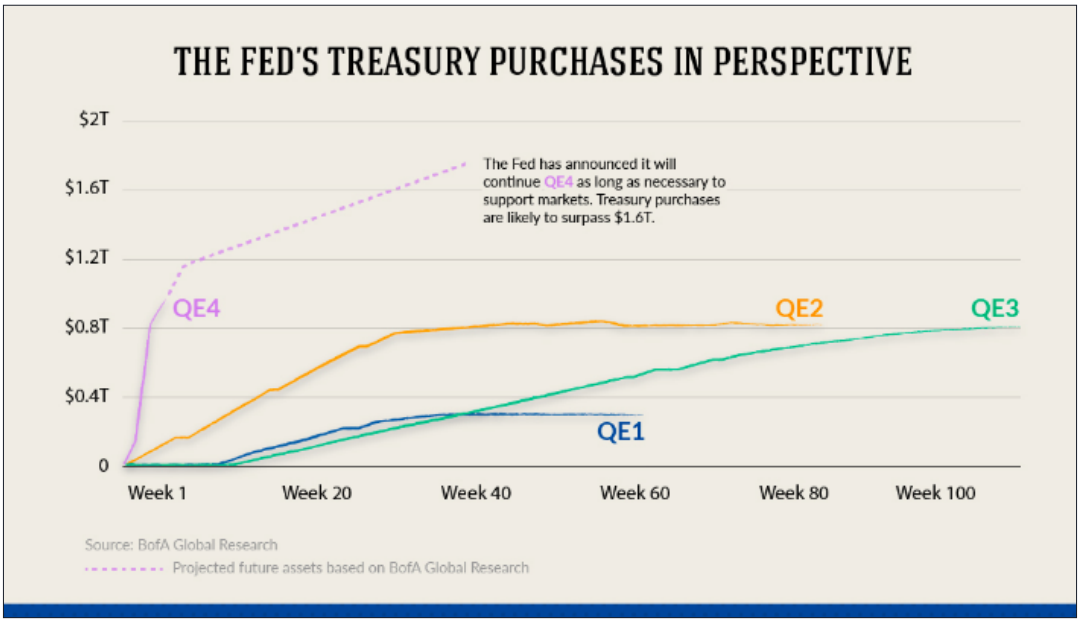
Things changed in 2008. At the onset of the biggest financial crisis ever, the Fed, constrained to save the financial system from a collapse, used a brand new set of monetary policy tools such as Quantitative easing (QE) and Zero Interest Rates Policy.



Over the subsequent years, those tools were used at scale, and zero interest rates became the norm in developed economies. But things went one step further during the COVID pandemic, where the Fed - driven by its dual mandate of price stability and maximum sustainable employment - utilised all its available tools to restore market confidence by printing money like never before. In less than four years, the FED Balance sheet doubled and crossed the $8 trillion mark.

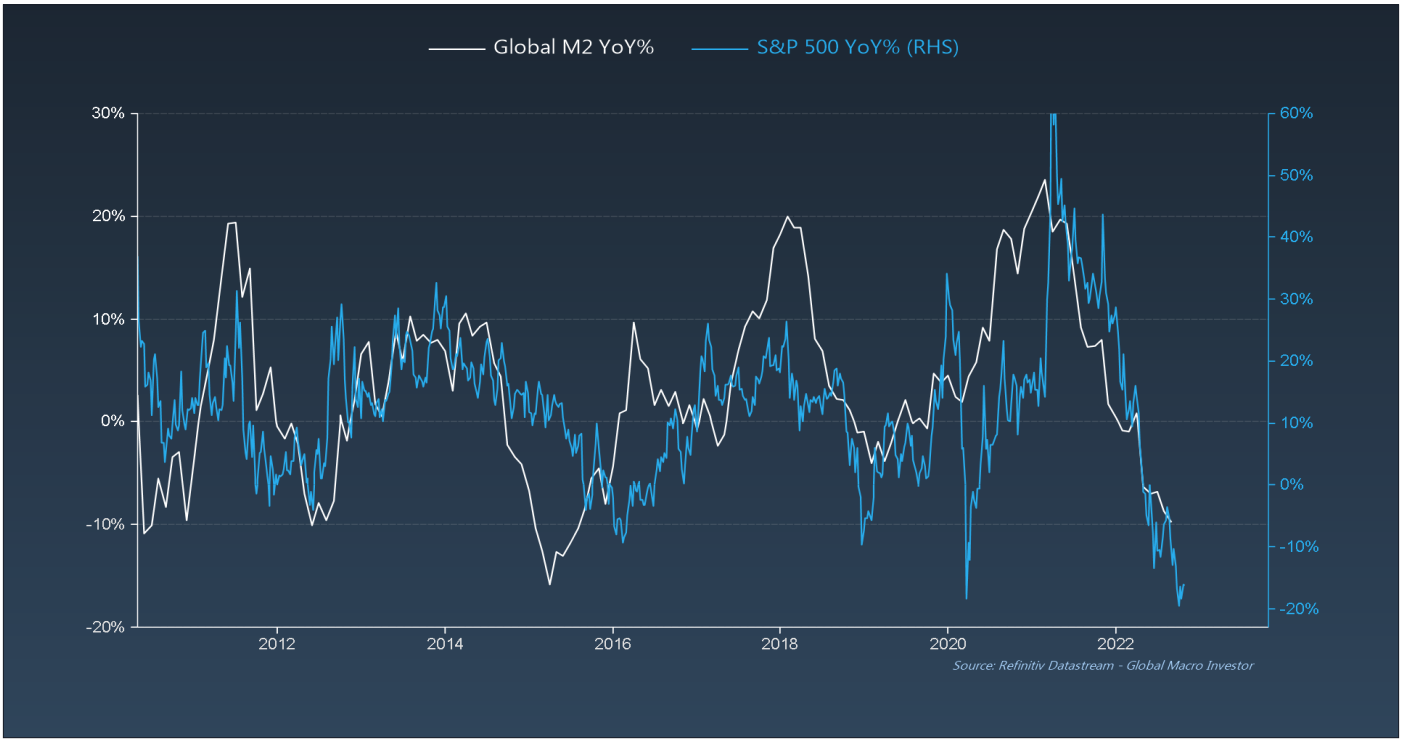


This is a pace that we've never witnessed before.



Liquidity is one of the most important drivers of financial markets, and in the current fiat era, the actions of central banks influence the latter. Therefore, the paradigm shift in 2008 and the amplified scale of intervention of central banks strengthened the link between monetary policies and financial markets, making the action of central banks one of, if not the most important drivers of asset prices, especially regarding risky assets.

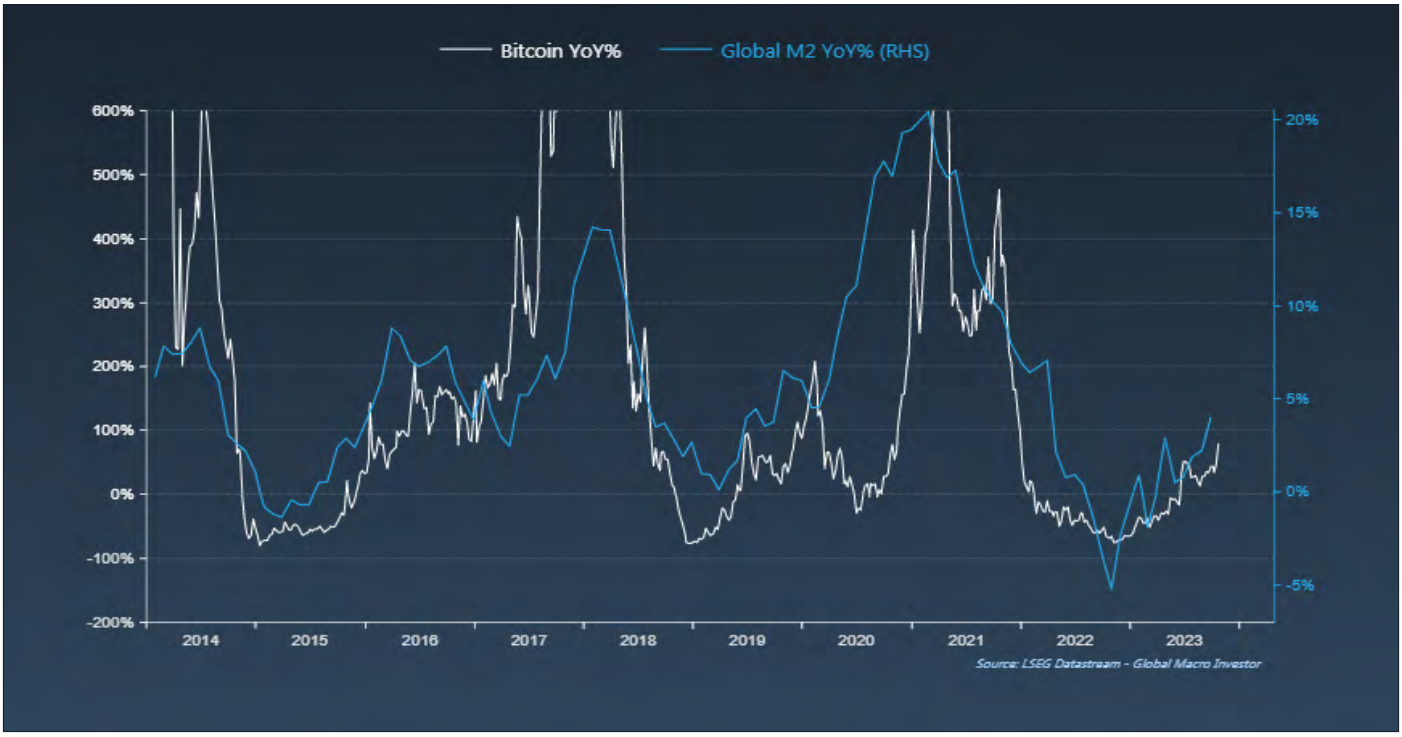
Here, we observe a tight link between the global growth in money supply and the S&P 500, a proxy measure for the US equity market:
M2 is one of the most popular liquidity measures. It is directly influenced by interest rates set by central banks and other unconventional monetary policy tools such as QE. We note that US equities tend to perform well and inversely whenever the money supply grows.



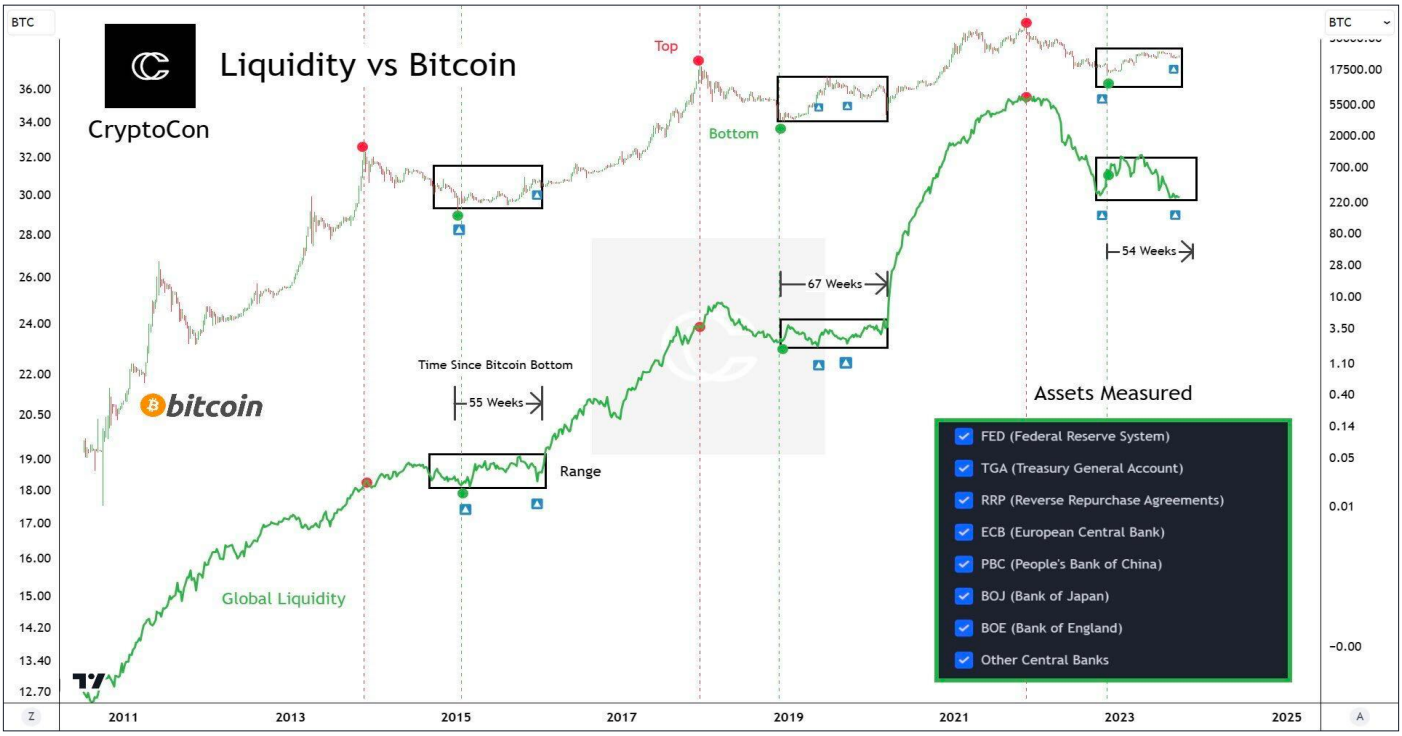
As the crypto asset ecosystem is also considered a “risk on” asset, it is also highly reliant on the liquidity environment set by the central banks.

We have a lot of empirical evidence supporting this:

Using a chart from GMI, we observe a strong positive relationship when we plot M2 YoY growth against Bitcoin. This shows that the rate of change in liquidity is what counts.

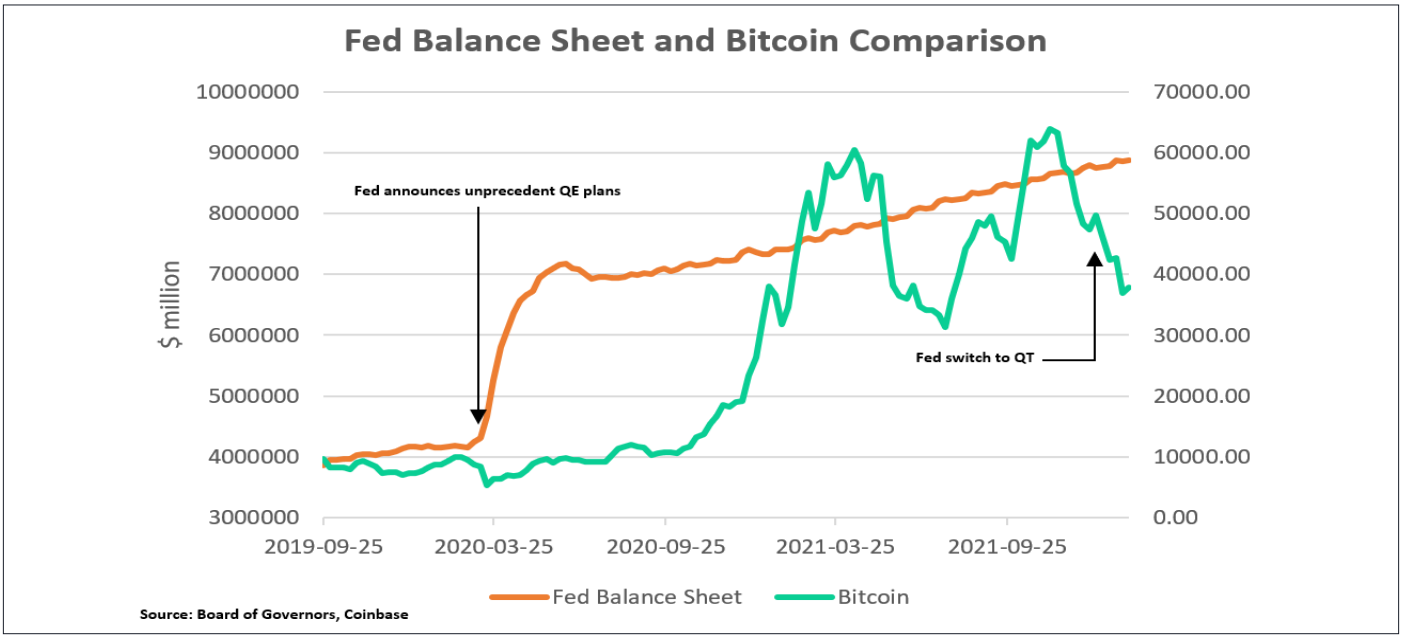


Further, we note that all previous Bitcoin bottoms coincided with a local bottom in global liquidity. All bull markets subsequently started when the rate of change of global liquidity was increasing.



Source: CryptoCon

In 2020, on the day that Jerome Powell publicly announced the beginning of QE, the crypto market started a bull run that lasted until the FED reversed their monetary stance. During this period, the FED balance sheet more than doubled, and Bitcoin appreciated by 1125%.



Given Crypto's strong relationship with how fiat money works, understanding the current monetary policy conditions is crucial to forming an opinion on the state of the market.

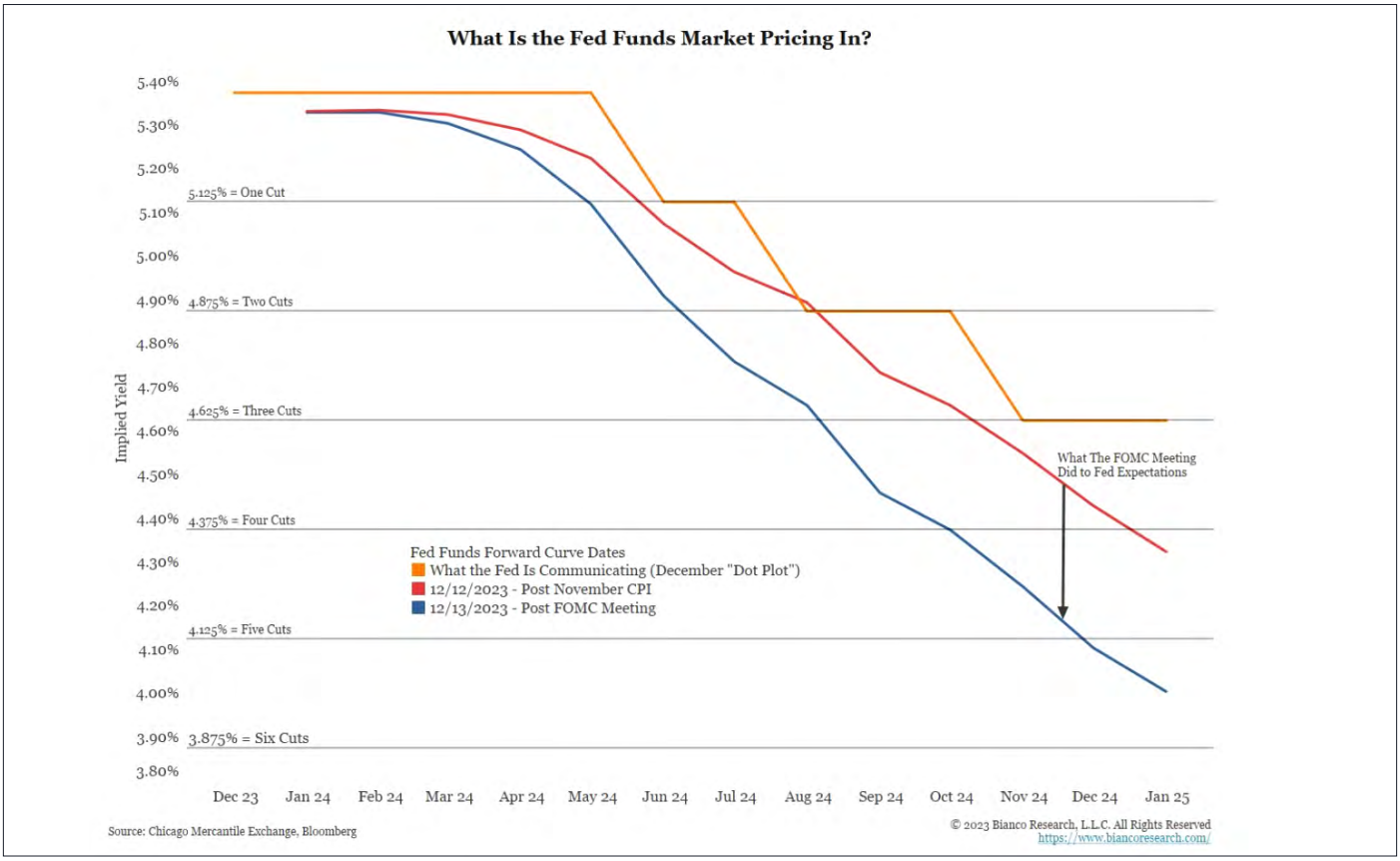
What to expect in 2024 for Monetary Policies

Forecasting future monetary policy is daunting as it hinges on the complex interaction of economic variables. One way to approach this is to aggregate different views from experts in the field before building our own in-house view.

Consensus view:

Interest rate is one of the most important variables to monitor to assess liquidity in the system. Two questions on every market participant's mind are: "When will the first rate cut occur?" and "How strong will the cut be?"

If we look at the market consensus view, we see that the first rate cut is expected in Q2 2024.
Further, the easing cycle is expected to be progressive, with small cut increments of 0.25% over the subsequent months.



As markets are forward-looking, we can assume that this is what they have already priced in. Now the question is: what is our view relative to this?

What the different experts say:

Raoul Pal - Founder and CEO of Global Macro Investor & Real Vision:

> "Since 2008, the economy has been going through a **3- 4 year debt refinancing cycle.** We are currently at the **beginning of a new cycle** where trillions of dollars of debt need to be rolled. In this environment, the US cannot afford to do that. In my view, the more likely scenario is that **the FED has managed to overshoot inflation expectations** through its tightening measures and will soon be in a position to **cut rates by Q2 2024** as demand collapses, the economy weakens, and inflation turns negative. This will be a cover to roll over the debt and avoid an undesirable alternative: yield curve control. In any case, this means that **more capital will enter into the system** and flow into technological assets like crypto."

Arthur Hayes - Co-founder of BitMEX & CIO Maelstrom

> "**The debt situation is bad in the US.** Over the last decades, debt levels have increased massively. Around $14 trillion worth of debt is expected to mature before 2026 in the US only. But as the US is not in a financial situation to reimburse its debt, it will need to roll over the latter. While we don't know which road it will take to pay that back, my view is that **the only way for the US to avoid a sovereign debt crisis is to print more money**, which is a way for the government to mutualise the debt through debasement. I believe that the most likely scenario is **yield curve control.** This makes me bullish crypto for 2024 as we could be heading into the biggest bull market ever."

Daniel Lacalle - PhD in Economics, Fund Manager & NYT Bestselling author

> "The Federal Reserve is likely to start **cutting rates by the end of 2024** as the economy weakens, but it is unlikely to start purchasing bonds. If anything, there will be a **modest rate pivot** but not an entire U-turn in policy if the economy remains in growth territory. Only a significant slump in aggregate demand, which means a recession, would make the Federal Reserve reverse the normalisation of the balance sheet.
> Regarding the US debt, I agree that **the situation is unsustainable.** The administration estimates that the accumulated deficit between 2023 and 2032 will reach $14 trillion. **Public debt is rising much faster than GDP**, and this is a symptom of a much larger problem. This does not mean that the U.S. will default, but it certainly means that **the fiscal situation needs to be corrected as soon as possible."**

Goldman Sachs Economics Research:

> "The global economy has **outperformed our optimistic expectations in 2023**. GDP growth is on track to beat consensus forecasts from a year ago by 1pp globally and 2pp in the US. Most major DM central banks are likely finished hiking, but under our baseline forecast for a strong global economy, **rate cuts probably won't arrive until 2024H2.** When rates ultimately do settle, we expect central banks to leave policy rates above their current estimates of long-run sustainable levels. **The market outlook is complicated** by compressed risk premia and quite well-priced markets for our central case. We expect returns in rates, credit, equities, and commodities to exceed cash in 2024 under our baseline forecast."

Our In-House view

If we had to describe the last 12 months of Monetary Policy decisions in three words, it would be, "postponing the inevitable".

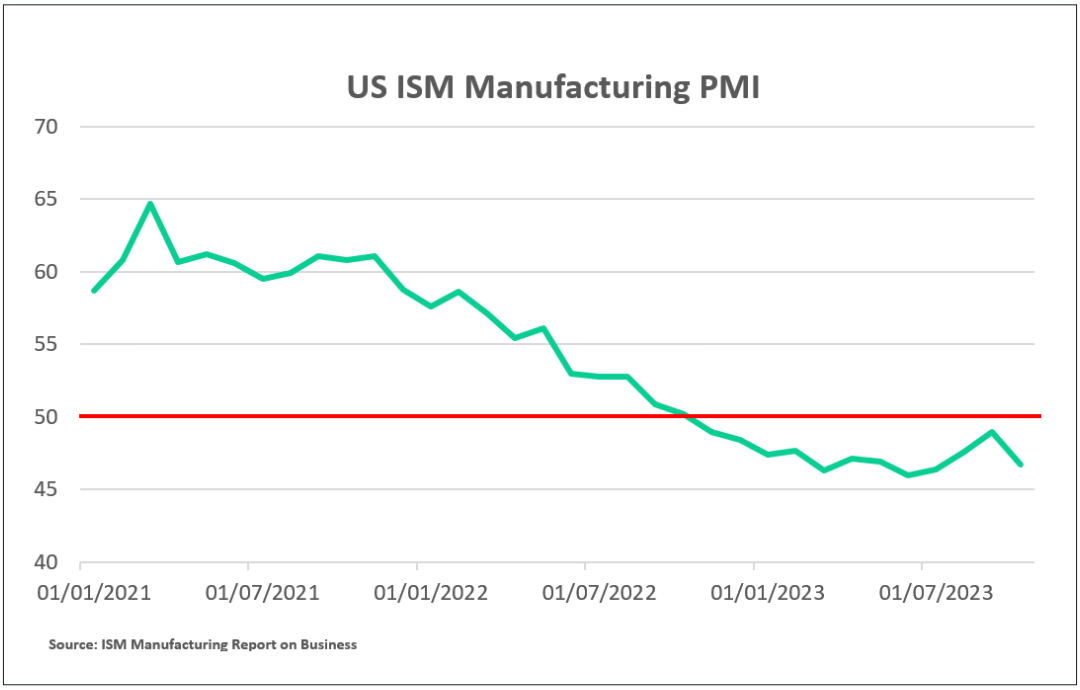
Our core view is in line with experts such as Arthur Hayes and Raoul Pal. We think that it is clear that all central banks will have to act and are already working to inject liquidity into the system. They simply have no choice.

However, it remains uncertain whether they will use traditional tools like rate cuts and QE or innovate entirely new strategies, which will require constant monitoring. Here, we will focus on the US and the Fed because they lead the dance. These factors influence our view:

The worst of inflation is behind us

We believe the Fed has managed to contain inflation with its aggressive rate hikes.
Further, despite the re-ignition of the Israeli-Palestinian conflict, the commodity market (as shown by the DP index) has stayed stable. It has resumed its secular downtrend. While concerning, the current situation in the Middle East is not expected to heighten volatility in the commodity market. Therefore, this makes deflation a more significant risk than inflation going forward.

The current ISM index reinforces our point: The index is a measure of economic activity and has been on a steady decline for over 2 years. After a quick rebound in Q2, it has resumed its downtrend. This suggests weaker economic conditions ahead.

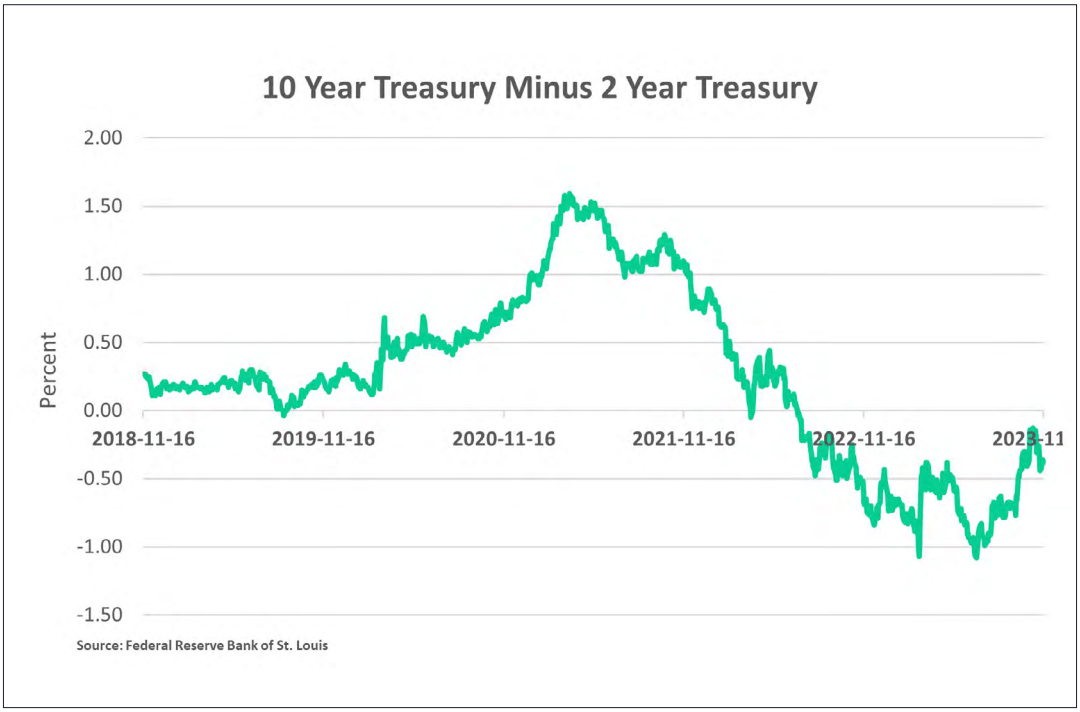


Weaker economic conditions destroy demand, fragilise the job market, and make prices shrink. This will discourage the FED from further tightening.

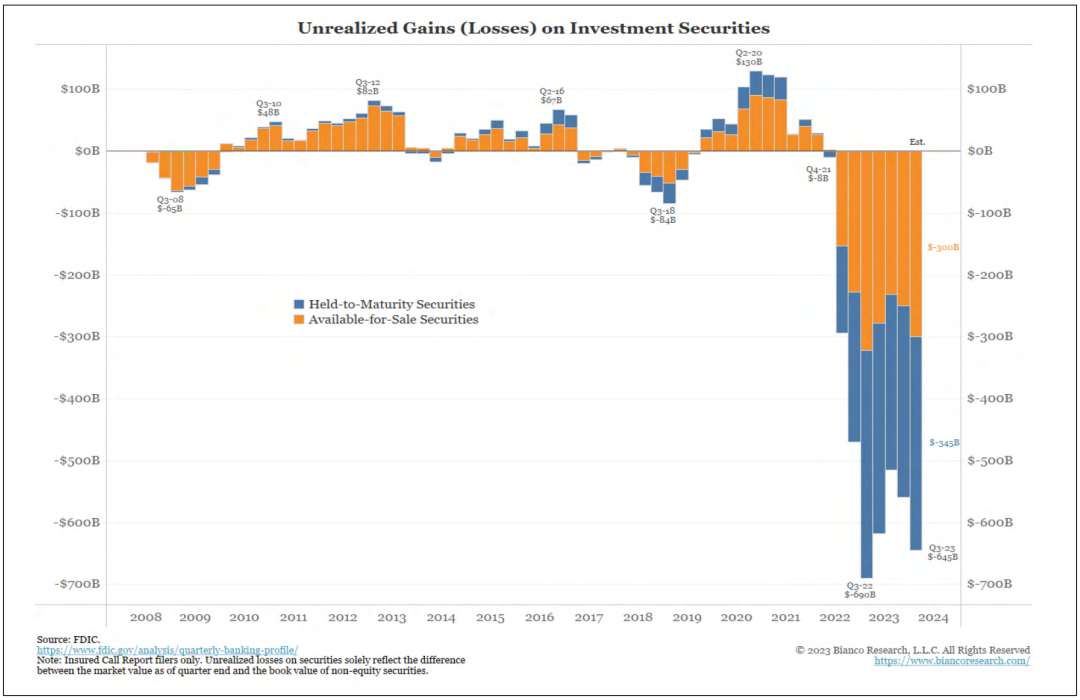
US Banks are still in a difficult position

The US Banking sector has been in a downtrend since 2022 and is currently trading at the same level as the post-banking crisis in March 2023.

Further, the yield curve is still inverted, as shown by the negative 10-year yield minus the 2-year yield. This puts pressure on the business model banks who borrow short and lend long:



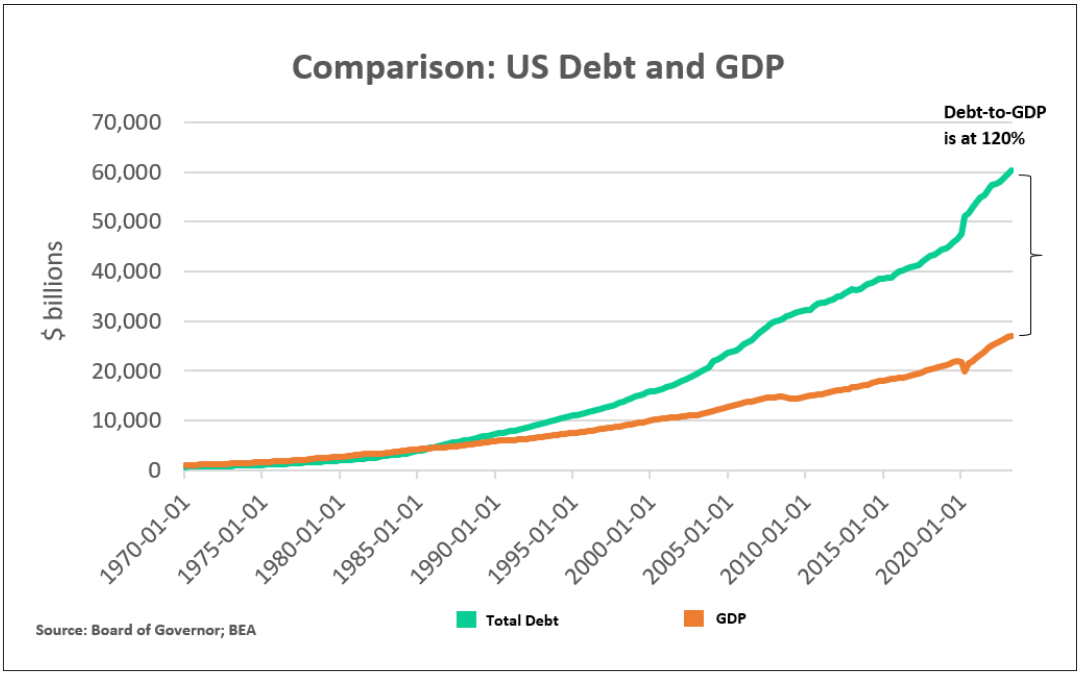
Banks are also sitting on a lot of unrealised loss:

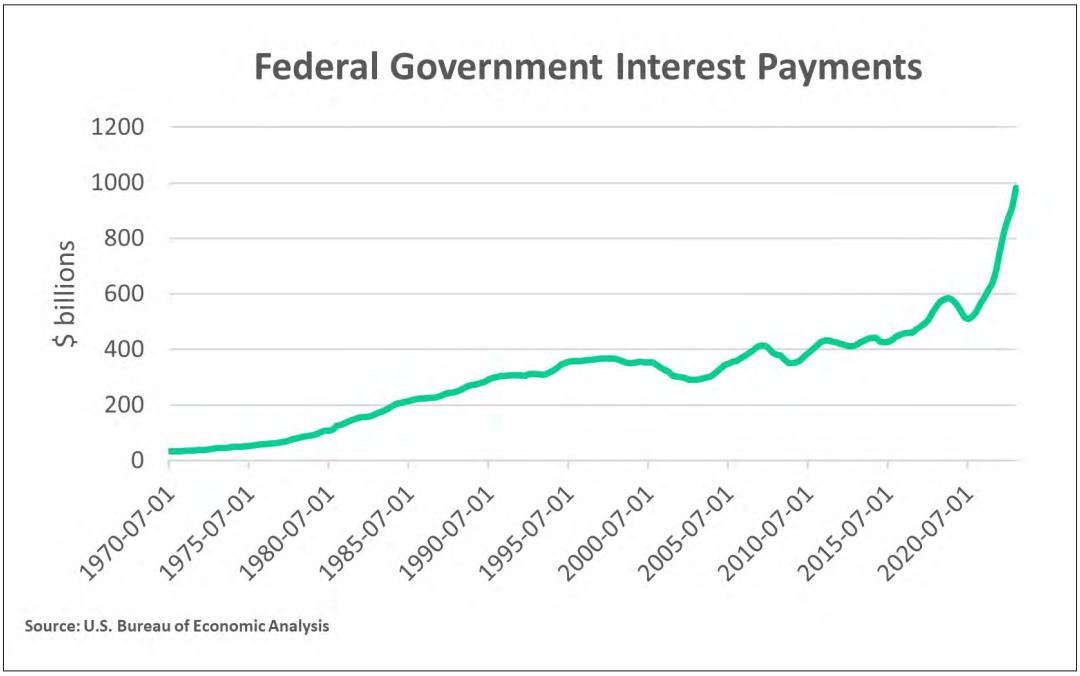


In our view, this confluence of elements is making the Fed less prone to further tightening as they understand the second-order impact that it could have on the banking sector.

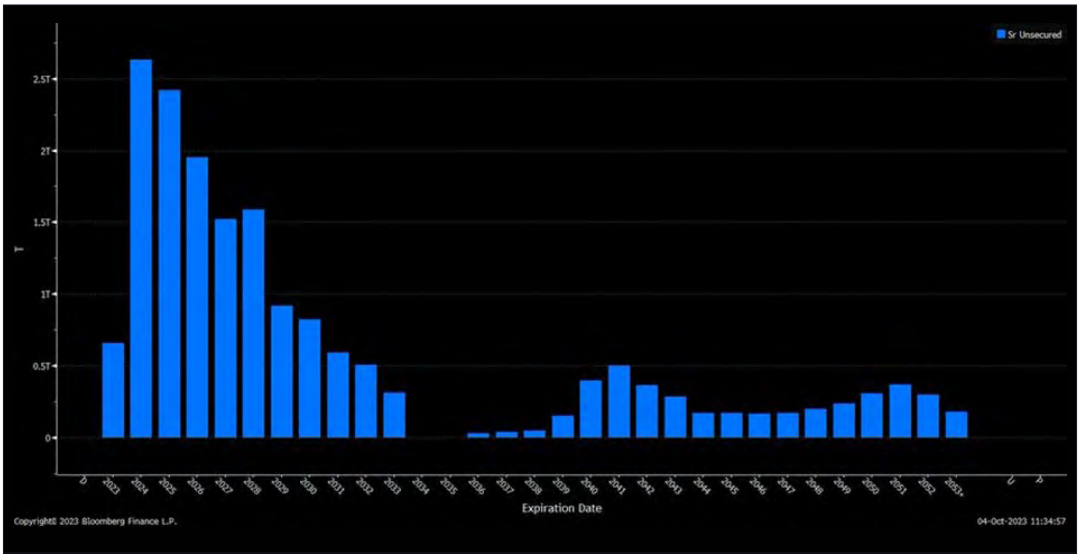
Global Debt is at unsustainable levels

Government debt worldwide has increased massively in the last decades. Using the US as a reference, we see that debt has grown at a faster rate than GDP over the previous 20 years to hit a debt-to-GDP ratio of 120%. As a result, interest rate expenditures have risen considerably, and this trend is expected to continue going forward.





However, the principal of those debts needs to be repaid at some point in the near future, and governments do not have the financial resources to repay them.



Arthur Hayes estimated that around $33.6 trillion of debt will need to be paid back before 2026, with $13.9 trillion coming from the US. This is roughly 1/3 of all outstanding treasuries that will need to be rolled over their debt with fresh new borrowing.

Line Item	Dollar (trillions)
2023 US Treasury Borrowings	**$ 1.85**
2024-2026 Maturing Debt	**$ 7.00**
2024-2026 Estimated Budget Deficit (Surplus)	**$ 5.05**
Total Debt to be Issued	**$ 13.90**

While governments can find natural buyers at higher rates, this is not feasible as interest expenditure will rise considerably, and the deficit will be unmanageable. The consequence of this is a sovereign debt crisis.

Therefore, we believe governments will do their best to roll over their debt in the markets. For that to happen, yield needs to decrease. If they can't do that, they will be forced to find alternatives, and it seems that the only choice left is for central banks to fully monetise their debt with full yield curve control (YCC). Both scenarios inevitably lead to the same outcome: another cycle of monetary debasement and, hence, liquidity injection in the market.

Geopolitical tension

Last but not least, geopolitical tensions also increase the probability of further US Government borrowing. This puts even more pressure on the US Treasury Market and reinforces our previous point: central banks will be constrained to ease their policy in some way or another to finance their debt.

Conclusion

Our macro outlook for 2024 is there will be a calling for another cycle of liquidity injection in the market. That being said, the biggest question remains: Will the new cycle of QE be done using tools employed during the past 10 years (i.e., rate cuts), or will central banks innovate and use new tools, as we have seen with the RRP recently? This is a crucial element to watch for investors in 2024.

As crypto is the most sensitive asset to liquidity and is starting to be perceived as a flight-to-quality asset that offers a hedge against the monetary debasement trap from central banks, we believe that this asset class is one of the most asymmetric bets for 2024. As the saying goes, "We shall not fly against the wind," it seems the wind will be favourable for crypto in 2024.

III. Evolution of the Adoption and Usage of Crypto

Florent Koenig Twitter: @tradetheflow_

"All revolutions are impossible until they happen - then they become inevitable" - Albie Sachs

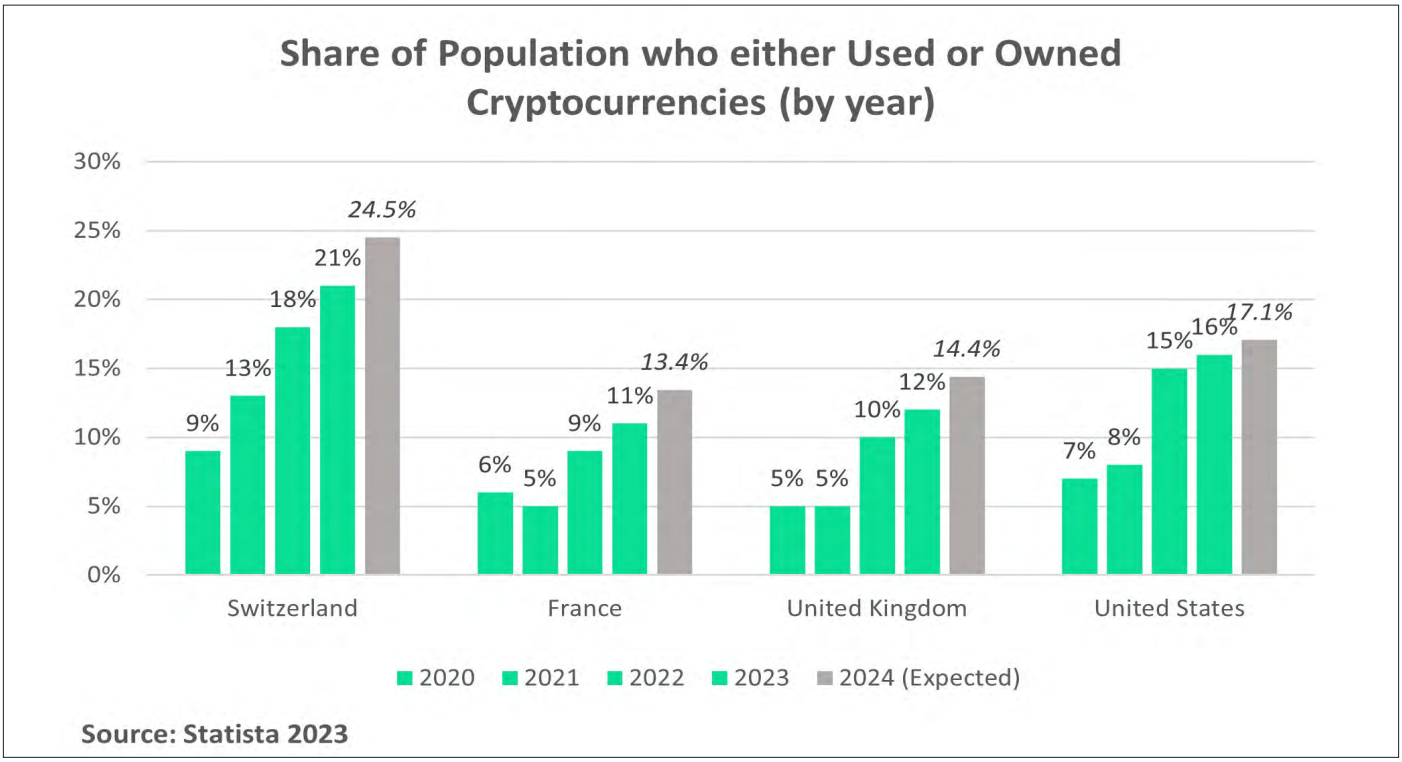
Sometimes, it is hard to realise that fifteen years ago, this industry that we love to call "crypto", did not even exist. Yet, today, this rapidly evolving sector is starting to challenge the status quo of many industries, and it seems increasingly evident that it is bound to have a lasting impact on society.

Crypto Adoption Trend

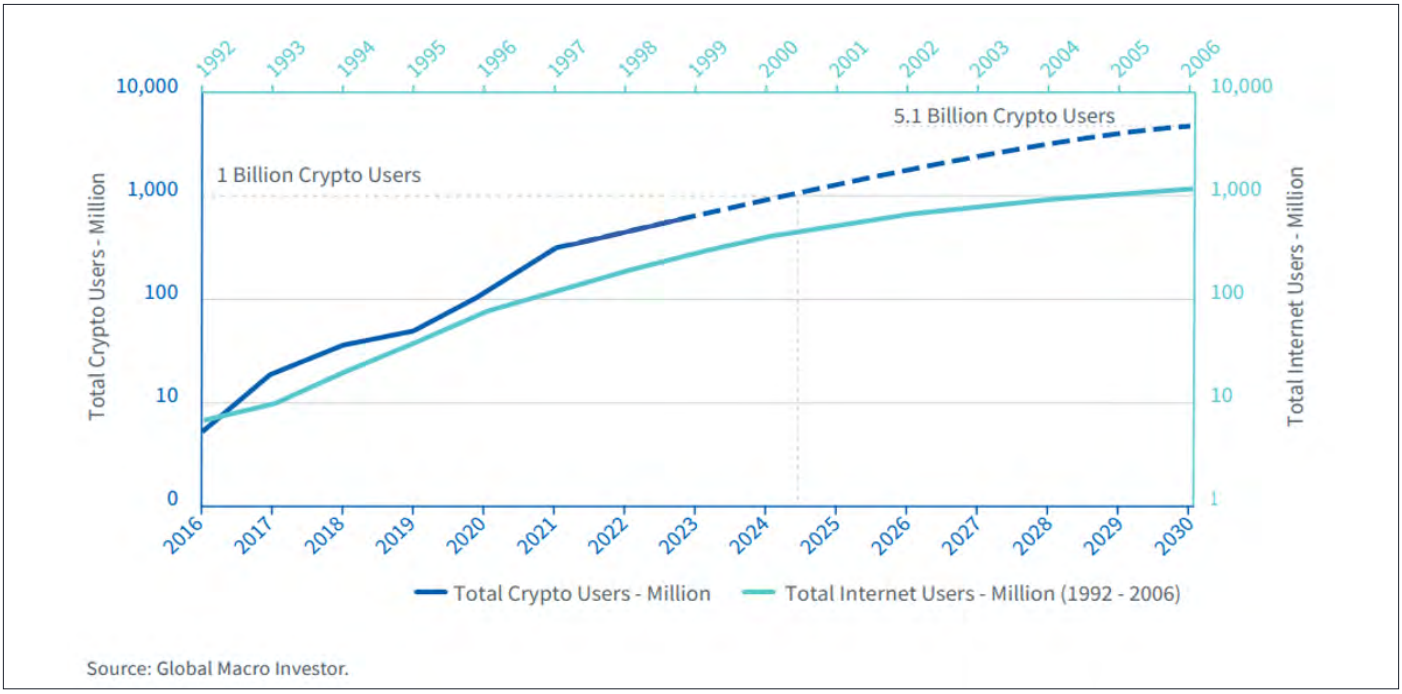
Crypto adoption is growing fast. It is estimated that, as of 2023, there are over 420 million crypto users worldwide, with 31 million users in Europe.

Source: Triple-A

Zooming in, we observe that in Switzerland, 21% of the population has used or owned cryptos in 2023, representing a 16% increase from last year. On the other hand, France is still a bit behind, with an adoption rate of 11%, up from 9% in 2022.



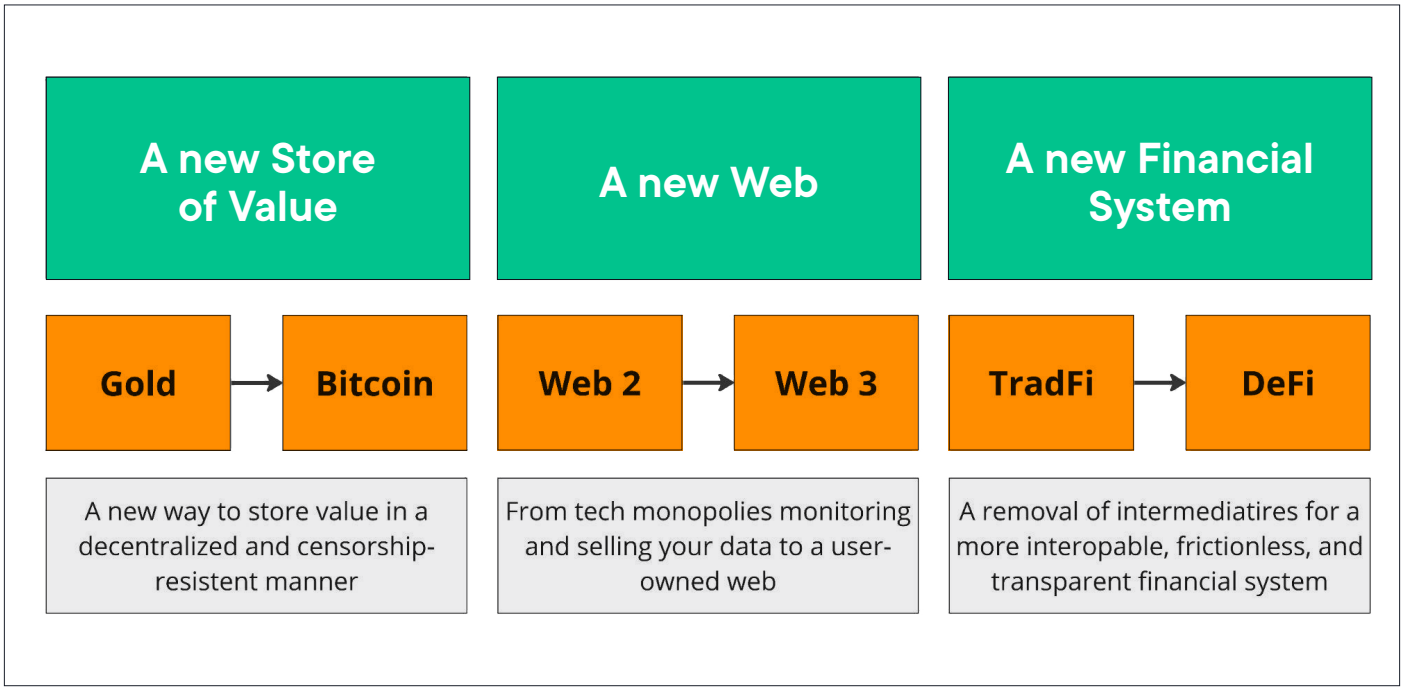
Crypto users are growing similarly to the internet's 1990s growth.



While we are still early, adoption is expanding fast. At this rate, it is estimated the 1 billion users milestone will be reached towards the end of 2024.

Crypto: Towards A Tridimensional Revolution

Crypto infrastructure is at the core of three main paradigm shifts in society.



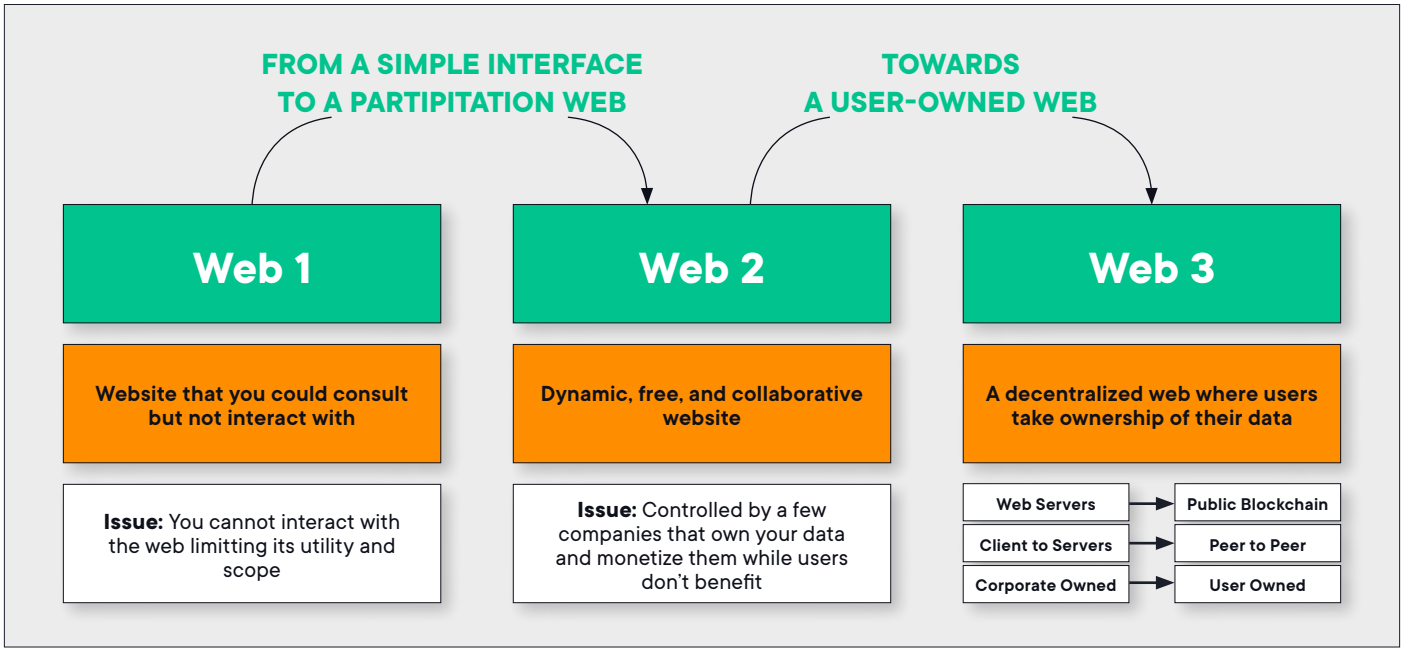
A New Store of Value

In the fiat era, where money can be printed in infinite amounts, the need for a proper and effective store of value increases as fiat purchasing power diminishes yearly. Even though Gold has played this role for the last thousand years, Bitcoin has emerged as a promising alternative to fight the monetary debasement trap. Its property as a store of value is superior to its competitors.

	Bitcoin	Gold	Fiat
Scarce	★★★★	★★★	★
Fungible	★★★	★★★★	★★★★
Divisible	★★★★	★★★	★★★★
Portable	★★★★	★★	★★★★
Censorship Resistant	★★★	★★	★
Established History	★	★★★★	★★
Usability	★★★	★★★	★

A New Web

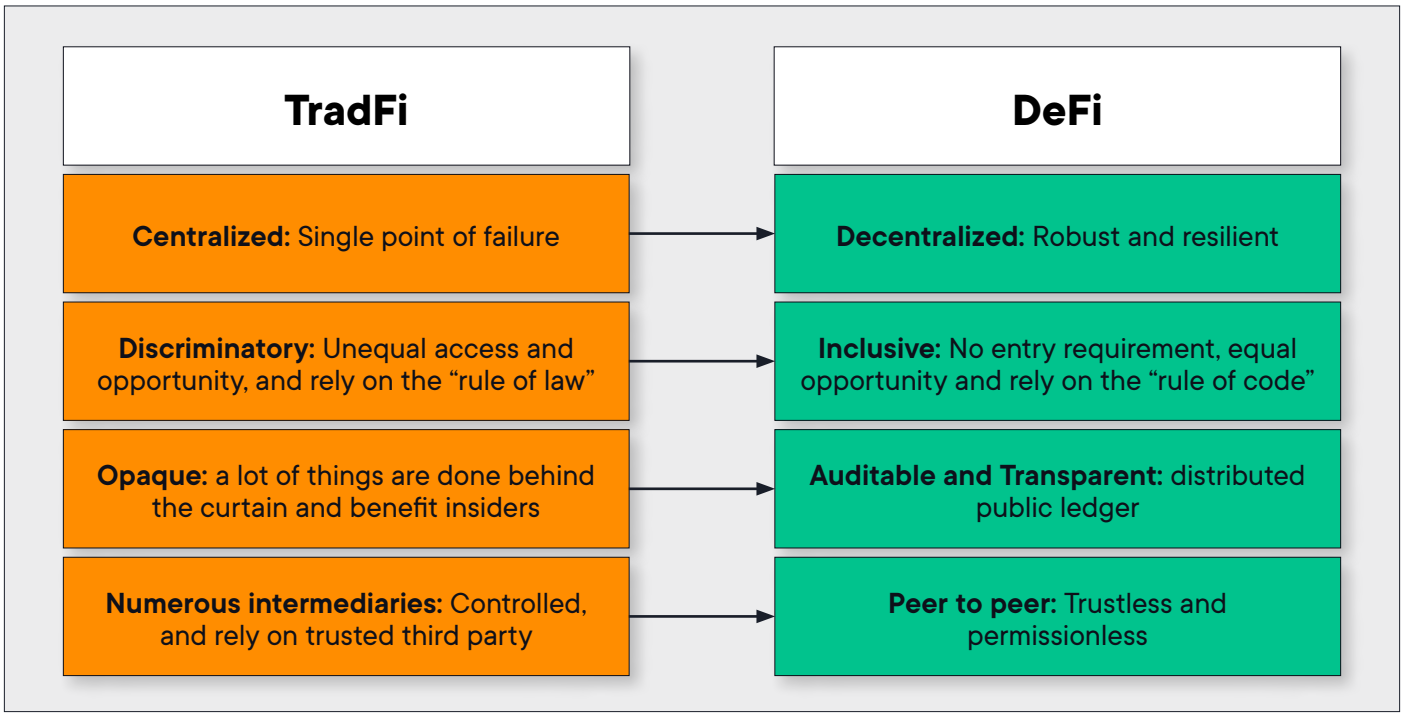
Public blockchains and smart contracts are pioneering a new era for the internet.



We are transitioning to a user-owned web operated on decentralised protocols. This shift is creating many new use-cases for the internet and crypto is at the center of this initiative.

A New Financial System

The future of finance is inclusive, permissionless, and collaborative. Crypto is core to making this vision a reality. The shift from traditional finance (TradFi) to decentralised finance (DeFi) can bring significant improvements within the financial realm.



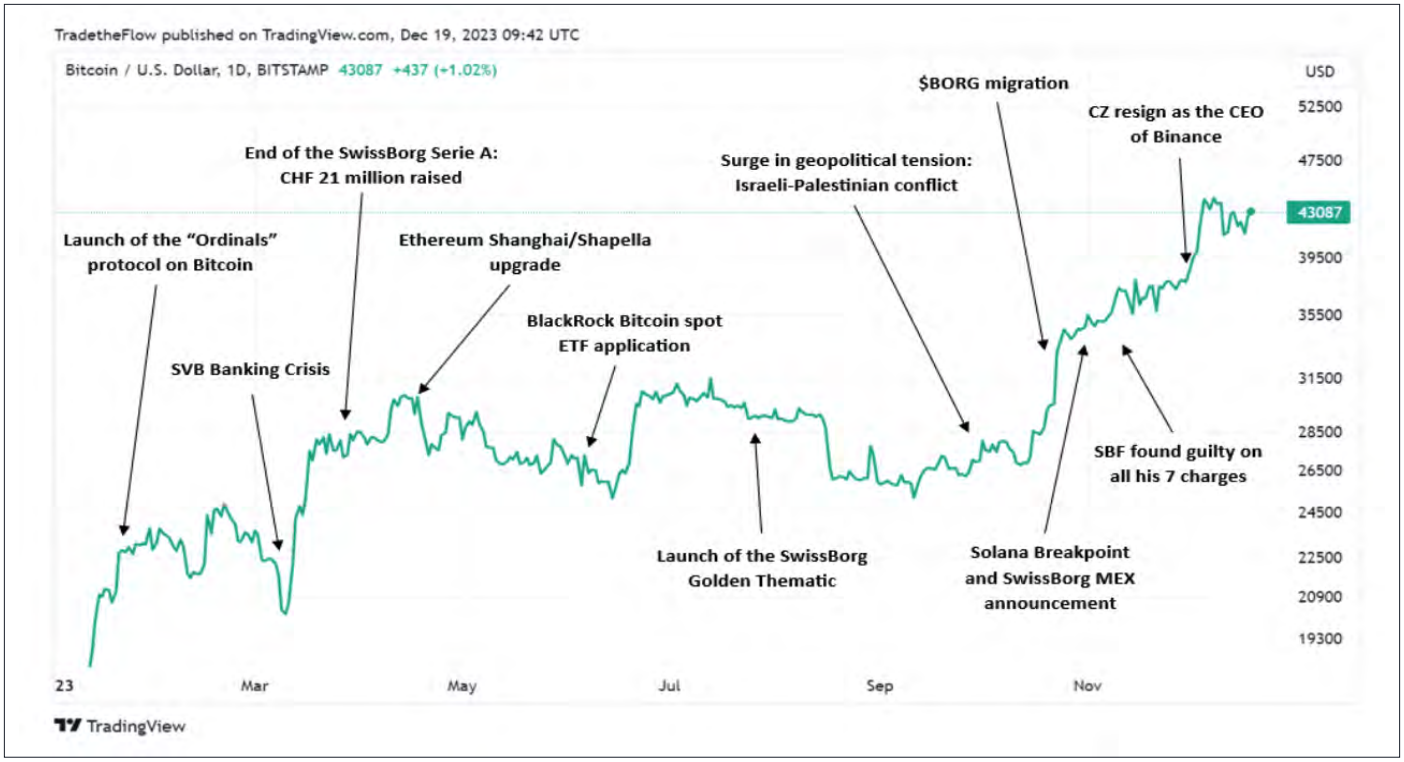
Current Challenges of DeFi:

- **User experience:** Interacting with DeFi is not user friendly. It necessitates a good understanding of blockchain, smart contracts, and wallets. For DeFi to reach mass adoption, it needs to be easy to use.
- **Lack of regulatory clarity:** While one of core ethos of Blockchain has always been: "rule of code before rule of law" DeFi cannot challenge TradFi at scale without being regulatory compliant.
- **Security risk:** Code is not always perfect and DeFi is prone to smart contract issues.

DeFi use-cases are still relatively new and it faces many challenges. Nonetheless, it represents one of the most important use-cases for crypto and has the potential to challenge the status quo of finance.

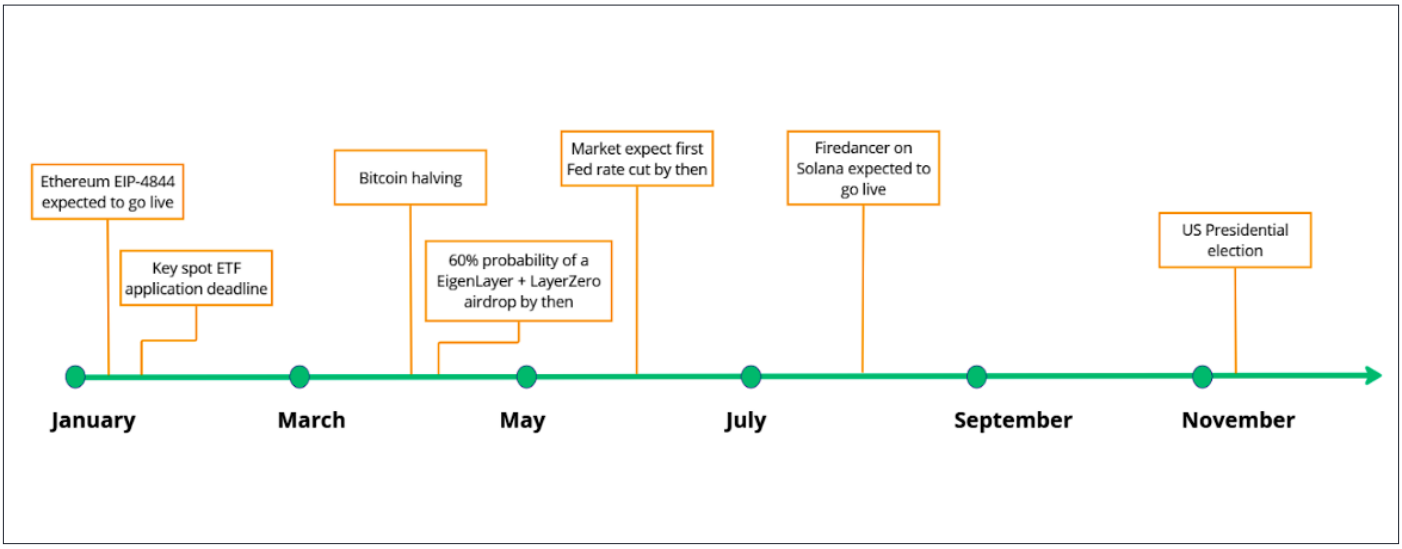
IV. Timeline of Major Events in 2023

Florent Koenig Twitter: @tradetheflow_



V. Calendar for 2024

Florent Koenig Twitter: @tradetheflow_



VI. Is Bitcoin a Real Leading Indicator: An In-depth Analysis of Previous Cycle

Stefan Duprey Twitter: @stefanduprey
Florent Koenig Twitter: @tradetheflow_

Bitcoin is the oldest crypto, Bitcoin is still king, and Bitcoin is inevitable. Yet, many Altcoins have had their moment of glory, and during each of the preceding cycles, the best performance always came from them. Therefore, it is imperative to a better understanding of its price dynamics, especially relative to Bitcoin.

The current word on the street suggests that Bitcoin leads the market and that Altcoins subsequently follow its price action. We wanted to test this hypothesis with an empirical data-driven analysis and answer the question: Is Bitcoin a true leading indicator? To address this, we employed two distinct approaches.

Bull Market Analysis

In the first study, we focused on identifying the start and end of the bull market during the 2017 and 2020 cycles.

Defining the start of the bull market was a challenging task, and we noted that using a 100% quantitative approach was not the best option. In this analysis, we decided to use a mix of quantitative and qualitative analysis and define the beginning of the bull market as follows:

- Price > 50 days Moving Average > 150 days Moving Average
- Price breakout from the first consolidation phase after the cycle bottom

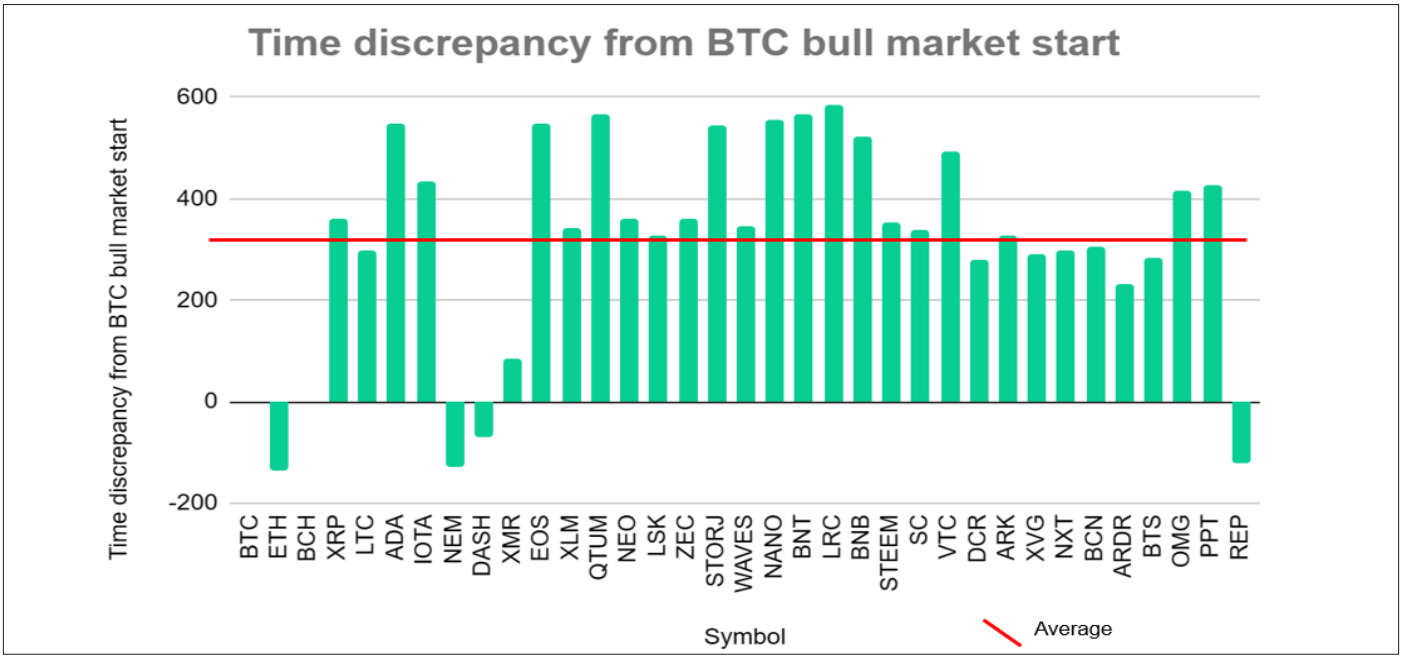
However, we sometimes had to adapt to the individual situation of some token, especially when they were new and just launched.

In contrast, the end of the bull market for each token was set as the date when they reached their local all-time highs, ATHs (no other ATH was reached for at least 365 days).

2017 Cycle:

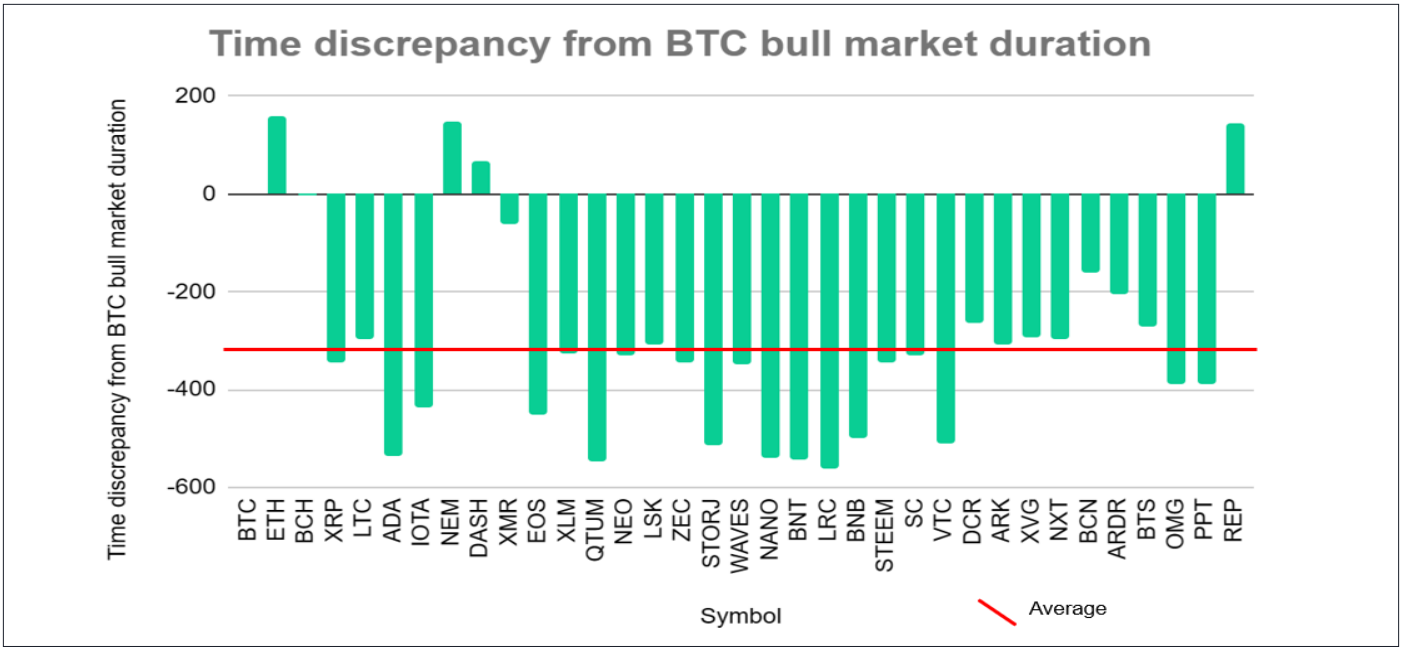
Analyzing the 2017 cycle was not always straightforward. For this first study, we used a small sample of tokens that had a bull run and were on top of the CoinMarketCap list when Bitcoin hit its all-time high in 2017.

Our first observation is pretty clear: Altcoins are laggers compared to Bitcoin.



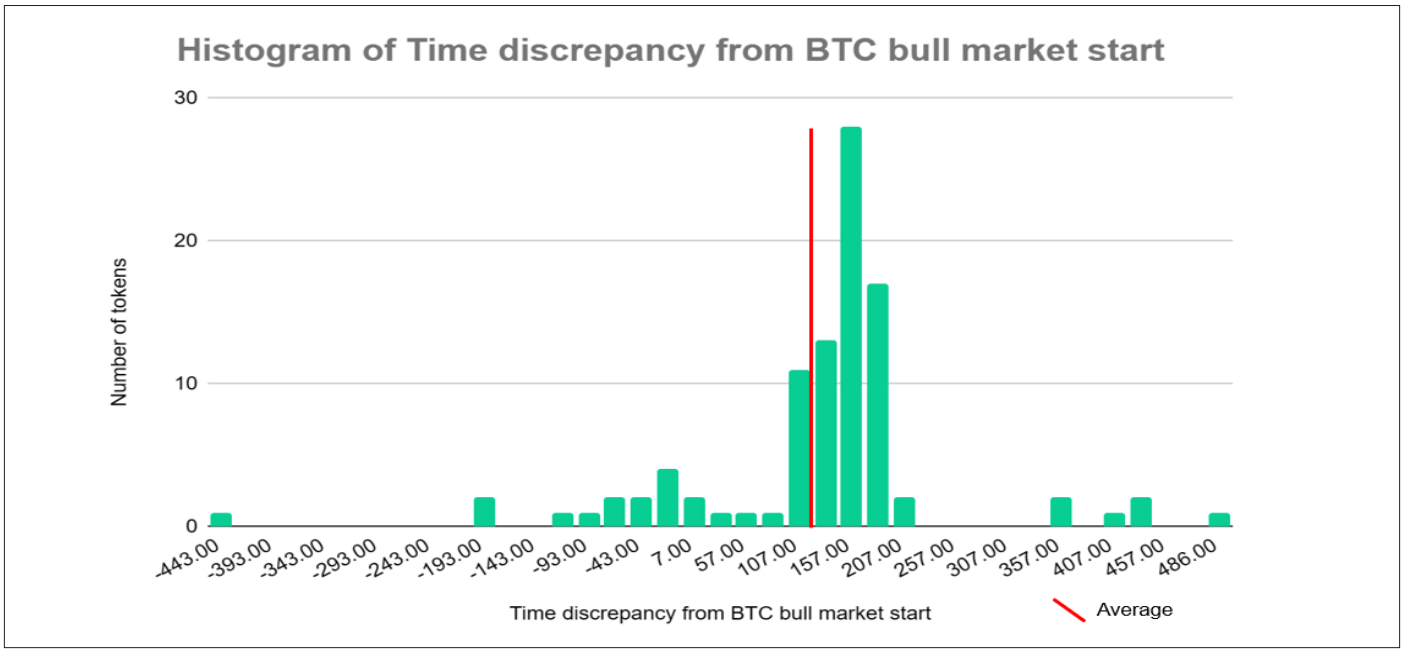
The bull market on Altcoins started on average 1 year later than Bitcoin. However, it is important to note that the 2017 cycle was long for Bitcoin, and when it began its bull run in 2016, the ICO mania had not yet started. This can explain why Altcoins were lagging.

Furthermore, the duration of the bull market on Altcoins was also, on average, 300 days shorter than the Bitcoin one, meaning that the 2017 cycle was a fast one for Altcoins.



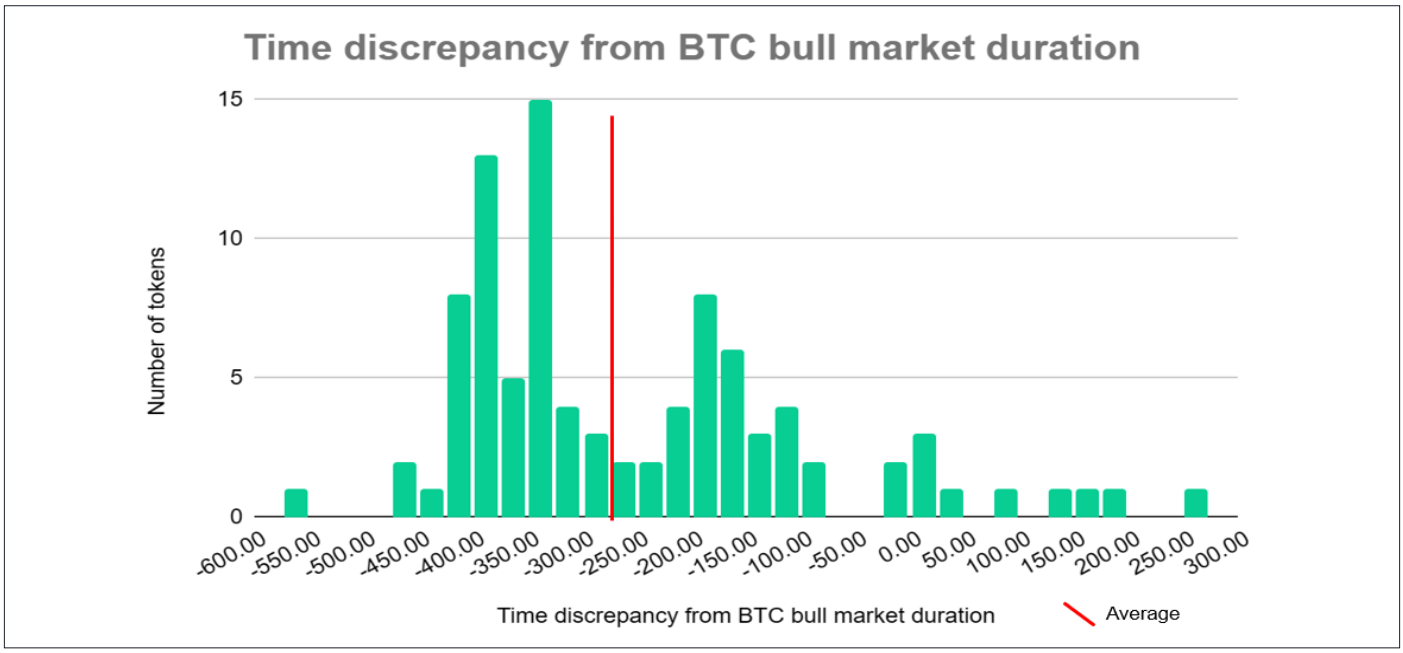
2020 Cycle:

For the 2020 cycle we used a bigger sample size that corresponded to the top 100 tokens of CoinMarketCap when Bitcoin reached its cycle high. Looking at the histogram below, we observe that most of the coins were also lagging with respect to Bitcoin.



We note that most coins in our sample centred around a 100 to 250-day lag with an average of 140 days and a 95% confidence interval of 114- 166.

On the other hand, we also note that the bull market duration was generally shorter than the Bitcoin one.



We see a strong cluster of coins from -450 to -50 days, and less than 10% of our sample of coins had a longer bull market than Bitcoin. Moreover, the average time discrepancies in the 2020 bull market was around negative 256 days, and the 95% confidence interval is tight: -290 to -222.

In both the 2017 and 2020 cycles, there is supporting evidence that Bitcoin was a leading indicator. However, we note that the time discrepancies between 2017 and 2020 got shorter. One possible explanation could be that as crypto became more mainstream and more efficient in 2020, investors realised the bullish case for Altcoins and acted accordingly.

Drawdown Analysis

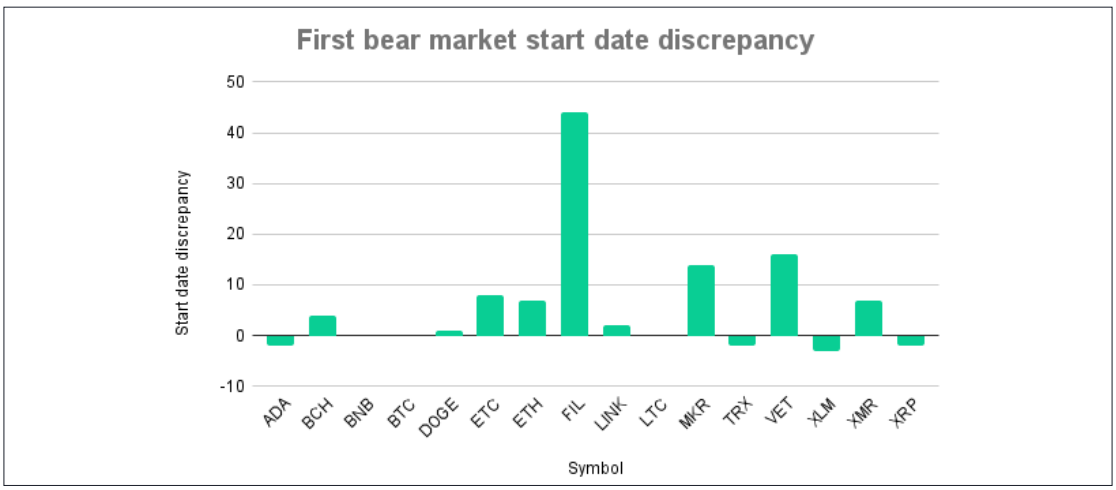
The second analysis centred around drawdown to identify different market phases. A drawdown is a peak-to-trough decline during a specific period : it refers to how much a cryptocurrency is down from the peak before it recovers back to the peak and is typically quoted as a percentage. Here, a return to 100% means that the coin has reached an all-time high again. A drawdown of 25% means the coin has lost 75% from its latest peak at 100%. The cryptocurrency market value is normalised at 100 at the start of the investigating period. Here is an example for three randomly selected coins and BTC:

We extracted the two longest-lasting drawdowns from 2018 onward to find the two bear market stages for Bitcoin since 2018. The beginning of those drawdowns is the bear market starting date, and we defined the date of the maximal loss as the start of the bull market. Then, we computed the same drawdown profiles for all the Alternative coins and computed the discrepancies of their bear/ bull market starting date with those of BTC.

First Winter Analysis:

In this first study, we used a narrow list of tokens of the 50 biggest market capitalisations as the crypto market was not as diverse as it is today.

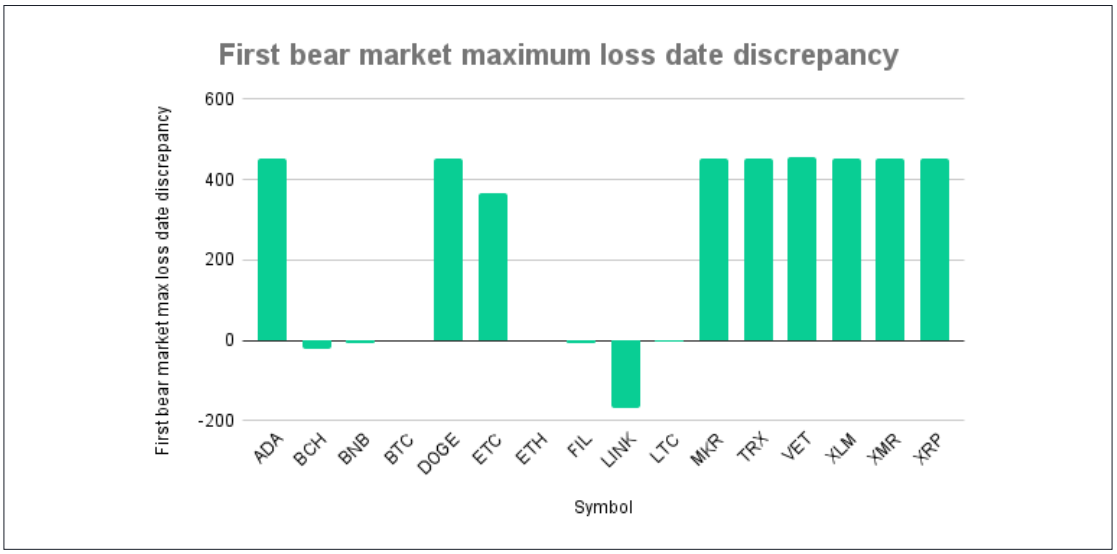
Starting Date



This chart shows that most coins left their ATH only a few days after Bitcoin: Bitcoin is an early indicator of the global crypto market turnover.

Maximum Loss Days

Now, looking at the maximum loss date, we observe again that most coins are lagging in their recovery relative to Bitcoin.



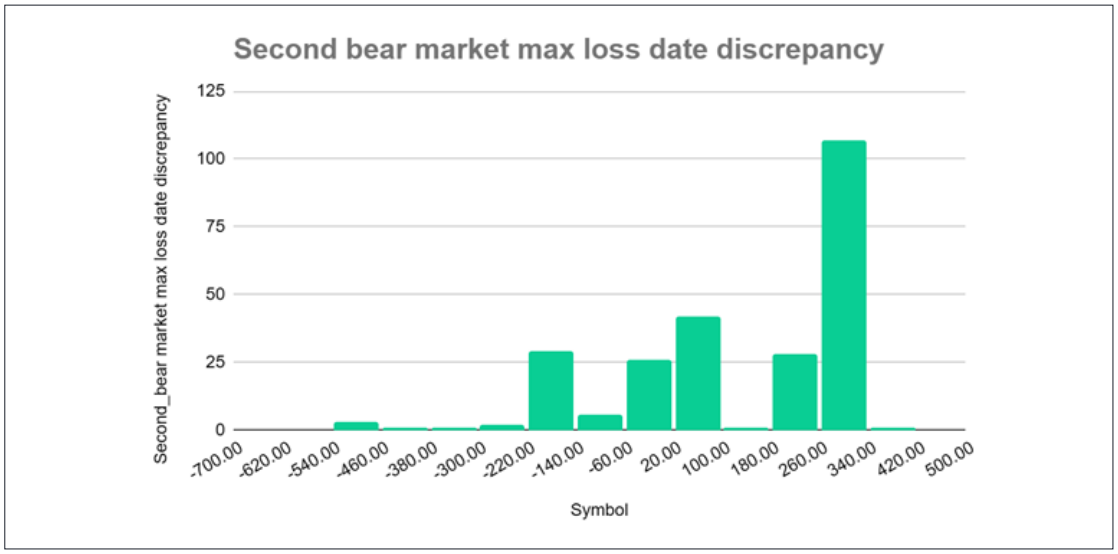
Second Winter Analysis:

Here, we used a bigger sample of the 500 biggest capitalisation tokens as of today to understand the time discrepancies between Bitcoin and Altcoins in their second winter (since 2018).

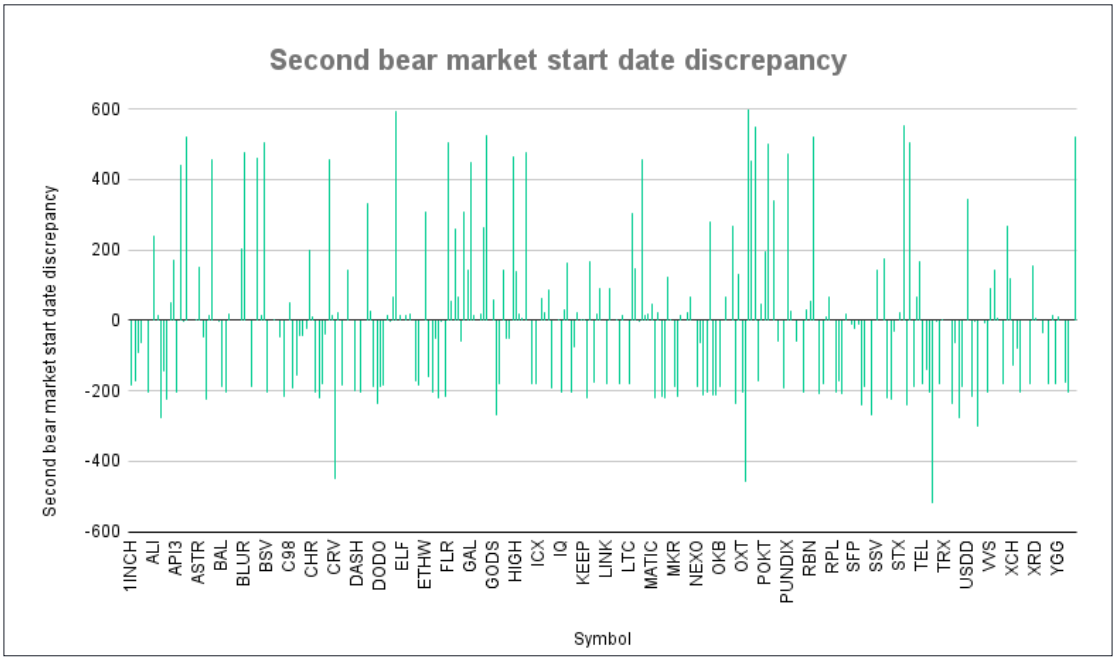
Starting Date

Looking at the starting date of the second bear market, we can identify three relevant clusters:

- The ones that never recovered from the last ATH when Bitcoin began the second bear winter (negative ones)
- The ones which moved close to BTC with a slight lag: BTC is an early indicator of a bear market threshold.
- The lagging one lagging by a vast number of days: Bitcoin is an early indicator.



For a more compact visualisation, we can restrict our graph to include only 50 tokens (same histogram as above).



We find the two distinct groups again:

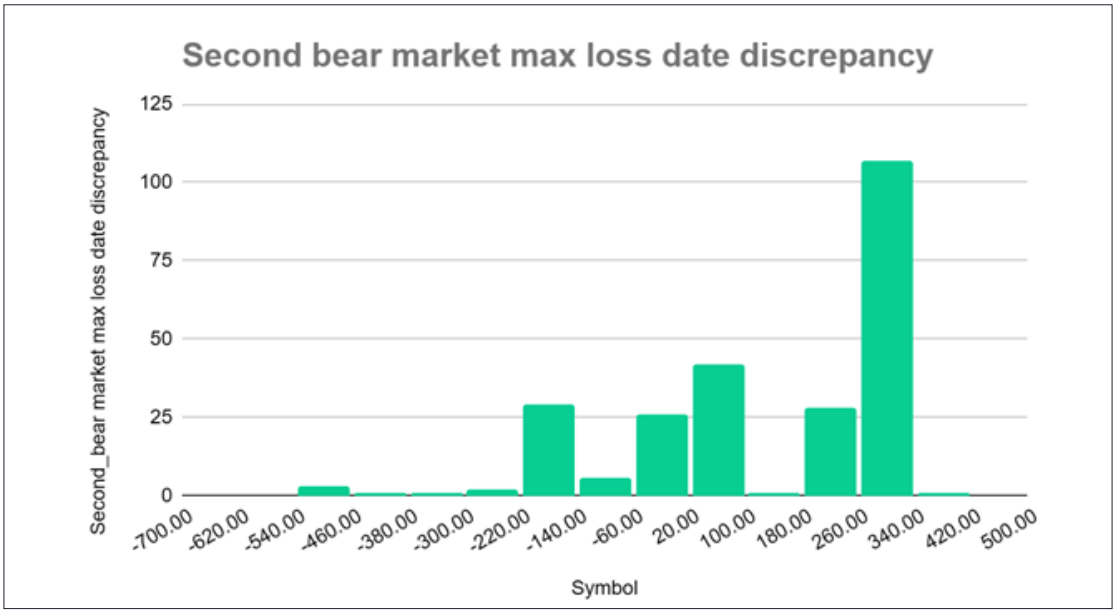
- The ones that never recovered from the last ATH when bitcoin began the second bear winter (negative ones)
- The ones that move close to BTC with a slight lag: BTC is an early indicator of a bear market threshold.

Maximum Loss Days

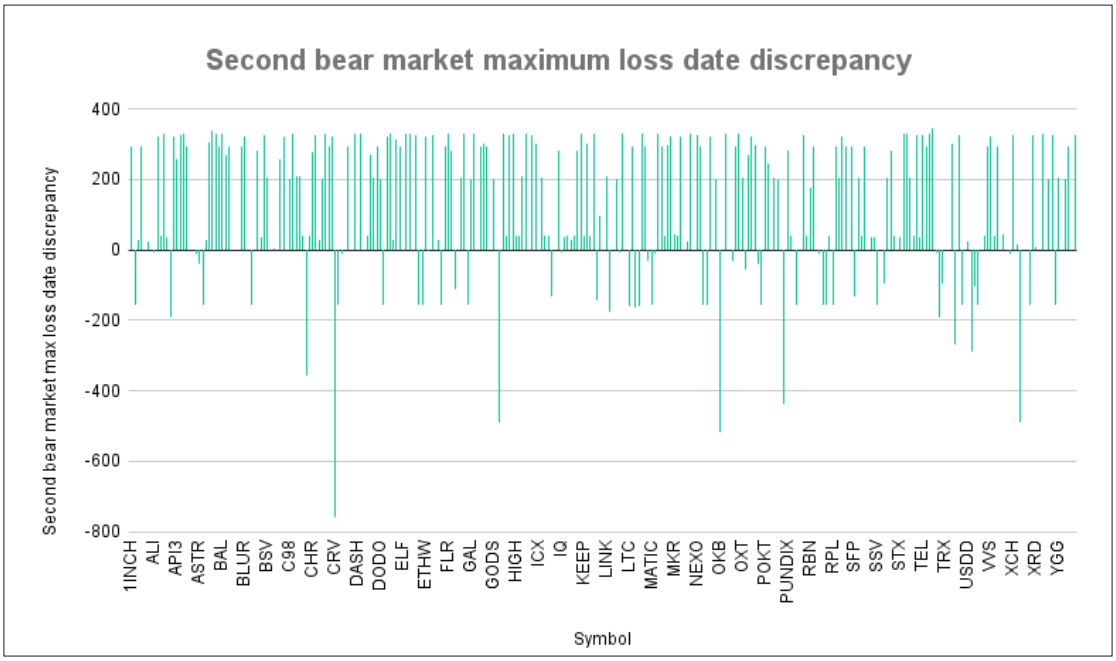
Now, switching to the maximum loss days, we can also separate the study into two different categories:

- Negative ones (underwent an earlier market push spring than BTC)
- Lagging ones: Bitcoin is an early indicator

We note that the percentage of lagging Altcoins are vastly superior to the leading ones showing that indeed Bitcoin reached its bottom earlier than most Altcoins



Again, if we plot only 50 token for the sake of better visualisation, this is what we get:



We again see the two distinct groups:

- Negative ones (underwent an earlier market push spring than BTC)
- Lagging ones: Bitcoin is an early indicator

Conclusion

The two studies reach the same conclusion: Bitcoin has been a reliable leading indicator in the past, both for the start and end of a bull market. Further, Bitcoin generally had a longer cycle duration than Altcoins in the previous cycles.

While it seems probable that we could observe a similar lagging pattern in the next bull run, as market efficiency has increased, this lag could be shorter.

VII. Themes to follow in 2024

1. Bitcoin: The Digital Gold Rush is just Starting

Florent Koenig Twitter: @tradetheflow_

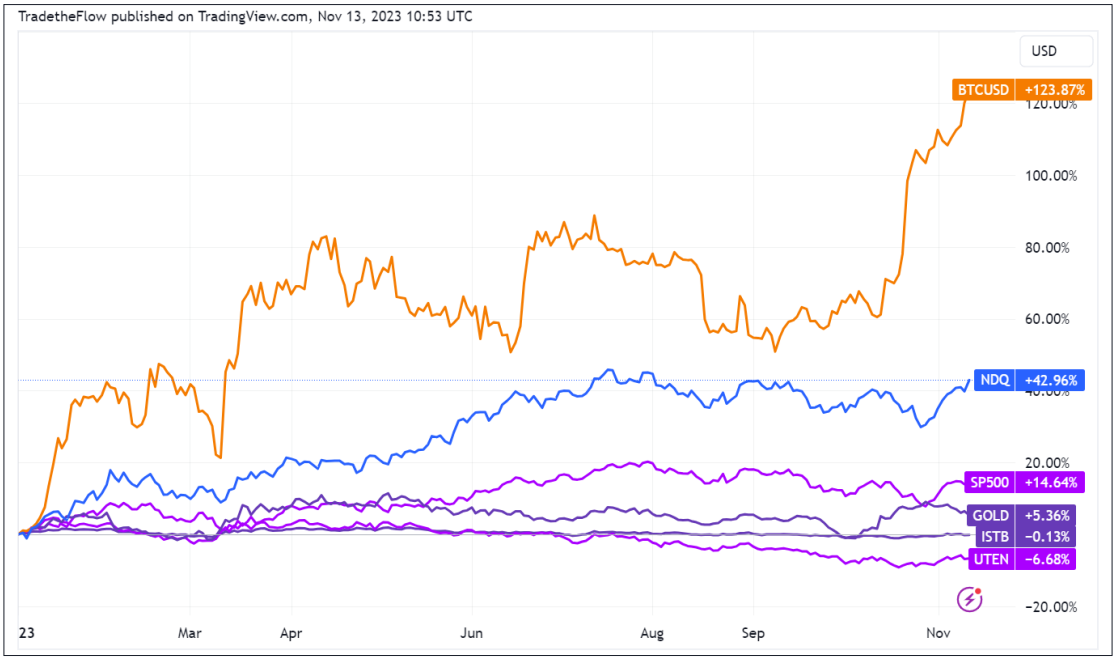
In 2009, a mysterious whitepaper was released. At the time, no one could imagine that this would be the spark of a financial revolution. But here we are, fifteen years later, where, despite the many battles and hardships, Bitcoin is still standing strong and currently sits at a $600 billion market cap.

Let's explore in more detail what Bitcoin has been up to in 2023 and provide a taste of what it has in store for the following year.

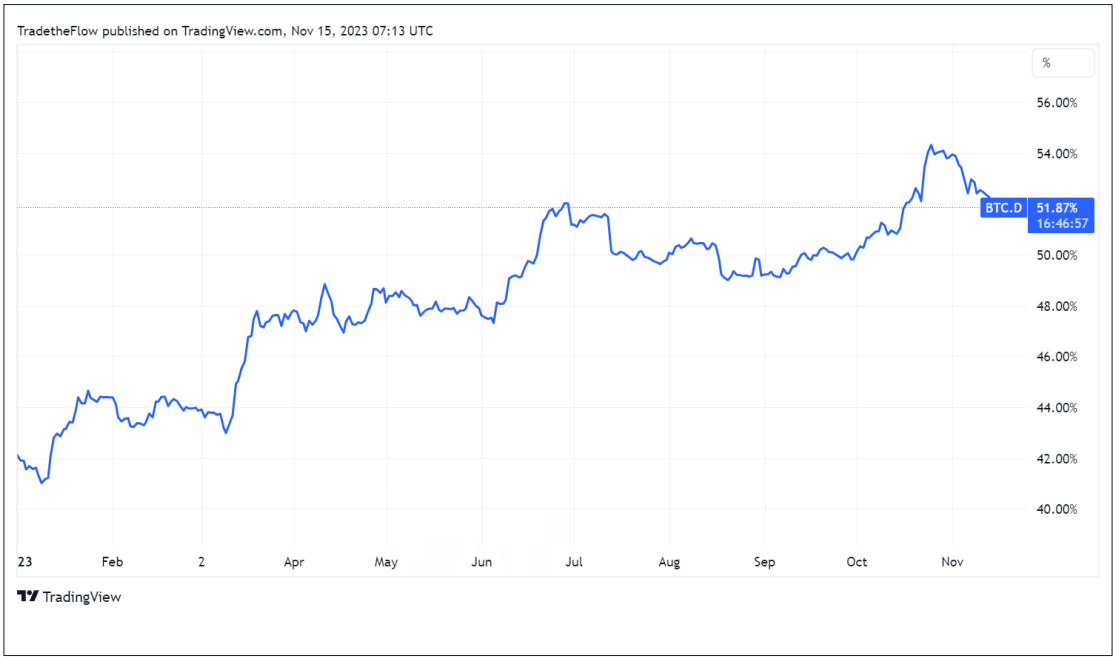
Bitcoin Performance in 2023

In 2023, Bitcoin soared 120%, outperforming all major asset classes. But what's truly remarkable is that Bitcoin achieved this despite a poor economic outlook at the onset of 2023, quantitative tightening, sticky inflation, a banking crisis, and geopolitical tension.

While it is too early to call this a "new paradigm shit", its price performance this year demonstrates that Bitcoin is starting to be seen as the new "fly to quality" with a clear and compelling utility within any portfolio.



Further, Bitcoin has maintained its position as a leader within the crypto landscape; despite the fast, creative destruction process taking place in this industry, Bitcoin's value proposition continues to make sense. This is reflected by its rising dominance within the crypto ecosystem throughout 2023.



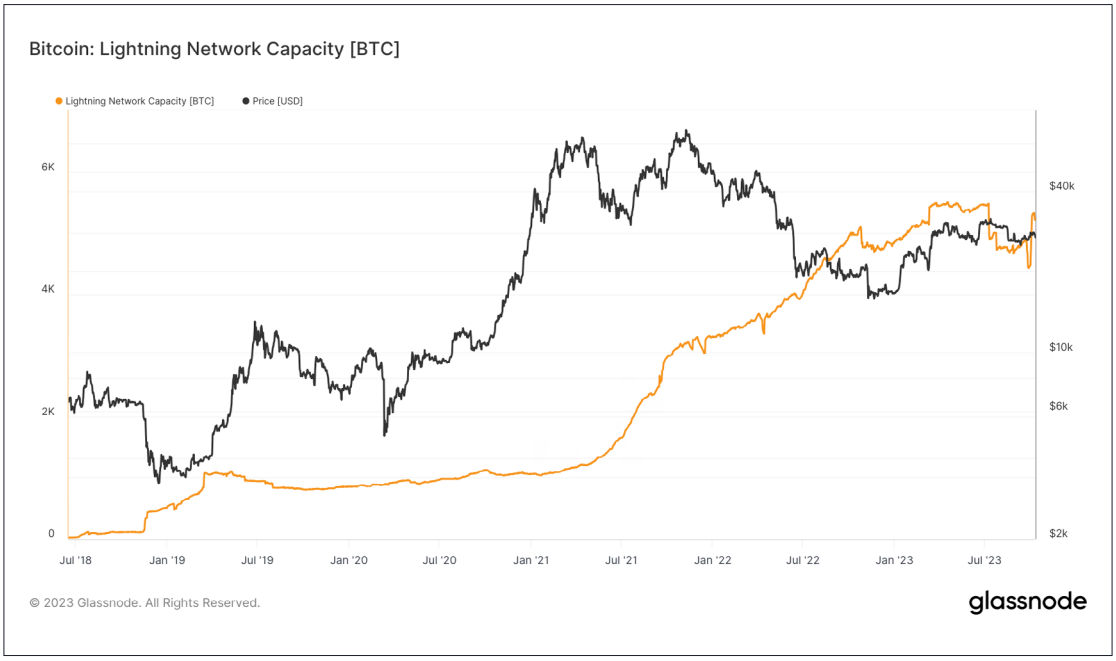
Bitcoin Fundamentals Continue to be Strong

"I'm sure that in 20 years there will either be very large transaction volume or no volume." – Satoshi Nakamoto

While we all have an obsession with the bitcoin price, we often forget to go beyond and look at the network as a whole and its underlying value. This can be seen with different on-chain data.

First, we see that Bitcoin's hash rate and mining difficulty are in a strong uptrend and are constantly setting new all-time highs.

Lightning Network, a "Layer 2" payment protocol built on Bitcoin, is also near its ATH capacity.



The “Halving Story” is Still There

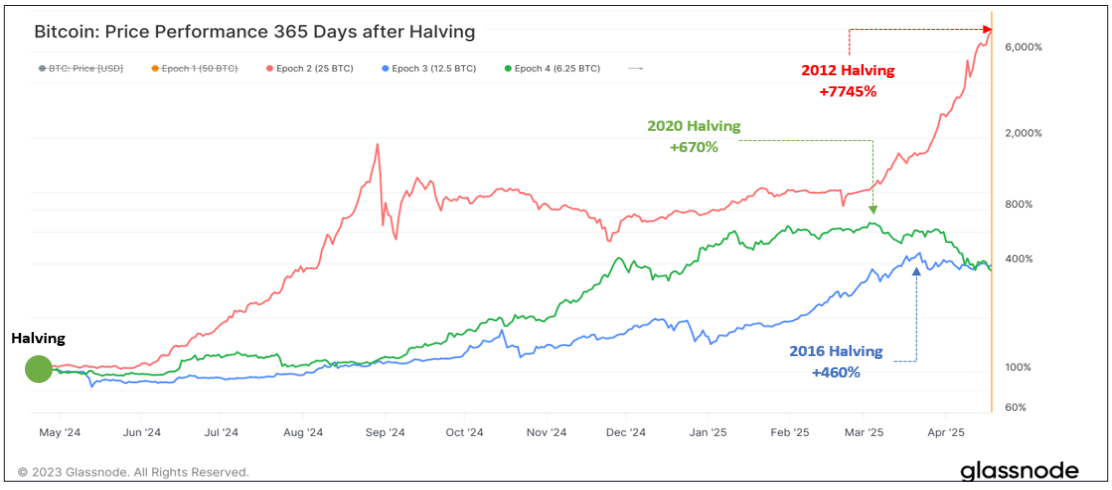
An important characteristic distinguishing Bitcoin from other assets is its limited supply. By design, there will never be more than 21,000,000 million bitcoins. This is achieved through a process known as the “Bitcoin halving”, where every four years, the number of new bitcoins produced is cut in half.

Historically, Bitcoin has always followed a 4-year price cycle where the beginning of each cycle coincided with the halving. So far, Bitcoin seems to follow precisely the same path as during the preceding pre-halving period.

The halving acts as a supply shock, and while this event is known and predictable, prices always reacted positively.

	New BTC per block before halving	New BTC per block after halving	Price on halving day	Price 150 days later
2012 Halving	50 BTC	25 BTC	$12.00	$127.00
2016 Halving	25 BTC	12.5 BTC	$650.00	$758.00
2020 Halving	12.5 BTC	6.25 BTC	$8,821.00	$10,943.00

The next halving is expected in April 2024. We feel confident that this will be one of many factors contributing to increased buying pressure on Bitcoin.



However, the magnitude of the price movement will be influenced by many other factors.

A New Wave of Adoption is Coming

"I do believe that crypto is digitising gold in many ways" — Larry Fink.

We witnessed some key moves from the traditional world of finance in 2023. In June, BlackRock, the largest financial institution in the world, filed for a Spot Bitcoin ETF. A few days later, we saw many other big names from "TradFi" following. Now, all they are waiting for is the green light from the SEC.

Issuer *(Ticker)*	Company	Filing Date	19b-4 Ack Date	Replication Strategy	Highest Act	Status	SEC Date
"Physically" Backed							
VanEck Bitcoin Trust	VanEck	12/30/2020	3/15/2021	Coin-based (Bitcoin)	1933 Act	Denied	11/12/2021
Wisdomtree Bitcoin Trust	Wisdomtree	3/11/2021	4/9/2021	Coin-based (Bitcoin)	1933 Act	Denied	12/1/2021
Kryptoin Bitcoin ETF	Kryptoin Invst Advisrs	10/28/2019	4/22/2021	Coin-based (Bitcoin)	1933 Act	Denied	12/22/2021
Valkyrie Bitcoin Fund	Valkyrie Investments	1/22/2021	5/6/2021	Coin-based (Bitcoin)	1933 Act	Denied	12/22/2021
First Trust SkyBridge Bitcoin ETF Trust	First Trust & SkyBridge	3/19/2021	5/21/2021	Coin-based (Bitcoin)	1933 Act	Denied	1/20/2022
Wise Origin Bitcoin Trust	Fidelity	3/24/2021	5/25/2021	Coin-based (Bitcoin)	1933 Act	Denied	1/27/2022
NYDIG Bitcoin ETF	Stone Ridge/NYDIG	2/16/2021	7/13/2021	Coin-based (Bitcoin)	1933 Act	Awaiting Approval	3/16/2022
ARK 21Shares Bitcoin ETF *(ARKB)*	21 Shares/ARK Invest	6/28/2021	8/6/2021	Coin-based (Bitcoin)	1933 Act	Awaiting Approval	4/3/2022
Teucrium Bitcoin Futures Fund	Teucrium	5/21/2021	8/5/2021	Futures/Derivs-based	1933 act	Awaiting Approval	4/8/2022
Global X Bitcoin Trust	Global X	7/21/2021	8/17/2021	Coin-based (Bitcoin)	1933 Act	Awaiting Approval	~4/20/2022
Valkyrie XBTO Bitcoin Futures Fund	Valkyrie Investments	8/23/2021?	9/2/2021	Futures/Derivs-based	1933 Act	Awaiting Approval	~5/6/2022
One River Carbon Neutral Bitcoin Trust	One River	5/24/2021	9/29/2021	Coin & Crbn Creds (Btc)	1933 Act	Awaiting Approval	~6/2/2022
Bitwise Bitcoin ETP Trust	Bitwise	10/14/2021	########	Coin-based (Bitcoin)	1933 Act	Awaiting Approval	~7/1/2022
Grayscale Bitcoin Trust(Re-file) Conversion *(GBTC)*	Grayscale	N/A	11/8/2021	Coin-based (Bitcoin)	1933 Act	Awaiting Approval	~7/6/2022
Wisdomtree Bitcoin Trust(Re-filing) *(BTCW)*	Wisdomtree	12/8/2021	--	Coin-based (Bitcoin)	1933 Act	Awaiting Approval	19b-4 Filed

Note: "?" Denotes confidential filing where date cannot be known for certain

Source: Bloomberg Intelligence, SEC.gov

Bloomberg

When we look at the track record of the Blackrock ETF application of 575 approvals to 1 rejection and encouraging views from different analysts, we are confident that the first Bitcoin Spot ETF will soon get approved. This represents a significant milestone because launching more regulated products will allow a new class of investors to enter the market.

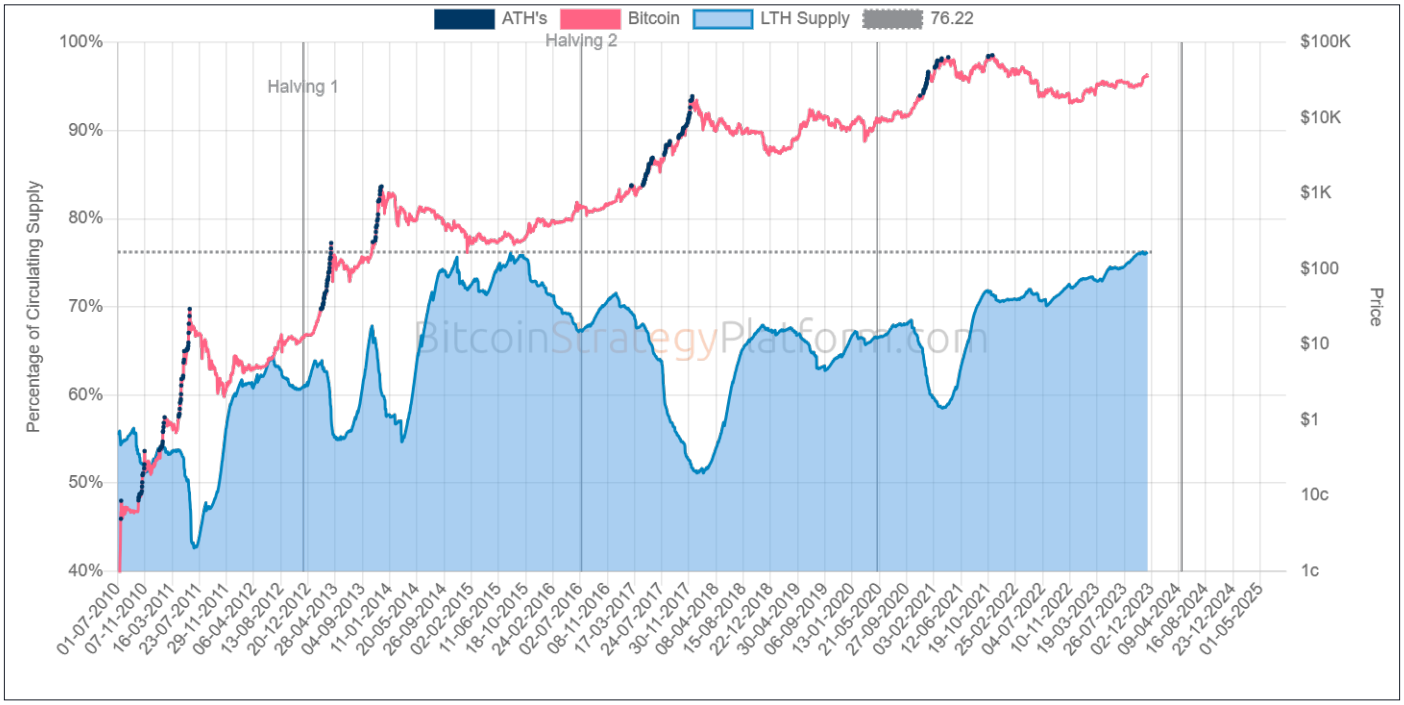
In our last paper on the implication of a spot ETF on Bitcoin price, we estimated that the approval of the first ETF in the US could bring an inflow of $100 billion to Bitcoin in the first year only.

	Conservative	Base	Aggressive
Portfolio Allocation in Year 1 (%)	2.5%	2.5%	2.5%
Adoption rate by Fund Managers (%)	5%	10%	15%
Inflow in Year 1 ($USD bn)	50	100	150

While it is difficult to estimate precisely how the price will react for a given dollar inflow, we are confident that this could well be the biggest catalyst for Bitcoin in 2024.

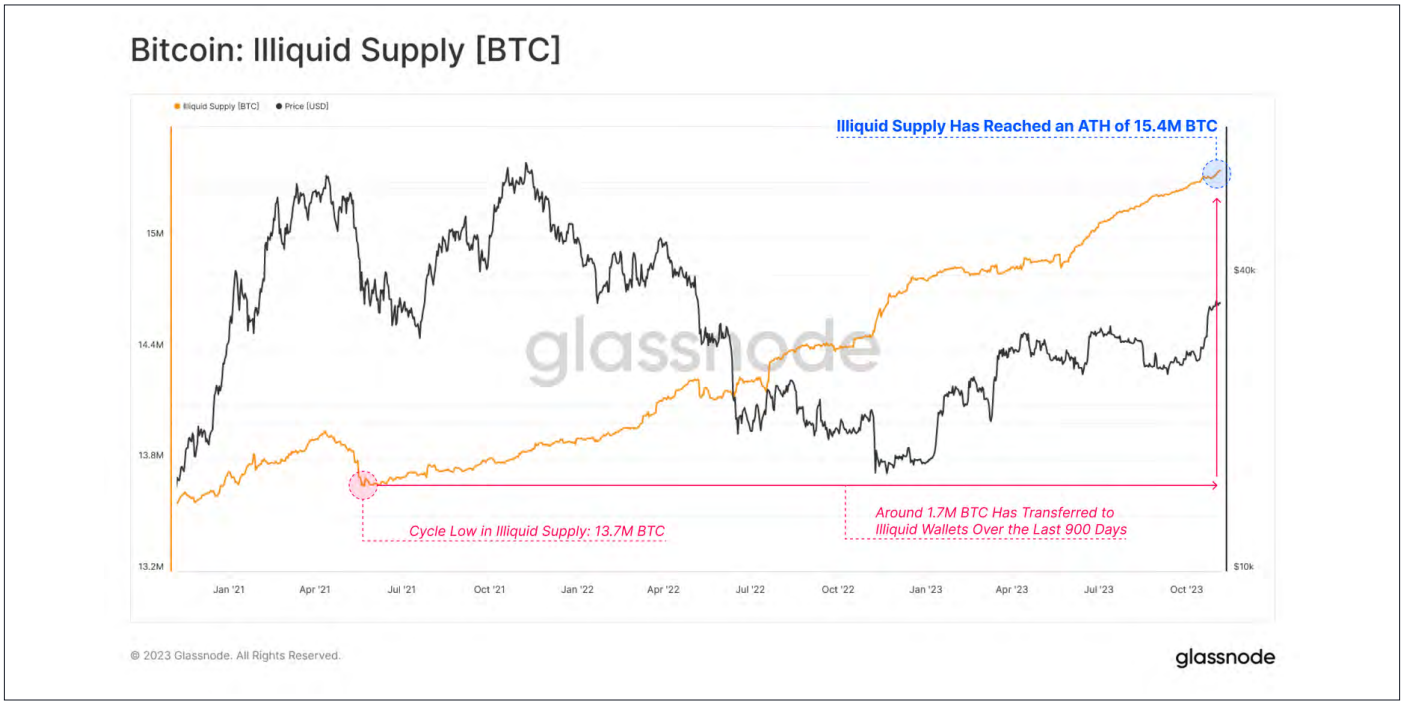
Towards a supply squeeze,

Our research suggests that the current circulating supply of Bitcoin is historically tight. In addition to the halving, which is fast approaching, the percentage of coins held by long-term investors is at its ATH.

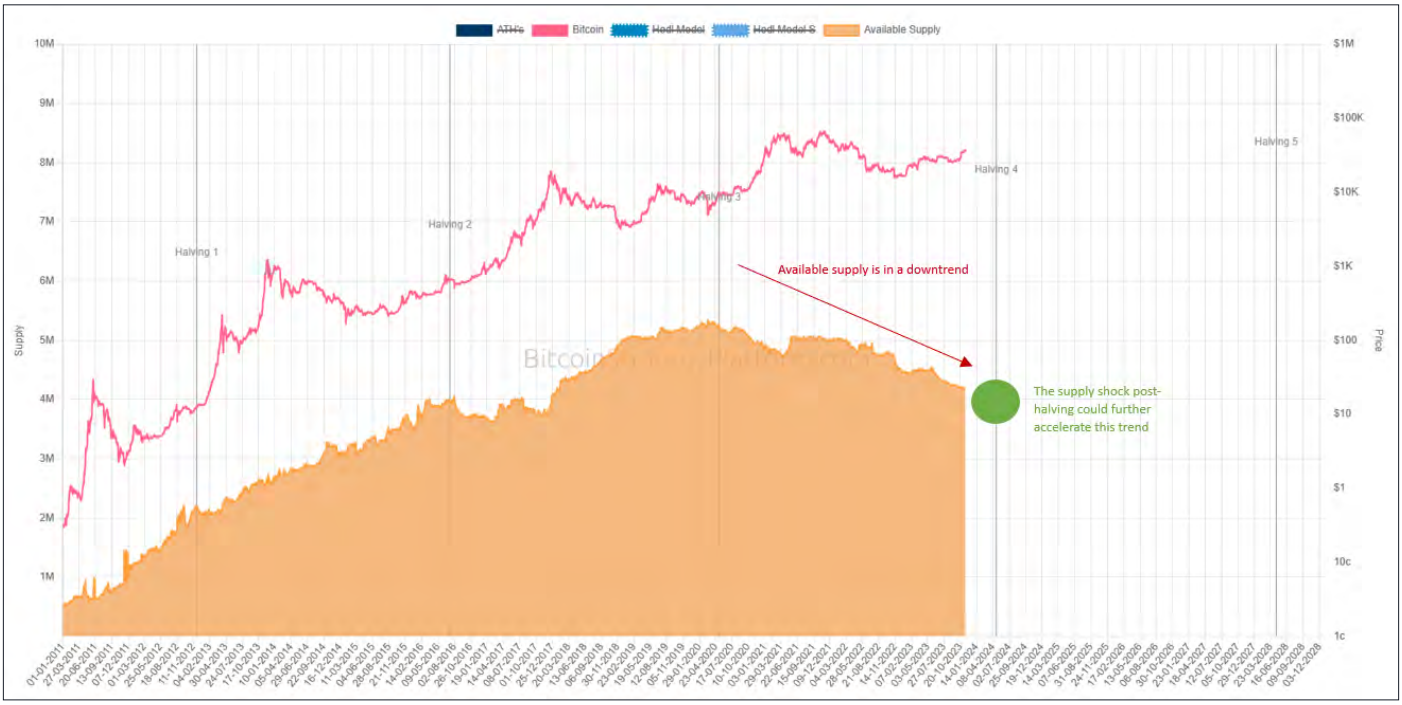


Source: BitcoinStrategyPlatform

Further, the illiquid supply metric, which measures the supply held in wallets with minimum spending history, is also at an ATH.



As a result, available supply has been in a downtrend since the 3rd halving in March 2020. This trend could further accelerate at the onset of the next halving in April 2024.



Source: BitcoinStrategyPlatform

This tight circulating supply suggests that Bitcoin could become more prone to more reflexivity in its price actions.

Conclusion

As we step into 2024, Bitcoin stands on firm ground and is becoming a robust asset with a clear value proposition. Given its strong track record, there is virtually no doubt that it deserves its place in one's portfolio.
Moreover, with liquidity coming back to the market (as explained in our macro outlook), the halving that is fast approaching, positive momentum around the approval of a spot ETF in the US, and a tight circulating supply, the following year should be an exciting one for Bitcoin investors.

2. Will Ethereum fulfil its prophecy?

Aurélien Giraudon Twitter: @spdy

Introduction

In the dynamic world of blockchain and cryptocurrencies, Ethereum has emerged as more than just a platform; it's a visionary's dream that reimagined what decentralised technology could achieve. As we stand at the threshold of 2024, a pivotal question looms: Will Ethereum continue to be the guiding force in this era of blockchains?

Genesis

Ethereum's journey began with a groundbreaking vision set forth by Vitalik Buterin in 2014. When Buterin released the Ethereum white paper, he didn't just introduce a new technology; he proposed a revolution. Ethereum was envisioned as a platform that would empower decentralised applications through smart contracts on a blockchain – a concept that was not just innovative but transformative at the time.

Over the years, the sector surrounding Ethereum has evolved at an unprecedented pace, paving the way for new developments and necessitating continuous adaptation from Ethereum. The rapid technological advancements in the blockchain space demand constant mental agility and foresight, traits that Ethereum has demonstrated amply. This agility is best exemplified by the updated roadmap released by the Ethereum team, which provided much-needed clarity on the network's future direction.

The Evolutionary Roadmap

As we fast forward to 2023, we witness a pivotal moment in Ethereum's history: the successful completion of 'The Merge.' This marked Ethereum's transition to a Proof of Stake mechanism, reducing energy consumption and allowing greater crypto-economic security - two critical aspects for any blockchain platform aspiring for mass adoption.

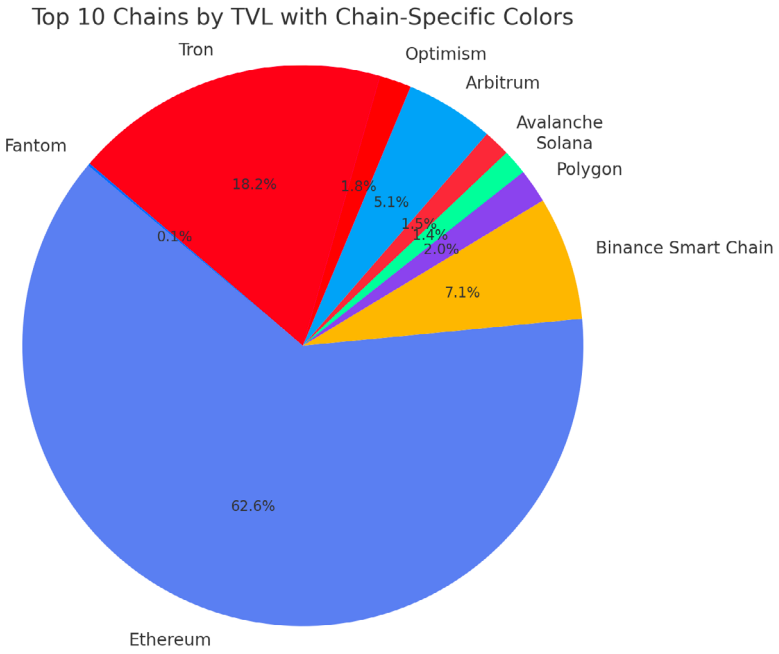
'The Surge' focuses on further scalability improvements through sharding and rollups. A vital element of this phase is the introduction of EIP-4844, known as proto-dank sharding, aimed at reducing gas fees - a significant pain point for Ethereum users. But will these improvements be enough in the face of increasing competition and evolving user needs?

Ethereum's roadmap outlines three additional phases, each aimed at refining and enhancing the network's efficiency and scalability. These phases represent critical steps in Ethereum's journey towards becoming a scalable, efficient, and user-friendly blockchain.

This roadmap helped create and foster the DeFi ecosystem as we know it.

DeFi Dominance

A significant aspect of Ethereum's dominance in the blockchain space is its stronghold in the Decentralised Finance (DeFi) sector. As of now, Ethereum and its rollups secure a substantial portion of the total value locked in DeFi. This dominance is a testament to Ethereum's technical capabilities and the trust and reliability it has built within the community.



Ethereum's influence in DeFi is expected to remain strong, but the landscape is dynamic.
In a world brimming with Layer 1 (L1) blockchains, each vying for dominance, Ethereum faces the monumental task of maintaining its position and proving its mettle as the most scalable, efficient, and user-friendly option available.
Can L2s be Ethereum salvation?

The Layer 2 Dilemma and Future Prospects

While sharding was part of the original scaling roadmap for Ethereum, the rapid development and advantages of Layer 2 rollups led to a shift in focus, making them the primary method for scaling the Ethereum network.
But the Layer 2 market is currently at a crossroads, faced with a dilemma:

- Should these platforms prioritise rapid rollout to capture market value?
- Should they adhere more closely to Ethereum's foundational ethos?

The initial vision of highly user-focused and decentralised L2 solutions seems to have been overshadowed by a rush to market and a lack of native cross-L2 messaging solutions, resulting in fragmented liquidity and a deviation from the original ethos.
Despite these challenges, the L2 space is far from stagnant. Innovations continue with platforms like Arbitrum, Optimism, zkSync Era, Mantle, zkEVM, and Scroll leading the charge. These platforms vary in their technological approaches - some leverage existing technology like Optimistic Rollups, while others explore newer methodologies like Zero-Knowledge Proofs (ZKPs).
The trade-offs between these technologies are significant. ZK Rollups offer fast and secure scaling with immediate transaction verification but currently need longer go-to-market, limited EVM equivalence, and developer adoption challenges. In contrast, Optimistic Rollups provide easier compatibility with existing Ethereum infrastructure but have longer withdrawal times (7-day challenge period).
As we look to 2024 and beyond, the question of whether ZK-based rollups will overtake the initial traction gained by Optimistic Rollups remains open. The Ethereum community eagerly awaits how these technologies will evolve and integrate with the broader Ethereum ecosystem.

These developments within the Layer 2 ecosystem are not just shaping the immediate future of Ethereum but also hinting at a broader, more profound evolution in its role within the blockchain world. This realisation leads us to an important juncture as we transition from Ethereum's technical and immediate challenges to a broader perspective of its future role and impact.

Ethereum as a Foundation Chain

I generally concur with the notion that Ethereum's future may not be exclusively tied to its main chain. Instead, it is gradually evolving into a comprehensive service, serving as the fundamental and secure backbone for an entire ecosystem of scaling solutions. The main chain of Ethereum increasingly functions as a decentralised foundational layer, supporting various scaling solutions such as Rollups, State Channels, and Plasma Chains.

To accentuate the above statement, this year marked the introduction of an innovative approach to leveraging Ethereum's security - a middleware layer known as EigenLayer.
EigenLayer can be envisioned as using your staked assets to secure one network (Ethereum, in this case) while simultaneously supporting other projects or protocols.
'Using' here refers to tapping into Ethereum's established network of trust, eliminating the need for these projects to establish their own set of validators, thus effectively turning decentralised trust into a commodity.

Consequently, this naturally fosters a marketplace for decentralised trust, where protocols can 'rent' security from Ethereum's validators. They do this by encouraging users to restake their liquid-staked assets in a process known as restaking. This innovative perspective further solidifies Ethereum's role as a foundational layer, extending beyond its identity as just a blockchain.

Conclusion

As we reflect on Ethereum's journey and its potential paths forward, it becomes clear that Ethereum's story is far from over.
The network continues to evolve, facing new challenges and seizing opportunities in the ever-changing blockchain landscape. Whether Ethereum becomes the ultimate scalable, efficient, and user-friendly blockchain remains to be seen. However, its role as a foundational chain in a broader ecosystem of interconnected blockchains seems increasingly likely.

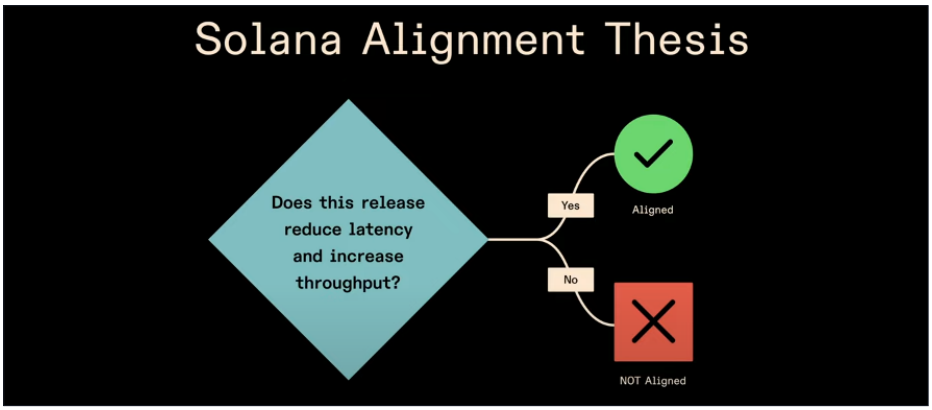
Ethereum's adaptability, vibrant community, and ongoing technological advancements suggest a bright future. The coming years will undoubtedly be pivotal in determining whether Ethereum fulfils its original prophecy. Still, one thing is certain: Ethereum will continue to play a significant role in shaping the future of blockchain technology and decentralised applications as it has done in the past.

3. The Rebirth of Solana

Florent Koenig Twitter: @tradetheflow_

Introduction

"What doesn't kill me makes me stronger"



Source: A Global State Machine, Breakpoint 2023

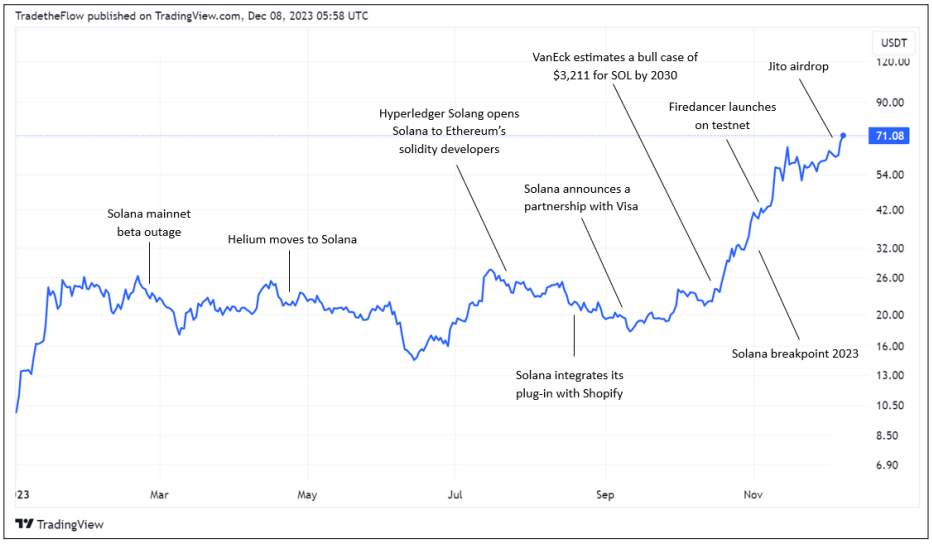
Solana's vision is, and has always been, to be the fastest and most efficient open-source blockchain.

Even though the network faced many hardships and criticism, especially in the wake of the FTX scandal, we witnessed a rebirth of the network in 2023. Its resilience and antifragility reinforced that Solana is here to stay and is well-positioned to be one of the key infrastructures enabling the internet and financial revolutions.

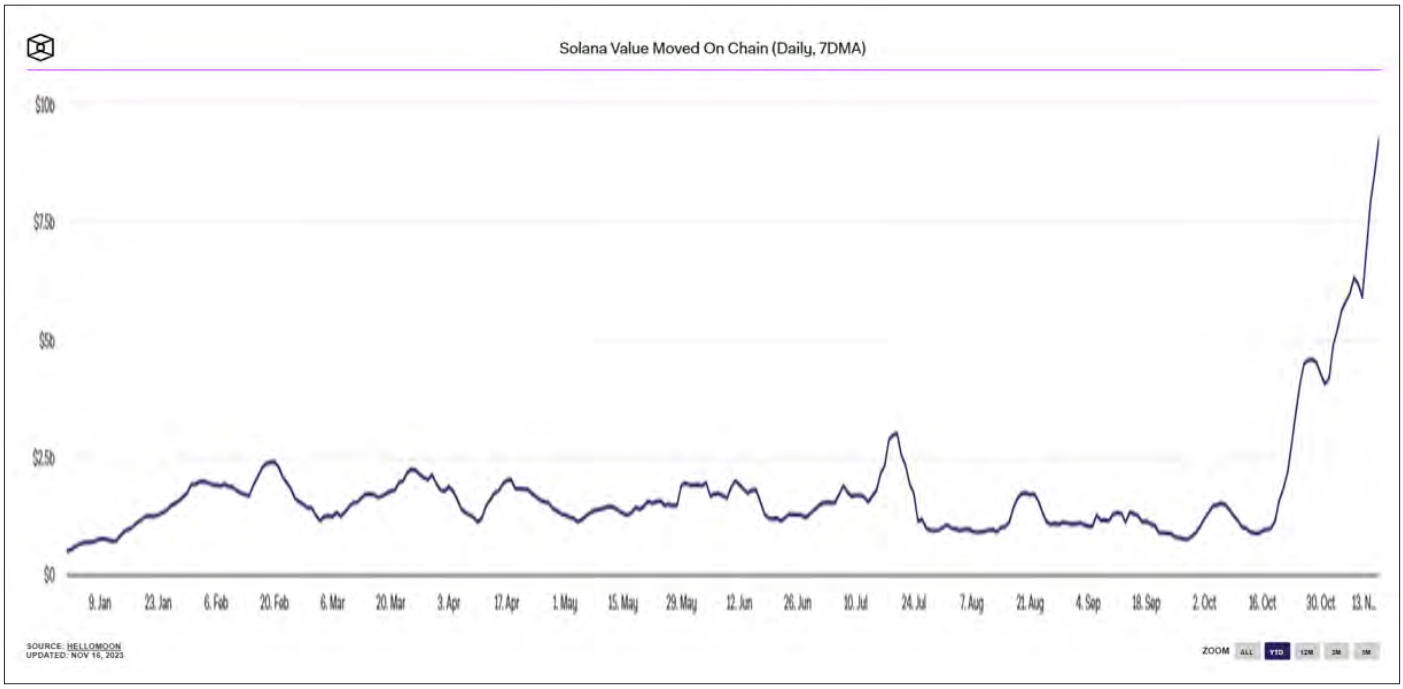
In the eyes of many crypto advocates, Solana is still the most likely project to solve the scalability trilemma of Decentralisation, Security, and Scalability.

Solana Hit New Yearly Highs Post-FTX Collapse

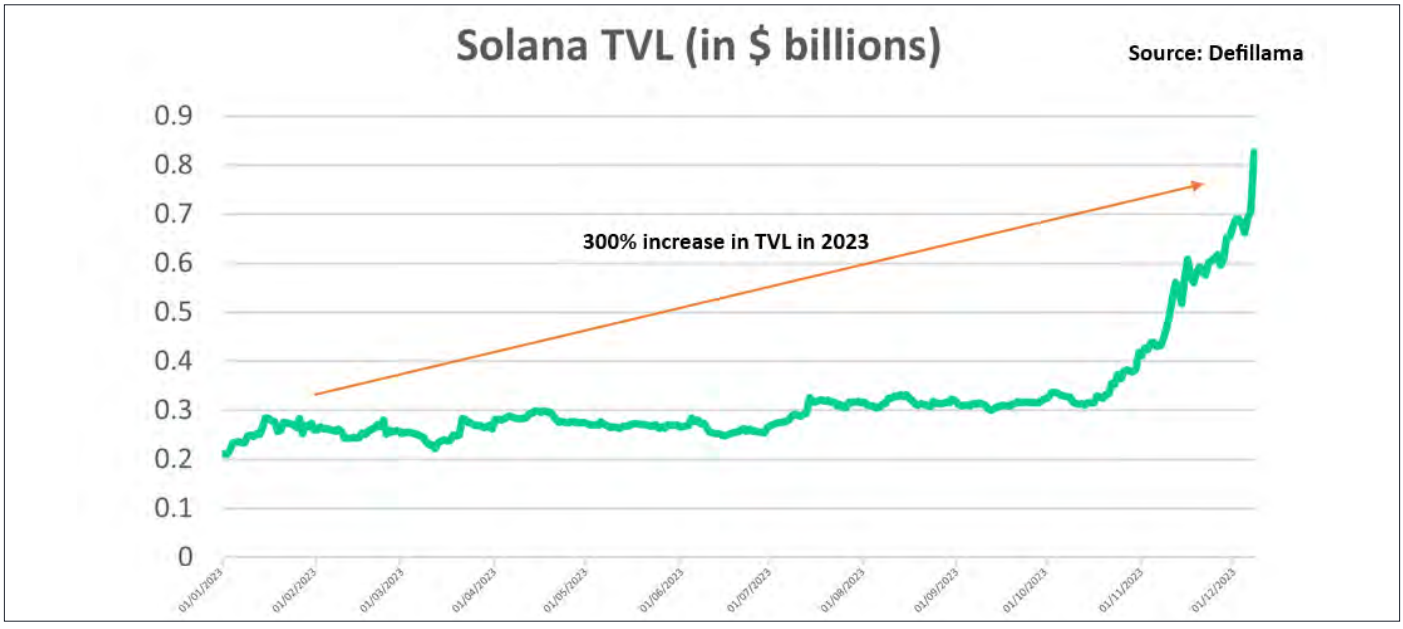
The best word to describe Solana during 2023 is "renaissance". While Solana was deeply impacted by the collapse of FTX in 2022, it seems to have left its past behind and recovered fully with a 635% price increase in 2023. Currently, Solana is the sixth biggest crypto project with a $30.4 billion market cap.



Numerous alternatives were deployed on the network, and Solana's activity rebooted considerably. The total value moved on-chain increased by more than sevenfold since the beginning of the year, with the majority of the rise concentrated towards the end of the year. This reinforces that Solana is indeed experiencing a powerful rebirth.



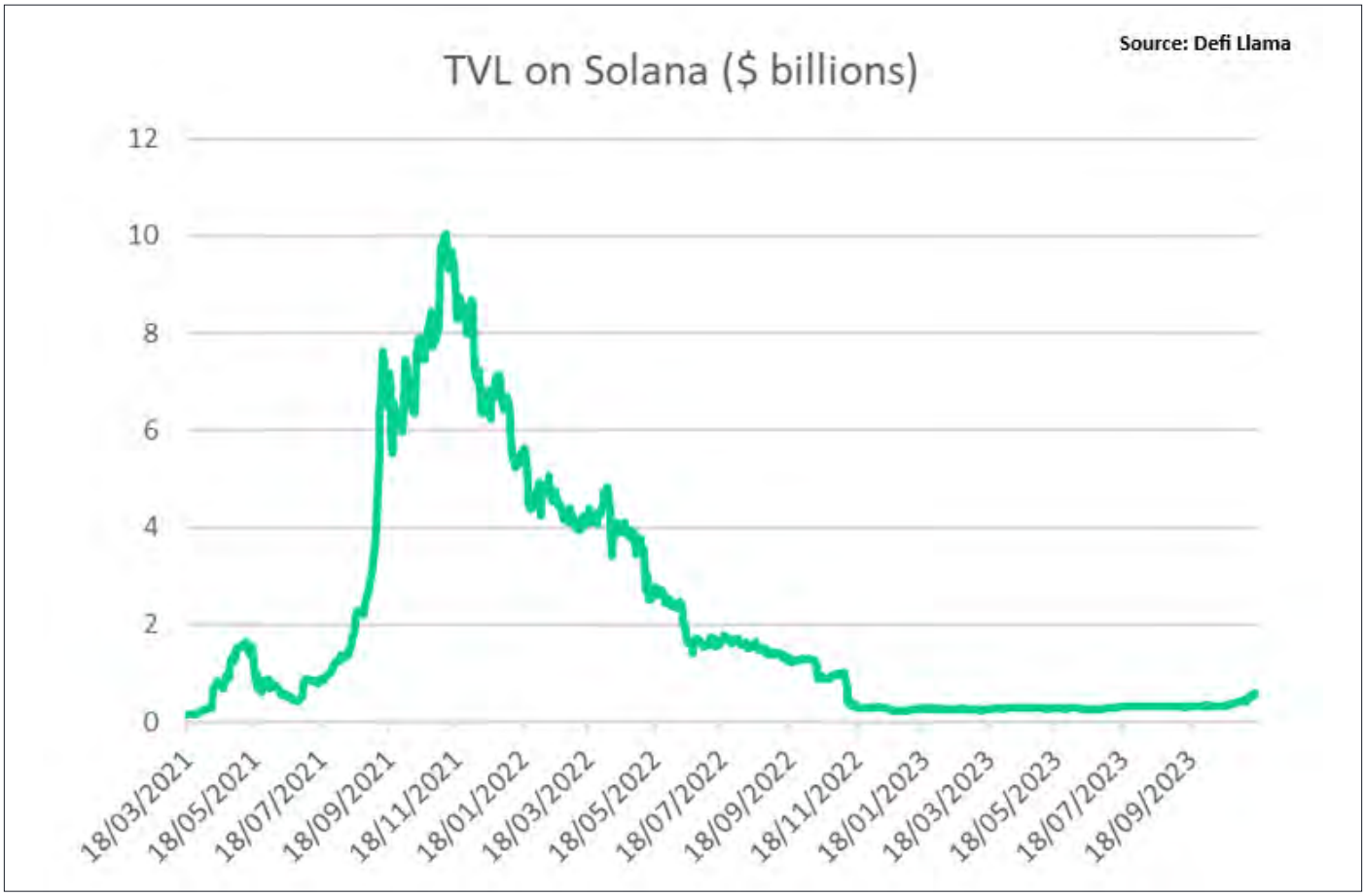
Further, DeFi on Solana saw a surge in 2023. Total Value Locked (TVL) on the blockchain quadrupled throughout the year.



Solana's robust network effect, showcased through successful projects like Kamino, Marginfi, Drift, Jito or Zeta, indicates a thriving ecosystem. Moreover, there was a recent surge of new project launches as suggested by the latest "Hyperdrive" Solana Hackathon, which saw a record of 907 projects submitted.

Another worthy boom for the Solana DeFi ecosystem was the Jito airdrop. It was undoubtedly the biggest Solana airdrop of 2023, and more than $225 million worth of Jito tokens have been airdropped to the community.

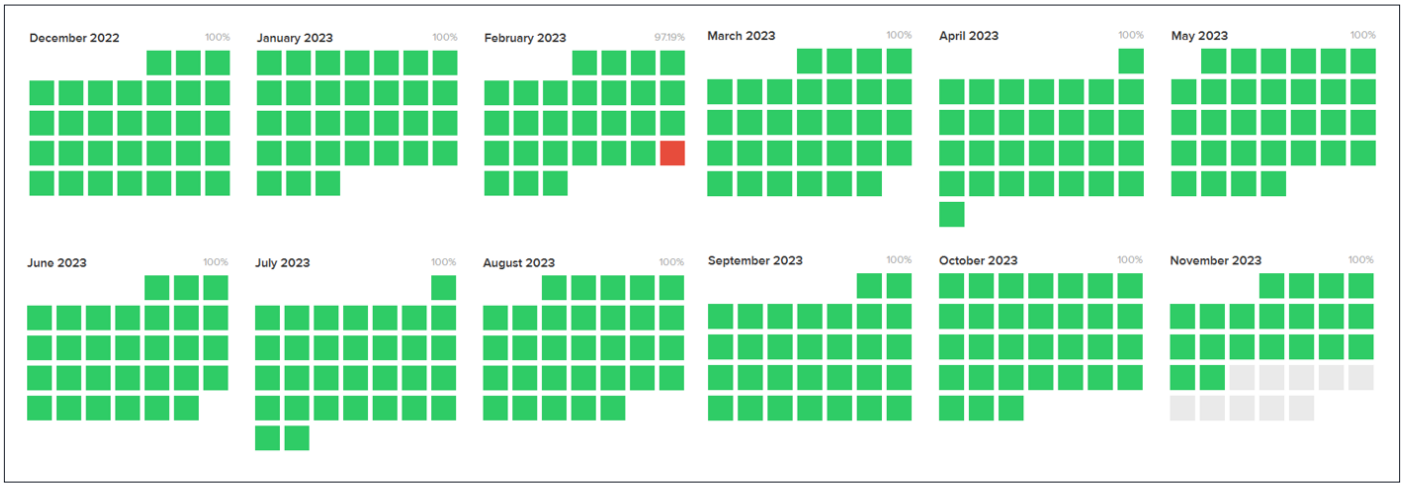
Those elements reinforce our view that TVL is set to continue on its upward trajectory. If Solana were to reclaim its ATH, we would witness a 10x increase in TVL.



The Current State of the Blockchain is Promising

Network Stability

Solana has suffered numerous criticisms over network outages and instability in the past. However, it seems that this is a thing of the past. After the last outage on the 25th of February, Anatoly, the co-founder of Solana, assured us that network instability issues have been fixed permanently. Since then, Solana has had a 100% uptime and has not experienced any problems.

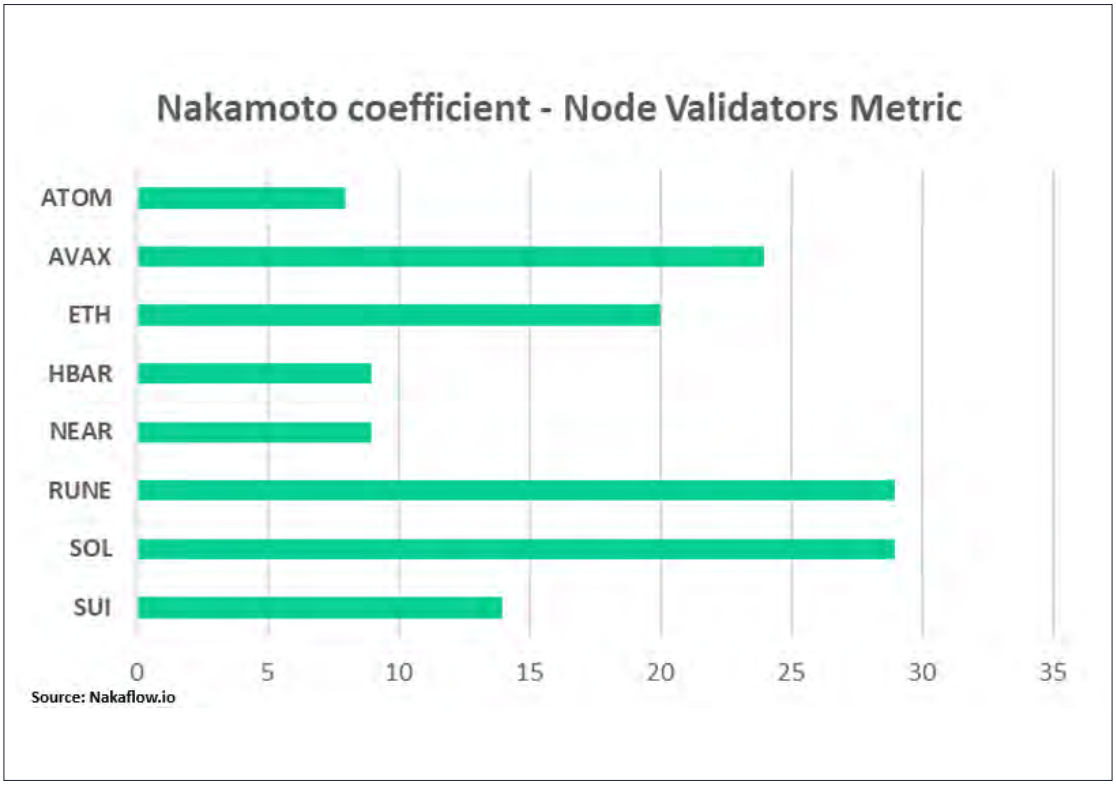


Source: Solana.com

Decentralisation on Solana

Decentralisation is a guiding principle of blockchain networks. It is a vital property because this is what distributes power, control, and decision-making across nodes, making it more secure and censorship-resistant. This can be measured by the Nakamoto coefficient, where the larger the coefficient is, the more decentralised the blockchain network.

According to data from NakaFlow, Solana, together with ThorChain and Avalanche, is one of the most decentralised networks.



A diversity of validators' clients is essential for the long-term robustness of a blockchain network. Currently, Solana has two different client validators and two other ones: Firedancer and Sig, are in the making. When completed, this will make Solana, together with Ethereum, the only chain to claim to have four different client validators.

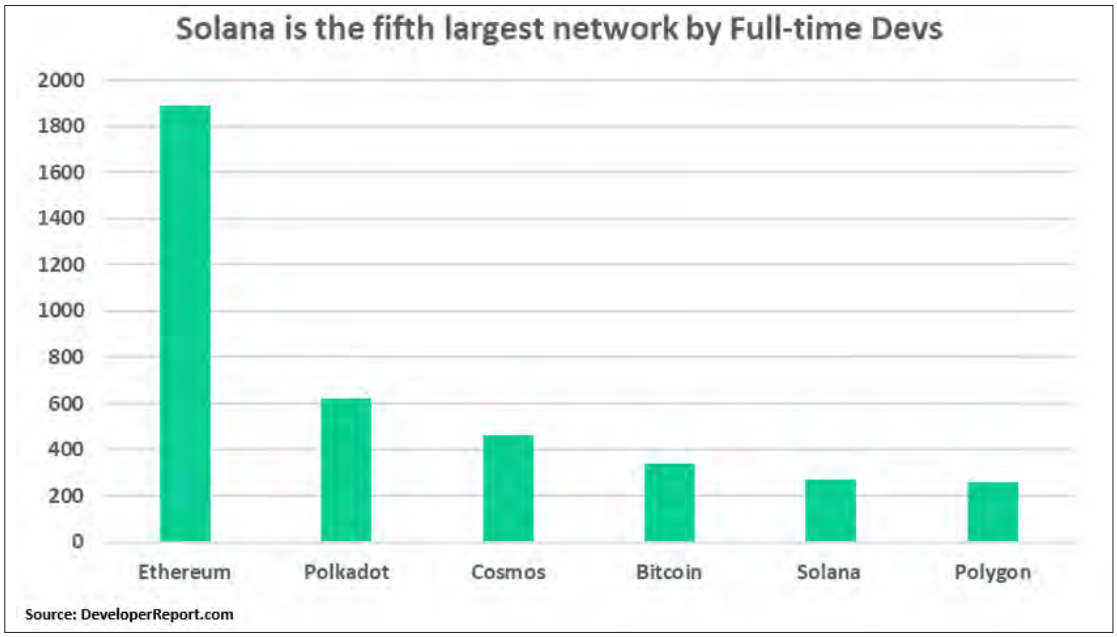
Moreover, Solana is also the chain with the second most number of block-producing validators.

Blockchain	Total Validators (Block Producing)	Validator Software Clients
Solana	1,961	2 (2 in development)
Ethereum	4,435	4
Avalanche	1,311	1
NEAR	218	1
Polygon	79	1

Source: Solana.com (as of 6th september 2023)

Monthly Active Developers

Monthly active developers is an important metric because ultimately, it is developers that are building the future of a chain. With around 270 full-time developers, Solana is currently the fifth most active chain in terms of development.



Major back-end tradeoffs

"We really love pushing boundaries, making Solana the fastest, most efficient, lowest latency, and highest throughput" - Anatoly Yakovenko.

As we've said, Solana is designed to be a high-performance blockchain. Comparing it to Ethereum and its major L2: Polygon, we see that Solana is a step beyond in terms of performance.

	Solana	Ethereum	Polygon
Data Throughput (MB/S)	125	0.1	1.3
Average transaction per second	4,500	15-25	255
Average transaction fees	$0.0002	$33	$0.01
Time to finality	1 sec	13 min	2-3 sec

Source: VanEck; Sygum; Polygonscan; SolanaCompass; Klaytn; Etherscan

While Ethereum has been the chain of choice for smart contracts so far, we believe its design has influenced a wave of developers to create applications with minimum interaction with the blockchain due to its cost. This has put a lot of limits on the spectrum of applications that can be built. Solana's design, on the other hand, enables developers to have full expression of what they can and want to create, unleashing true innovation and breakthroughs in the space.

Compressed NFTs are Pushing Solana Ahead of its Competitors

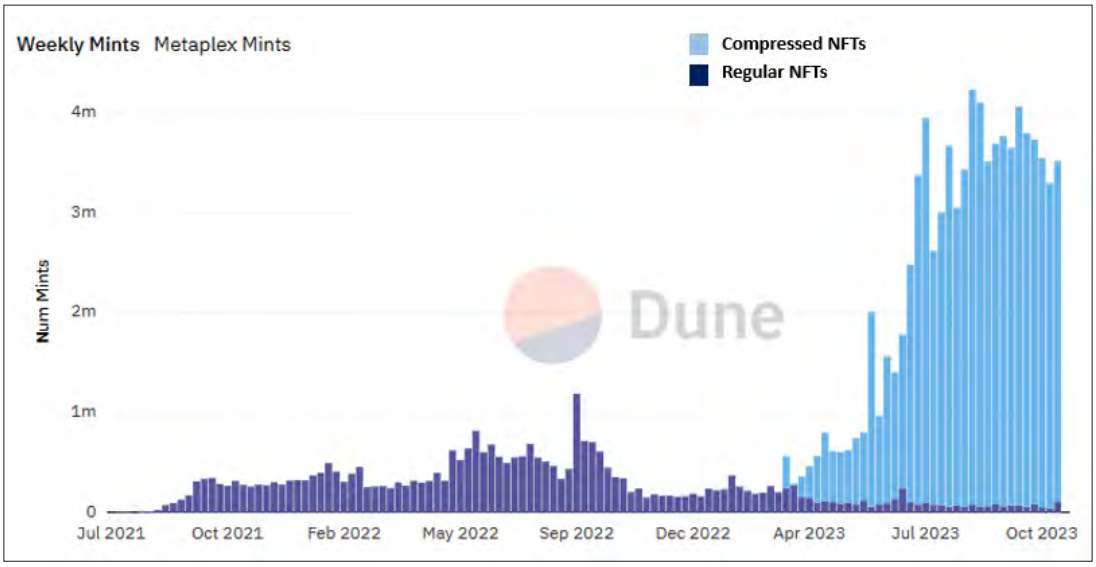
We are witnessing a big change in the NFT landscape. While the last NFT cycle was driven by "scarcity", where NFTs were seen as "luxury goods" and mainly produced on the expensive Ethereum network, we believe the next cycle will be about democratising digital assets to build web3 experiences at scale. The way to make this happen is to reduce the cost of producing and trading NFTs so that brand-new use cases can be made possible.

Solana is well positioned to be at the forefront of this shift with their "compressed NFTs", which are, by a significant order of magnitude, the cheapest way to produce an NFT.

Number of NFTs	Solana Today	Solana with Compression	Ethereum	Polygon
1,000	12 ◎ $253.68	2.57 ◎ $54.33	17.69 Ξ $33,774.35	28.94 $33
10,000	120 ◎ $2,536.80	3.49 ◎ $73.78	176.09 Ξ $336,231.23	288.14 $328.48
100,000	1200 ◎ $25,368	4.22 ◎ $89.21	1760.09 Ξ $3,360,800.03	2880.14 $3,283.36
1,000,000	12000 ◎ $253,680	5.35 ◎ $113.10	17600.09 Ξ $33,606,488.03	28800.14 $32,832.16
10,000,000	120000 ◎ $2,536,800	10.76 ◎ $227.47	176000.09 Ξ $336,063,368.03	288000.14 $328,320.16
100,000,000	1200000 ◎ $25,368,000	56.45 ◎ $1,193.35	1760000.09 Ξ $3,360,632,168.03	2880000.14 $3,283,200.16
1,000,000,000	12000000 ◎ $253,680,000	507.13 ◎ $10,720.73	17600000.09 Ξ $33,606,320,168.03	28800000.14 $32,832,000.16

Source: Solana.com (based on a snapshot taken on the 5th of April 2023)

Weekly mints on Solana rose sharply as compressed NFT launched in 2023. This implies that users and creators have clearly expressed their preferences.



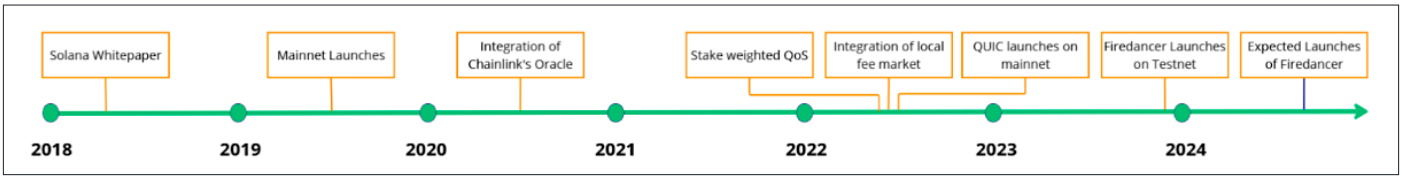
Source: Dune Analytics

One notable use case of "compressed NFTs" was during the Helium Network migration to the Solana Blockchain, where 1 million NFTs were produced for 5 SOL. On the Ethereum mainnet, this would have cost creators approximately $33.6 million.

"Compressed NFTs" redefine how we interact with digital assets and open up a myriad of new opportunities. This could well be what we were waiting for to reach mainstream adoption of crypto technology. From in-game assets to event tickets or music ownership, the opportunities are endless.

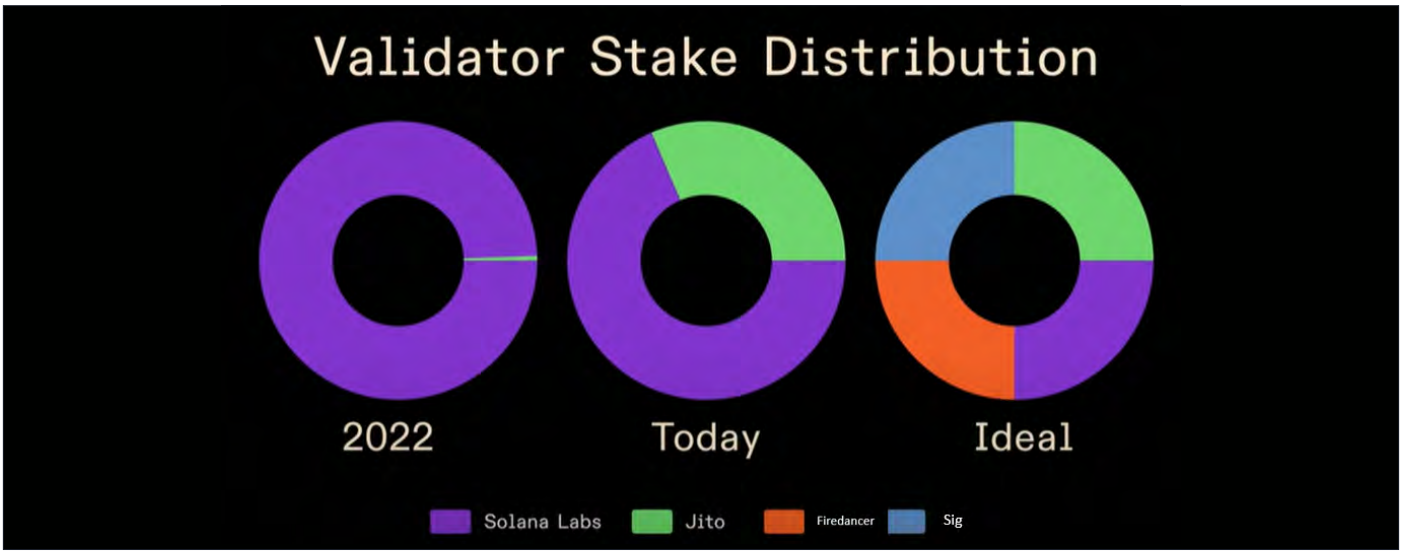
Ultimately, we believe that those elements altogether increase the propensity that Solana will be the first blockchain to host the first viral application that bridges the gap between web2 and web3.

A Work in Progress



The next milestone for Solana is the implementation of Firedancer on the mainnet. Following Solana Labs and Jito, Firedancer will be the third validator on Solana. However, it will be the second one with a fully independent consensus node client implementation. This represents a significant step in improving decentralisation by removing the single point of failure. Moreover, Firedancer is being meticulously monitored to be optimised at every level and provide maximum speed and reliability.

Firedancer Claim		Comments
Increase Validator Diversity	Multiple client implementation will remove the central point of failure. This is key for long-term resiliency and decentralization of the Solana Network	• Jito and the Solana Labs are currently the two only validators on Solana • Firedancer will be an important step in making sure no single validators control >33% of the stake • So far only Ethereum can claim this diversity in terms of validators
Maximum Performance	A validators that operates as close as possiple to the physical capability of the underlying hardware. This means that speed will be bounded only by hardware	• Even though Solana is currently the highest performing blockchain today, Firedancer will improve drastically the current vision of how fast a blockchain can be • They managed to achieve +1,000,000 TPS on testnet



Firedancer will be an important narrative to monitor throughout 2024. If successful, it will be a pivotal moment in Solana's performance and progress.

Solana is in the fast lane

- Solana has had a strong 2023 with a 540% price increase.
- It has shown great resilience as a network: activity has rebooted, and much growth has happened in the ecosystem.
- Its major back-end tradeoff makes it one of the most probable blockchains to bridge the gap between web2 and web3.
- An important improvement is expected in 2024: Firedancer. This could improve the performance of the blockchain by a significant order of magnitude.

All those elements make us bullish on Solana in 2024. We believe this blockchain is well-positioned to be a leader within the crypto ecosystem and that its strong price action in 2023 marks only the beginning of a much bigger move.

4. The SocialFi revolution

Ziga Potrebujes Twitter: @jpegco11ector

The past year has been a transformative period for the SocialFi and Digital Identity sectors. Integrating social interactions with financial systems has led to the emergence of a new paradigm in decentralised finance (DeFi), known as SocialFi.

As we have already discussed in this article, the misaligned incentives on social media today are a huge problem. Social Media operates with a business model that benefits from unethical social engineering.

A concept closely related to SocialFi is also decentralised digital identity, which has been gaining traction as it can potentially solve many fundamental problems in crypto. On the other hand, it is a delicate subject since a trial-and-error approach could mean your personal data becomes available to bad actors or is even made public.

Proof-of-Personhood

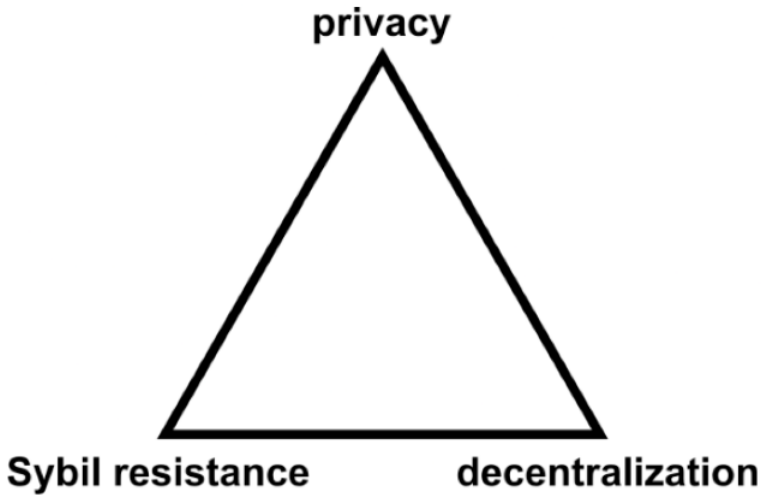
Solving this problem has proven difficult; however, in 2023, we have seen a renewed vigour to crack it. It's essentially having a system that would allow humans to create 1 account, prevent them from doing 2 and allow bots to create none. Solving it has the potential to thwart many of the concentration of power/spam attacks and would greatly add to any form of decentralised governance.

The two-sided risk we are facing a two-sided risk if we don't solve the P-o-P. Source: Vitalik's Blog

So far, we have seen two main ways to solve the proof-of-personhood problem: social-graph-based or biometric-based. The first relies on vouching - for example, other verified humans attest that you are one as well and potentially could be penalised if you are not.

Biometric-based means some institution scans a body part unique to you, like a fingerprint or iris. Notable risks with this approach include some serious privacy and security concerns about where and how your data is stored so that it is not misused by hackers, authoritative governments or other bad actors.

It seems that solving the proof-of-personhood problem is another trilemma. Blockchains we know today have mostly sacrificed Sybil resistance for decentralisation and privacy. But we should be careful with the fast-and-furious approach when it comes to sharing our private data.



The digital identity trilemma. Source: Molly White

As Molly White puts it in her article: "The technology industry, and the crypto industry especially, has long adopted a "move fast, break things" approach to innovation. Self-sovereign identity is not a field where such an approach is acceptable."

As Vitalik himself puts it, there are inevitable tensions between business interests and the needs of the wider community, so it's important to exercise a lot of vigilance.

SocialFi projects

We have seen several SocialFi applications gain significant traction and user bases over the course of last year. Here, we will look at some of the most notable ones.

The Lens Protocol, situated on the Polygon Proof-of-Stake blockchain, is a Web3 social graph designed to empower creators by granting ownership of the links between them and their communities. This results in a fully composable, user-owned social graph.

Cyberconnect is a native social graph platform that has emerged in the crypto space. It does not need an invite to access it; you merely connect your wallet and mint your profile. You can then use dApps built on top of these social graphs to produce content which is automatically shared across all dApps built on top of them.

The SocialFi application that probably received the most attention in 2023 was probably Friend.tech. It's a social application built on the Base blockchain, leveraging user data from Twitter. Arguably, one of the biggest reasons it took Crypto Twitter by storm is its polarising nature as it deals with the financialisation of relationships.

When you log into Friend tech, you first connect your Twitter account, deposit some ETH and mint your first share of "yourself", also called a key. This unlocks a chatroom that any of your shareholders (i.e., keyholders) can access and chat with you.

The price of shares is on a bonding curve, and the more minted, the more expensive they become. Conversely, you can burn your share and receive ETH. The dividing controversy stems from the fact that it financialises relationships and seems to empower the rich and already powerful more than the little guy looking to make a living on social media.

Opinion on where we are going next

According to our co-founder, Anthony Lesoismier, the first phase of innovation for socialfi projects needs to be centred around digital identity and allowing people to "export" their Web 2.0 identity, currently trapped in private databases, into an open yet anonymous Web 3.0 database.

"It is essential that all the achievements that are trapped in Web 2.0 are passported into Web 3.0. That's the beginning of the disruption of the Web 2.0 social media monopoly because it allows businesses and users to connect without playing the guessing game of the GAFA that costs them billions. Moreover, it's not a good strategy to ask users to give up their Web 2.0 identity that they have built for years and start from scratch in Web 3.0. With that regard, projects such as the 'Ethereum Attestation Service' (https://attest.sh/) are fundamental pieces of the socialfi landscape of tomorrow. EAS is an innovation as important, if not more, than ENS (Ethereum Name Service). Since digital identity will be key, we are also very much paying attention to the work of Karma Labs (https://karma3labs.com/)." Anthony Lesoismier

Conclusion

This area might not yet be discussed in the mainstream just yet, but we believe it's a hot topic that will soon get a significant amount of spotlight. I hope we find an option tailored to the privacy and security concerns mentioned above.

A measure of success should not be just convenience but also a technology used for the actual betterment of society instead of creating new means of control for oppressive regimes.

Sources

- https://www.youtube.com/watch?v=AVZIvnex4Gw
- https://blog.mollywhite.net/is-acceptably-non-dystopian-self-sovereign-identity-even-possible/#fn:fn1
- https://vitalik.eth.limo/general/2023/07/24/biometric.html

5. Cryptocurrencies & Artificial Intelligence

Stefan Duprey Twitter: @stefanduprey

Artificial Intelligence (AI) and its applications will usher in a new revolution which will dwarf the industrial revolution. Parallel to this, another revolution is taking place: the birth of the trustless world of blockchain, disintermediating any centralised entities into a secure, cryptographic, and trustless world. These two revolutions are bound to fusion

- Decentralisation of AI infrastructure leasing competing against the monopolistic big GAFAs
- Improvement of AI itself by applying proof of stake technology to a network of decentralised AI models
- AI services automation for retail investors access and intelligence provider rewarding
- More technically, AI will bring many of its algorithmic breakthroughs to improve blockchain technology (data compression by autoencoders, predictive analytics for consensus optimisation, ...)
- A more secure, private and customer-centric experience for the crypto industry as a whole

This symbiotic relationship will improve both industries symmetrically, as AI will improve blockchain overall and blockchain AI.

Smart big data

The artificial intelligence war is not in the algorithms (most of the state-of-the-art are open-sourced very early on) but in the data. Access to properly processed and organised big data is a key factor for Artificial Intelligence to be able to harness that data and build AI models on it. Big and cleanly processed data is a prerequisite to training deep neural networks on that data.
Constructing structured datasets by building features/predictors and mixing them with unstructured data (text, image) is an absolute necessity for any state-of-the-art predicting AI model.

Protocols for data indexing:
Smart blockchain data aggregation for rapid requesting and use.

Protocols are being built to allow people to access 'on-chain' data easily. This protocol allows users to define nodes and automatically specifically define on-chain data processing to process and index on-chain data and make it available for use. Nodes dedicated to specific decentralised applications (Uniswap and Aave apps, for instance, have dedicated nodes processing their transactions and aggregating their data at a macro level)) allow users to easily access decentralised application-processed data and build their own ai algorithm over macro-level data.

Some of the most prominent companies in that data structuring field are here to stay and will be vital data-feeding endpoints for any machine learning algorithms:

- The Graph: GRT The Graph is the most famous protocol for indexing and querying data from blockchains in a similar way that Google indexes and queries data from websites. Indexing blockchain data can be challenging, but The Graph aims to change that by organising data into smaller "subgraphs".
- Dune, Covalent, and shdwdrive: on-chain data indexing, structuring, SQL based on chain requesting and monetisation for enterprises

Decentralised network storage designed for Web3

Blockchains like Ethereum are designed at their very heart to be 'transactional' blockchains. They were not conceived from the get-go to store loads of data (NFTs store only a hash, not the whole image, for example).
Different alternatives try to usher in the era of big data directly on-chain by allowing decentralised unstructured data storage (text, website, images and videos).

This decentralised storage will allow for privacy-preserving, trustless, and censorship-resistant unstructured data storing.

- **FIL** is a decentralised network storage designed for Web3. It uses IPFS as a building block to share all the benefits of content addressing. IPFS is a modular suite of protocols for addressing, routing, and transferring data, designed from the ground up with the principles of content addressing.

  The verifiable, cryptographic proofs ensure that Filecoin storage providers store and keep data safe. Interoperability with Web3 compute networks unlocks novel applications and use cases.

- **Arweave:** The Arweave network is a permanent and decentralised web inside an open ledger. Permanent storage has many applications, from preserving humanity's most important data to hosting truly decentralised and provably neutral web apps.

  These decentralised data storage facilities will enable a fully decentralised World Wide Web.

Bringing safely external data via AI intelligent oracles

AI can enhance the accuracy and reliability of oracles, which provide external data to smart contracts. From external market price feed to IoT sensors data ingestion, oracle needs on the blockchain are ubiquitous. Defi has a strong need for external price feed, and in a decentralised world, insurance-triggered contracts could need any kind of external physical data.

AI intelligent oracle will ensure that smart contracts have access to real-time and trustworthy information, improving the overall functionality of decentralised applications (DApps) built on the blockchain through data curation, outlier detection, confidence threshold alerting and data reconciliation.

ChainLink is the poster company of the Oracle ecosystem. AI algorithms will enhance every aspect of their business:

- Cross-Chain Communication
- Market and Data Feeds
- Proof of Reserve
- Automation of smart contracts in a decentralised way with time or event-based triggers.

ORAICHAIN is the poster company for AI smart oracle with an IBC (inter blockchain communication protocol) enabled Layer 1 that verifies multidimensional trustworthiness of AI to allow it to build Web3 applications with unique AI oracles.

AI platforms

Deep learning infrastructure leasing

Appetite for GPU computation has become ubiquitous as needs for GPUs now do not limit themselves just to training deep learning models, a task that only big American companies (google/apple/facebook/amazon) could do (need of big data and multiple GPUs clusters). Generative AI still uses those pre-trained networks by big corporations, but now also inherently needs numerous network predictions : the 'Denoising diffusion probabilistic model' process of AI generation is iterative and costly in terms of computation. Retail users who need generative AI for their work (video/image creation and edition) do also need GPU computational power to achieve them.

GPUs infrastructure leasing: Render Network

The Render Network provides near unlimited decentralised GPU computing power for generative AI creation. The Render Network is the first decentralised GPU rendering platform, empowering artists to scale GPU rendering work on-demand to high-performance GPU Nodes around the world.
Through a blockchain marketplace for idle GPU computing, the network provides artists with the ability to scale next-generation rendering work at fractions of the cost and at orders of magnitude increases in speed when compared to the centralised GPU cloud.

AI Infrastructure Leasing

AI infrastructure is a business currently handled by big computational companies like Amazon AWS, Microsoft Azure or Google.
The addition of a blockchain infrastructure over the AI infrastructure eases the service payment and resource allocation process.

Here below are two examples:

- Akash gives you the power to own your cloud. You can deploy your own cloud infrastructure, and you can sell your unused cloud resources to others. You can also buy cloud resources from others, and you can use them to run your applications.
- Computable lab, Computable Labs is building new Internet infrastructure to democratise access to data and algorithms for artificial intelligence. The company aims to provide a token-curated registry (TCR) that serves as a hub for the buying and selling of data.

Sharing data: monetising data (datasets marketplace)

As previously mentioned, data is at the heart of AI. Dataset accessibility through marketplaces is an absolute prerequisite to building AI forecasting algorithms. Blockchain infrastructure can be used for data usage payment and data provider rewards in a seamless way.

- Enigma Data sharing and consumption with Enigma. The project is building a data marketplace that allows computation to be done securely off-chain. Right now, sharing data between parties is often a risky and irreversible process. Someone who buys data on a marketplace could distribute it, creating a free-rider problem. Recently released, Enigma's new data marketplace Catalyst allows companies and organisations to contribute data which users can subscribe to and consume via smart contracts.
- Datum On the more consumer-facing side is Datum, which is a global data marketplace that allows users to make money from their own data. The company boasts 90,000+ people using its app to monetise their consumer data with pre-screened partners.
- Ocean Protocol is an ecosystem for sharing and monetising data. It provides a tokenised service layer that helps to unlock data sharing, particularly for AI. It is designed for scale and uses blockchain technology that allows data to be shared and sold in a safe, secure and transparent manner.

Decentralised network of machine learning models: intelligence staking

At the very end of every data science forecasting competition, there comes a consolidation time where the different protagonists discuss together and merge into a bigger team after having successfully managed to ensemble their AI model into a better aggregated one. Decorrelated and differently built models almost always ensemble together into an altogether better-averaging model.

The brilliant idea that will put blockchain consensus technology at the heart of AI is to see AI models forecasting ability as a value that you can stake : intelligence staking.

Proof-of-Stake is a method to achieve a distributed consensus. It makes it possible to secure a blockchain by providing the right economic incentives to push for honest transaction validation.

The same PoS methodology can apply to rank and slash intelligence providers in a decentralised network of AI forecasting models by leveraging digital incentives and rewarding participants directly for their contributions of computational resources, expertise and innovation.

This pioneering approach will provide models in which the intelligence of the whole exceeds the intelligence of the parts by ensembling different models.

- **Bittensor:** TAO is the key reference for that innovation. Bittensor is a mining network, similar to Bitcoin, that offers censorship-resistant access to a decentralised network of machine learning models. Bittensor establishes a comprehensive, open-source ecosystem of artificial intelligence competencies. The native currency, TAO, constitutes both a reward and an access token in the network.

- **Numerai:** NMR is a famous AI tournament platform where algorithms try to predict the stock market in an AI tournament. They stake their tokens in the models they feel more confident: 'put your token where your conviction is'.

  Numerai data scientists stake their best models with the Numerai cryptocurrency: NMR.

  Watch the video introducing NMR to our community. Data taylored tournament Introduction to Numerai Signals, a tournament for data scientists who have their own data.

  Numerai uses crowdsources machine learning problems to a global community of data scientists and then combines the results into an AI meta-model to trade as a long/short equity hedge fund. Incentivising and rewarding in its native Numeraire token, the company allows data scientists to bet on how effective they believe their model is and shares the earnings measured against the staked amount.

Intelligence sharing (AI Marketplace, AI Publisher and AI automation): monetising AI

Another business around AI is how to bring its results to the retail users, may they be predictive, generative or explicative. A whole range of offers tries to cover that business through an extensive offer of published models and easy automation of those through API endpoints. Below are some examples.

- **SingularityNET (AGIX)**
  SingularityNET is a blockchain platform that allows anyone to build, share and monetise AI services. It has an internal marketplace where users can browse and pay for AI services in the platform's native cryptocurrency – AGIX. Developers can make money from AI solutions and models without having to build apps for end users fully. Similarly, developers can buy AI solutions and models to use in their applications. Search, trial, and select from an ever-growing library of AI algorithms created by a community of service providers. Integrate AI services into your own applications. Our publishing infrastructure provides both a central hub for creating, editing, and managing your AI services and the tools to launch those services to a global market. AI - Domain Specific Language (DSL) A deep self-organising network of AI agents running on the SingularityNET Platform can outsource work to each other dynamically — leveraging AI functions, exchanging input/output data, negotiating payments, and enhancing the agent reputation system.

- **Synapse AI**
  Synapse is creating a decentralised data and intelligence marketplace of re-searchers, trainers, processors, and contractors that can be accessed programmatically at any time.

- **Fetch.ai (FET)**
  Fetch.ai is an AI and machine learning platform based on the blockchain. Fetch.ai is all about automating business tasks such as data processing and trading. Its native cryptocurrency, FET, is used to pay for transactions on the network.

AI for scalability, robustness and interoperability enhancement

Consensus and network usage optimisation

AI algorithms can discover bottlenecks, optimise consensus protocols, and improve network scalability by analysing network data and transaction patterns for a more dynamic consensus.
AI algorithms can analyse network conditions, transaction loads, and other factors like validator reliability in real-time to dynamically adjust consensus algorithms. This can lead to improved scalability, faster confirmation times, and more energy-efficient consensus mechanisms.
AI can analyse historical data and predict future network usage patterns, helping blockchain networks anticipate scalability challenges. This information can be used to optimise parameters such as block size and transaction fees to accommodate increasing demand.

Interoperability

AI can facilitate the development of intelligent cross-chain interoperability solutions, allowing different blockchain networks to communicate and share data seamlessly. By integrating AI with blockchain technology, developers can create more adaptive, secure, and efficient systems that are better equipped to handle the complexities and challenges of real-world applications.
The synergy between AI and blockchain has the potential to unlock new possibilities and drive innovation across various industries.

Data compression

The recent progress in AI compression algorithms will benefit the blockchain industry. As cryptocurrency adoption is on the rise, it is important to have an arsenal of tools to scale and optimise the blockchain network. AI can help blockchain technology improve its efficiency and performance..
Auto-encoders become alternative compressing algorithms which outperform standard wavelet compression formats like jpeg and mpeg. AI models are then built over those compressed ‘embeddings’ rather than the whole image or text, leading to significant gains for both training and predicting.
AI-driven compression algorithms can optimise the storage of blockchain data, reducing the overall storage requirements. This is particularly beneficial for nodes that need to store the entire blockchain history, contributing to improved scalability and resource efficiency.

AI for customer-centric crypto

Helping customers through crypto with AI

Better web3 marketing

AI can improve the entire consumer experience in the cryptocurrency business in many ways:

- **Better targeted marketing for web3 retail users**
  The birth of targeted marketing platforms for web3 users like AdEx will usher in a new marketing revolution. Through blockchain interactions, blockchain users' behaviour will be analysed, and the more fitting will be devised to suit them.

- **Blockchain savvy AI bots**
  Blockchain savvy bots will drastically improve retail customer experience.

  Crypto-focused AI models can offer personalised investing recommendations, alert consumers to market movements, trigger customised news/price alerts and provide insights into their portfolio performance by analysing user behaviour and preferences. This level of personalisation has the potential to empower crypto users and help them in making better-informed decisions.

  The poster boy company here is **ChainGPT**: a blockchain intelligence chatbot IQ GPT, powered by the IQ token for service payment (https://www.chaingpt.org/)

- **Generative AI-powered NFT avatar**
  Generative AI will impact NFT and allow for more artistic creations. See delysium for an example (https://www.delysium.com/).

Decentralised AI-guided governance

AI can assist in the decision-making processes of decentralised autonomous organisations (DAOs) by providing data-driven insights. This can contribute to more efficient and transparent governance within blockchain communities.
BAD, for instance, is a new decentralised experiment that combines Blockchain, AI, and DAOs in a risky, meme-worthy concoction.

AI enhancing security and privacy

Enhancing Security and Detecting Fraud: Anomaly detection to secure wallet transactions

Security is a key worry in the crypto ecosystem because of the increasing frequency of hacking attempts, fraud, and identity theft.

AI-powered systems can monitor blockchain networks for unusual patterns or suspicious activities, thus helping in identifying potential security threats, including fraud and cyber attacks, and allowing for timely responses to mitigate risks. Machine learning algorithms can examine past transaction data for suspicious trends or anomalies that signal fraudulent behaviour. AI systems can adapt and grow to avoid emerging risks by continuously learning from fresh data and detecting in real-time fraudulent patterns in complex transactions as well as scam detection through a quick analysis of massive on-chain code through machine learning.

Forta is the poster company for blockchain security-enhancing technology for retail users.
Web3 wallets can use the Scam Detector to screen pre-signed transactions and protect users from scams to screen decentralised applications and pre-signed transactions, and protect users from scams. Security and compliance teams can supplement their existing data with the Scam Detector, staying up to date on the latest threats.
DeFi can develop and run custom detection bots on Forta to monitor protocols for security and operational health.
Asset managers can subscribe to existing bots like the Attack Detector to receive real-time alerts for threats impacting their positions and enabling capital preserving actions.

Privacy

AI can be used to implement and optimise homomorphic encryption, allowing computations to be performed on encrypted data. This enhances privacy and confidentiality by enabling secure data processing without exposing sensitive information.

Nucypher
Similar to privacy layers like Enigma, NuCypher provides proxy re-encryption for distributed data networks, bringing private data to public blockchains and facilitating secure data sharing between enterprises.

6. The State of GameFi

Anthony Lesoismier Twitter: @AnthoLGSB

It is impossible to ignore games. More than simple entertainment, games have become an integral part of our lives, and as virtuality takes up more and more space in our daily routines, this trend is not going to stop. The video game industry has come a long way since its humble beginnings in the 1970s, with Atari credited for releasing the first game, Pong. As of mid-2020, Fortnite had been played for a cumulative total of 3.8 billion days, which equates to 10.4 million years. Gamers have spent more time playing Fortnite than humans have existed.
The gaming industry is thriving, with 3.38 billion gamers globally and an expected revenue of $184 billion in 2023. Generations Z and Alpha, who increasingly view digital and physical experiences as one, devote considerable time to gaming, averaging around 15 hours per week.
According to Pitchbook, $3.475 billion of venture capital has been invested in web3 games from the start of 2019 through the end of June 2023. So, how is Web 3.0 intended to disrupt one of the biggest industries in the world, and why should an investor care?

Gamefi & Play To Own - The Chapter 4 of Gaming

Gaming Industry Transformation Timetable:

Period	Era	Business Model
1978-1985	The Arcade Era	Chapter 1 - Pay as you go to Play
1978-2000	The Console Era	Chapter 2 - Own & Pay to Play
2000-2010	The Computer & Beginning of the Online Era	Chapter 2.5 - Pay to Play & Free To Play
2000-2021	The Mobile & Cloud Era	Chapter 3 - Free To Play
2021-Present	The Web3.0 Gamefi Era	Chapter 4- Play to Own

In the early days of gaming, player engagement was directly tied to the revenue an arcade unit generated. The birth of the gaming console shifted the economics of game development away from smaller revenue streams towards larger upfront purchases. Massive game studios and publishers vied to be among the select few worthy of a consumer's $60 spend on a single game.

The birth of the internet and the rise of mobile have shifted the business model of gaming back towards small, ongoing transactions. Game developers deliberately engineer their in-game economies to facilitate player spending throughout their entire lifetime as a customer.

Gaming goes through paradigm shifts every 15/20 years, and in our view, the rise of Play-To-Own is similar to the rise of Free-to-Play 15 years ago. We believe that over the next 10 years, most games will incorporate web3 tech and ownership layers, enabling web3 to become a dominant business model in gaming.

Understanding business models is key to understanding the status quo and opportunities and therefore making assumptions about where an industry will go next. In our view, the current business model of games has strong ownership problems. Therefore, it's simply inevitable that gamers will eventually demand ownership of virtual gaming goods. Web3 is the response to that change.

~3.38b global gamers

~$184b in revenues 2023

75% of gaming revenue comes from in-game purchases of virtual goods

> 11.5m creators designed over 62m virtual clothing & accessory items on Roblox

3.2b virtual transactions on Roblox, 1.8b avatar items on Roblox's Marketplace

75% of kids ages 9–12 in the US, UK, Canada, New Zealand, and Australia play Roblox, which has 70.2m DAUs

Fortnite's revenues from 2018–2022 ~ $20b, primarily cosmetics, Fortnite has 200m MAUs

Time spent in video games ~12 hours / week; for Gen Z in the US, 15 hours / week

84% of US consumers engage with video games

Games are #1 entertainment for Gen Alpha & #3 for Gen Z

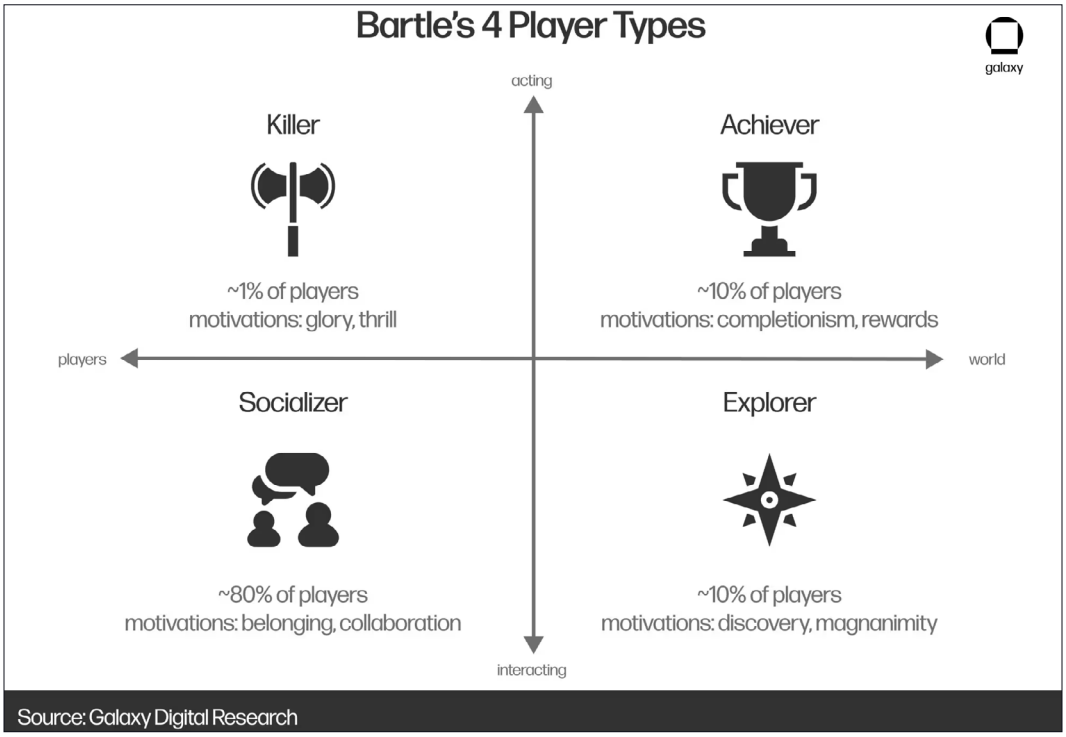
90% of Gen Z engage with games & virtual worlds

50% of Gen Z and Gen Alpha spend money in games

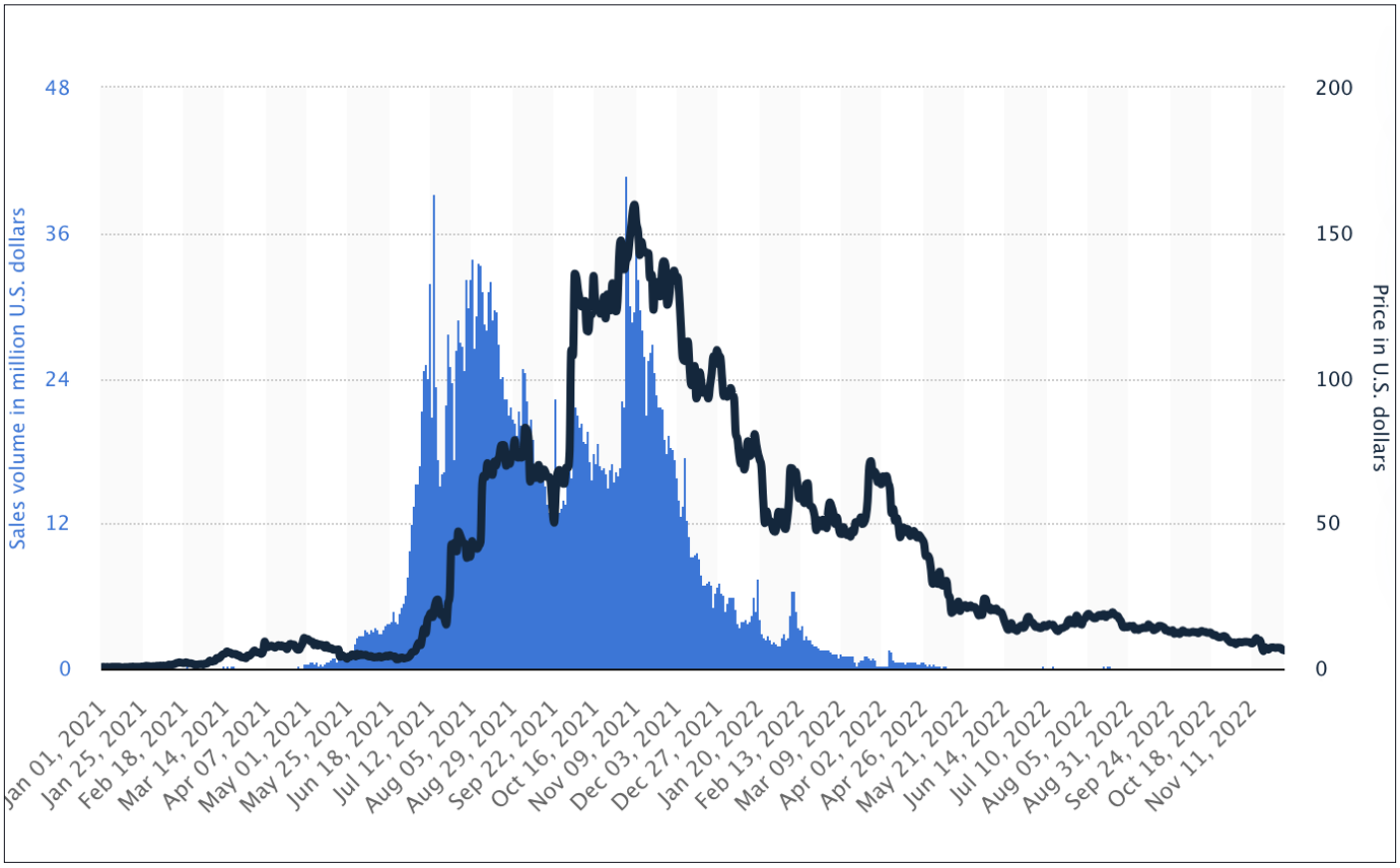
Gen Z has $360b in spending power

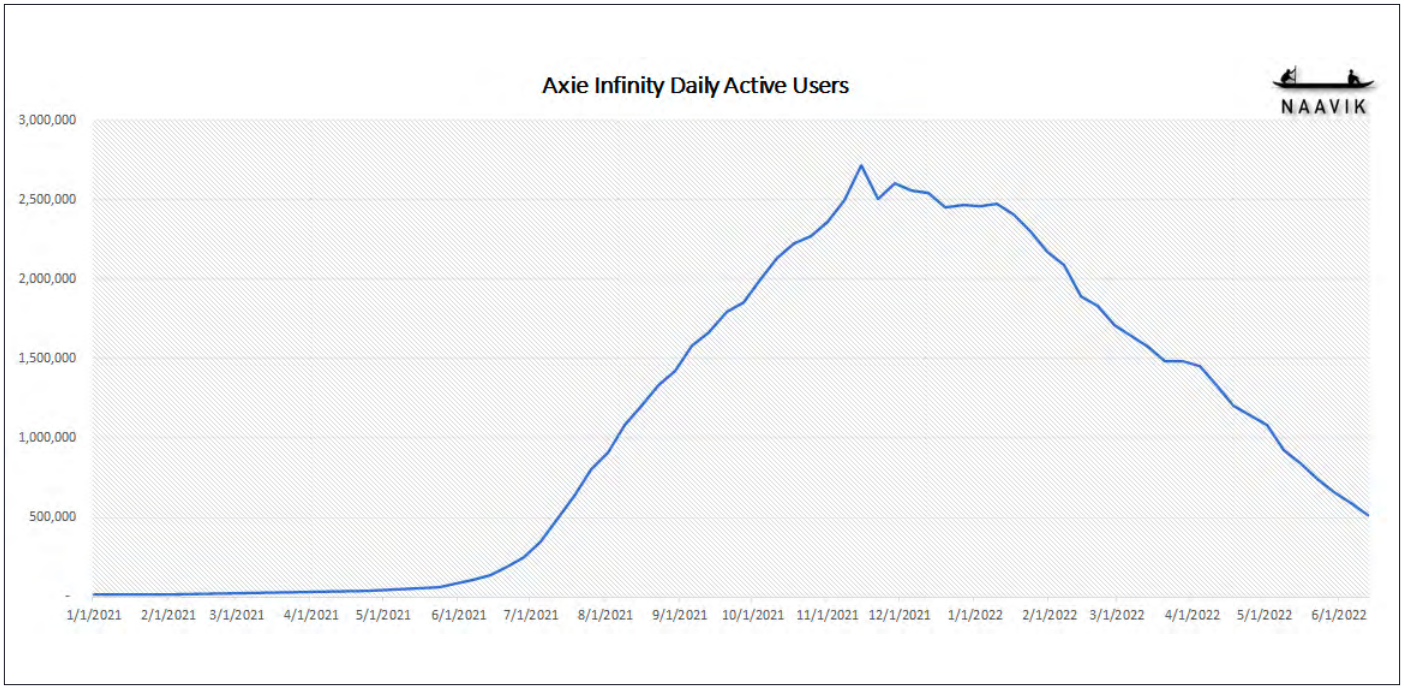
Crypto Adoption is mainly driven by speculation, is Gamefi a game changer?

It is fair to assume that the vast majority of the population thinks about money in the first 10 minutes after waking up every morning. Making enough money to live is a common goal and obsession. Gamers play first and foremost to forget about their daily problems; they are looking for fun and a moment of entertainment and distraction.



The first wave of gaming adoption failed in that regard, as the number one motivation for players to engage with games like Axie Infinity or other Play-to-Earn titles was making money rather than having fun.



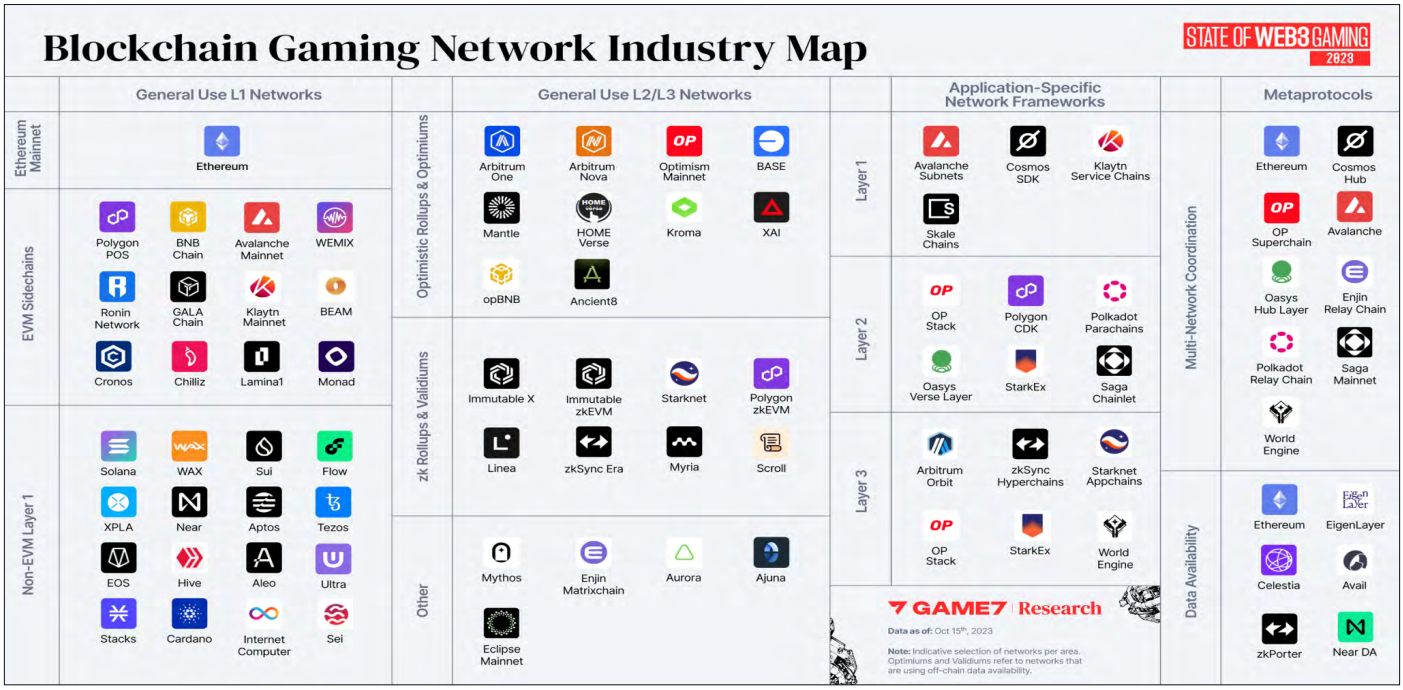


https://www.deconstructoroffun.com/blog/2022/6/17/axie-infinity-part-2-redemption-or-ruin

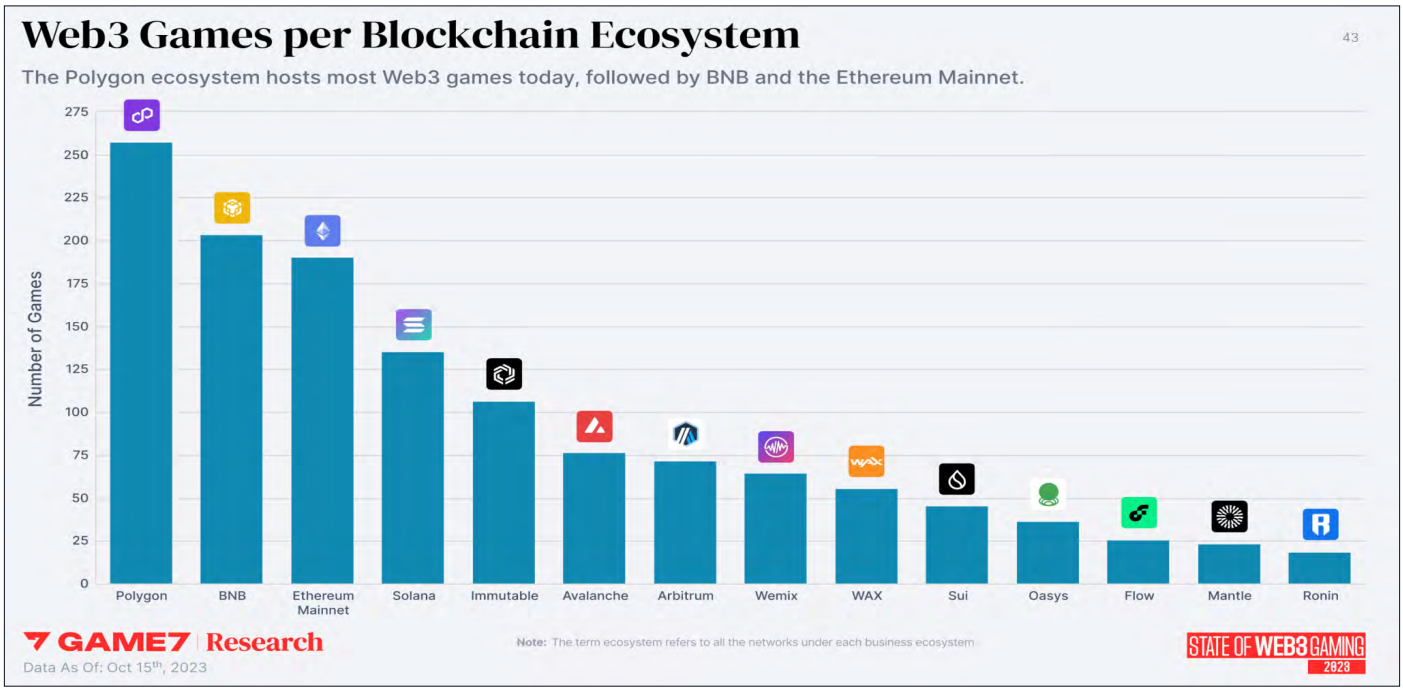
The first generation of GameFi projects in the 2020/21 era was designed for DeFi traders, not gamers, but this is changing. As an illustration of this phenomenon, the Epic Games Store in the United States now offers 69 blockchain games.

Driving Forces & Top Web3 Gaming Projects Ahead

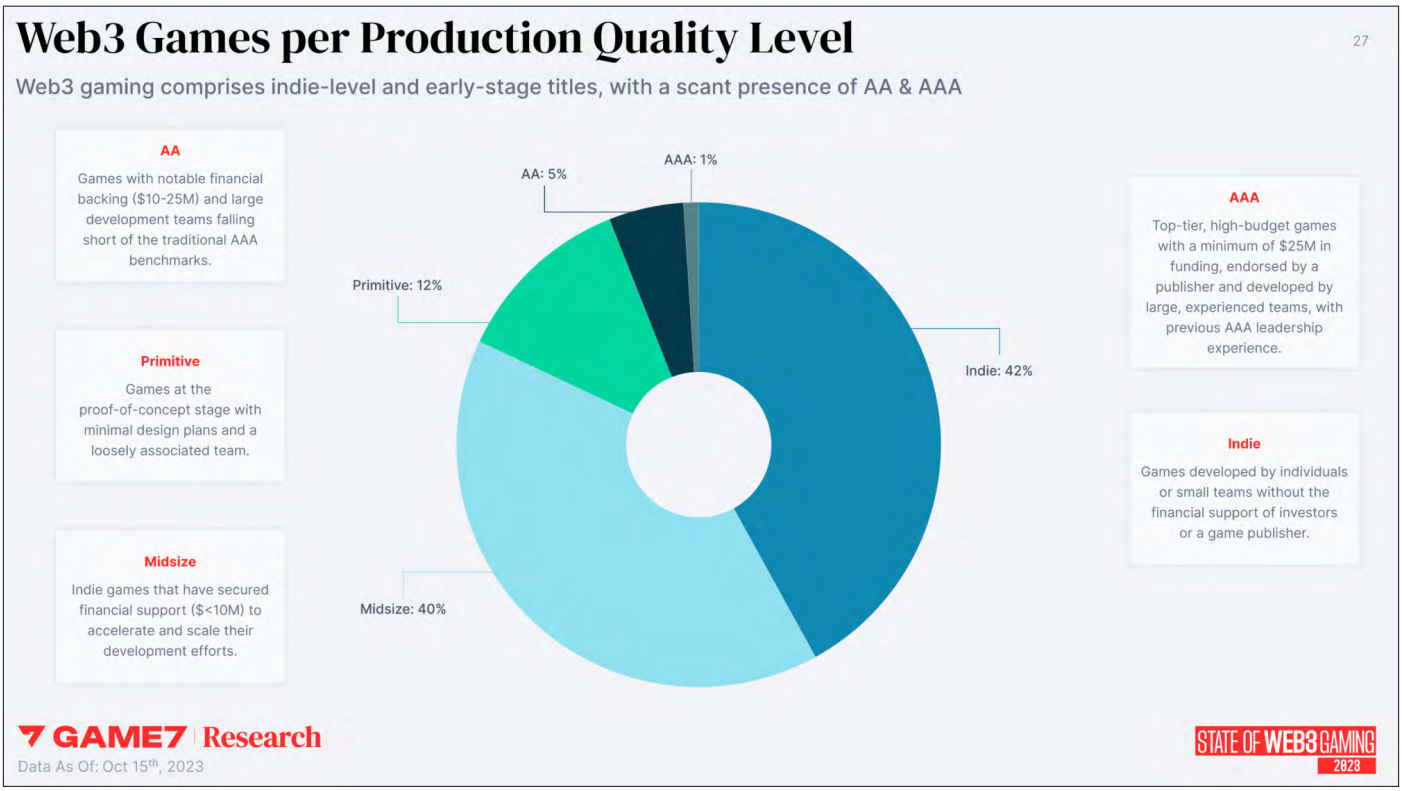
It is important to understand that gaming is at the centre stage of the strategic development for most blockchains, as there is a global consensus that gaming will be a driving force in crypto adoption. Therefore, the competitive environment at the infrastructure level is intense, as a lot is at stake. The blockchain ecosystem that successfully attracts the next Fortnight or League of Legends will emerge as a big winner in traffic and on-chain activity.



As investors, it is, therefore, important to monitor which blockchain is leading in terms of gaming adoption. Now that you understand, an indirect way to play the emergence of GameFi is to invest in the right infrastructure. Nonetheless, in games, much like in cinema, it takes only blockbusters to win; quantity matters less than quality.



Are we ready to compete with Web2 Games? That's the burning question on everyone's minds at the moment. At SwissBorg, we feel that we are increasingly approaching this moment. Indeed, there are encouraging signs, such as big investments (Parallel 50M Funding, Shrapnel > 20M funding) or the recent partnership between ILV and Team Liquid (https://illuvium.io/news/illuvium-x-team-liquid).



Nevertheless, as the graph above shows us, Triple-A games can still be counted on the fingers of one hand. For this cycle, we will prioritise infrastructure and, notably, gaming projects focusing on digital identity with the goal of enhancing the experiences of fans, gamers, and other video game enthusiasts by bringing a Web 3.0 layer to their favourite Web 2.0 games. We cannot be more confident that XBorg will emerge as a GameFi reference in 2024.

Our favorite picks going into 2024:

Category Triple AAA Games

- **Prime:** https://echelon.io/
- **Shrapnel:** https://www.shrapnel.com/
- **Big Time:** https://bigtime.gg/
- **Illivium:** https://illuvium.io/

Category Infrastructure

- **XBorg:** https://www.xborg.com/
- **IMX:** https://www.immutable.com/
- **BEAM:** https://onbeam.com/

Sources:

- https://messari.io/report-pdf/7fbdbf30ec57b9a7fa0d4df8b4193b768597a88b.pdf
- https://www.galaxy.com/insights/research/history-of-gaming-and-web3-future/
- https://research.game7.io/state-of-web3-gaming-2023

7. DeFi is here to stay

Matteo Bonato Twitter: @mattiboySB
Aurélien Giraudon Twitter: @spdy

The end of the bears

Yes, we can say that with a great degree of confidence: the crypto bear market is over!
Why all this confidence?
Because nearly all indicators point to this.

First, after losing ground in 2022, Bitcoin has rebounded 130% in 2023 and is on track to be one of the best-performing major assets of the year.
The crypto recovery continued in November as financial markets relaxed about a variety of macro risks. Within digital asset markets, this resulted in a shift in market leadership away from Bitcoin to an increasingly broad array of crypto market segments.
Tensions in the Middle East have eased and over the last month, financial markets appeared to relax about a variety of tail risks, which helped previously underperforming assets to rebound.
Central bank rates seem to have reached their peak.
Last but not least, crypto trading volumes are again on the rise.
And who is benefiting more from increased trading volumes if not DeFi?

DeFi Trends 2023

Perpetual and derivative protocols

A brand new financial product introduced by crypto is that of the 'perpetual futures', or 'perps'.
Perpetual futures are derivative contracts without an expiration date, allowing traders to speculate on asset prices indefinitely. By design, they also allow traders to take on leveraged positions.

2023 definitely was the year of perps protocols. Just to name a few of them: dYdX, GMX, Hyperliquid, VertexLevel Finance, Kwenta, and Gains Network.
Decentralised perpetual protocols have gone through multiple evolutions to provide more efficient trading and other earning possibilities for users:

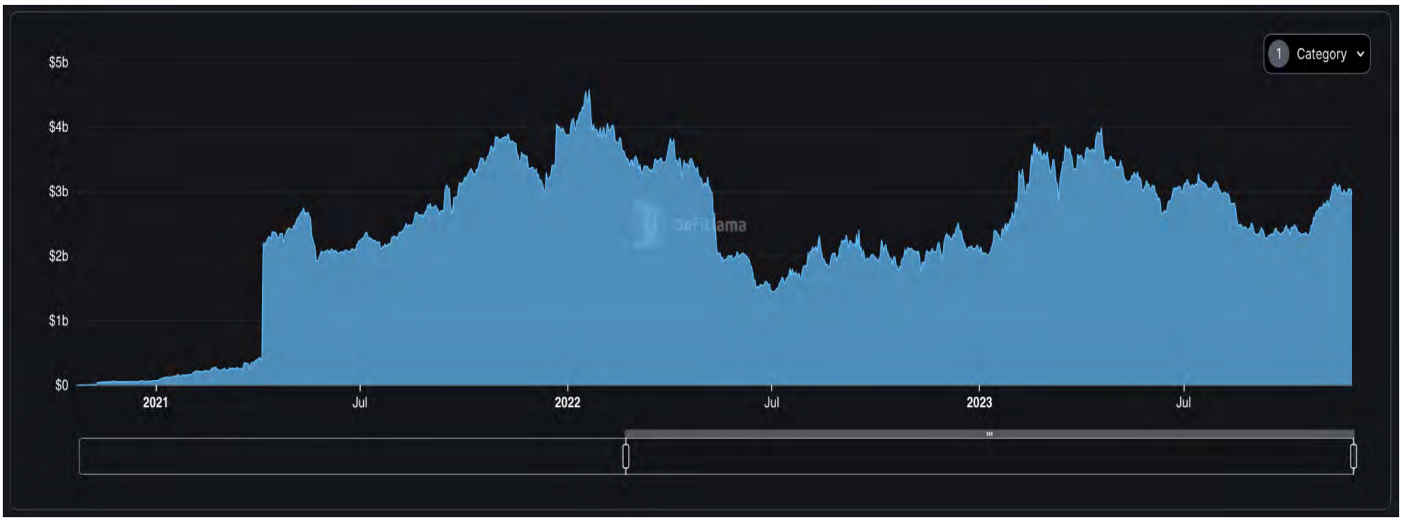
- **Central Limit Order Book:** The first iteration of decentralised perpetuals largely mimic those of centralised perpetuals by aggregating orders and matching buyers and sellers on an order book. While trades and liquidations are executed and settled on the network, the order book and order matching are handled off-chain.
- **Virtual AMMs (vAMMs):** From off-chain order books, Perpetual Protocol introduced vAMMs that utilise the same constant product formula as traditional AMMs such as Uniswap. No real assets are stored on the vAMM. Instead, they are stored in a smart contract vault which then acts as the collateral backing the vAMM.
- **The Great Layer 2 Migration:** The first decentralised perpetuals were built on Ethereum, but network congestion and high transaction fees hampered their viability, which relied on high throughput and lower costs. As such, protocols began deploying to alt chains and Layer 2 rollups.
- **Hyper-product oriented L1 are the new cool:** Running a CLOB dex on-chain is tricky; for now, only high-throughput chains were able to offer that experience (Solana being one of them), some protocols like DyDx and Hyperliquid decided to build their own chain by using various technology (Cosmos and Tendermint) and optimising it, creating true app-chains.
- **Protocol Fees to Token Holders:** As trading volume grew, these protocols also earned trading fees which could be distributed to governance token holders via staking mechanisms as a form of incentive. These fees were paid out in ETH or stablecoins instead of more governance tokens, offering a "real yield" to holders unlike other typical DeFi yield farms
- **Liquidity Pool Model:** Popularised by GMX, newer decentralised perpetuals began utilising a liquidity pool model, allowing liquidity providers (LPs) to become the counterparty for traders. If traders profit from their trades, losses are socialised by the liquidity pool, and vice versa. LP tokens increase in value as trader losses are added back into the pool.

The table below shows some basic features of the main perpetual protocols.

Decentralized Perpetual Protocols	Native Tokens	Liquidity Model	Supported Networks
dYdX	DYDX	Off-chain messages establish orderbook	Ethereum, Native blockchain on Cosmos
GMX	GMX, GLP	Liquidity Pool	Avalanche, Arbitrum
Gains Network	GNS, gDAI	Liquidity Pool	Polygon, Arbitrum
Kwenta	KWENTA	Synthetix Debt Pool	Optimism
Level Finance	LVL, LGO, LLP	Liquidity Pool	BNB Chain
Perpetual Protocol	PERP	Virtual AMM Model (vAMM)	V1: Ethereum V2: Optimism

Source: coingeko.com

Looking at Perpetuals and Derivatives protocols TVL over the last 2 years, it seems they did not suffer much during the bear market. TVL is back at its long-term average, notwithstanding the fall of crypto asset prices observed during the period.

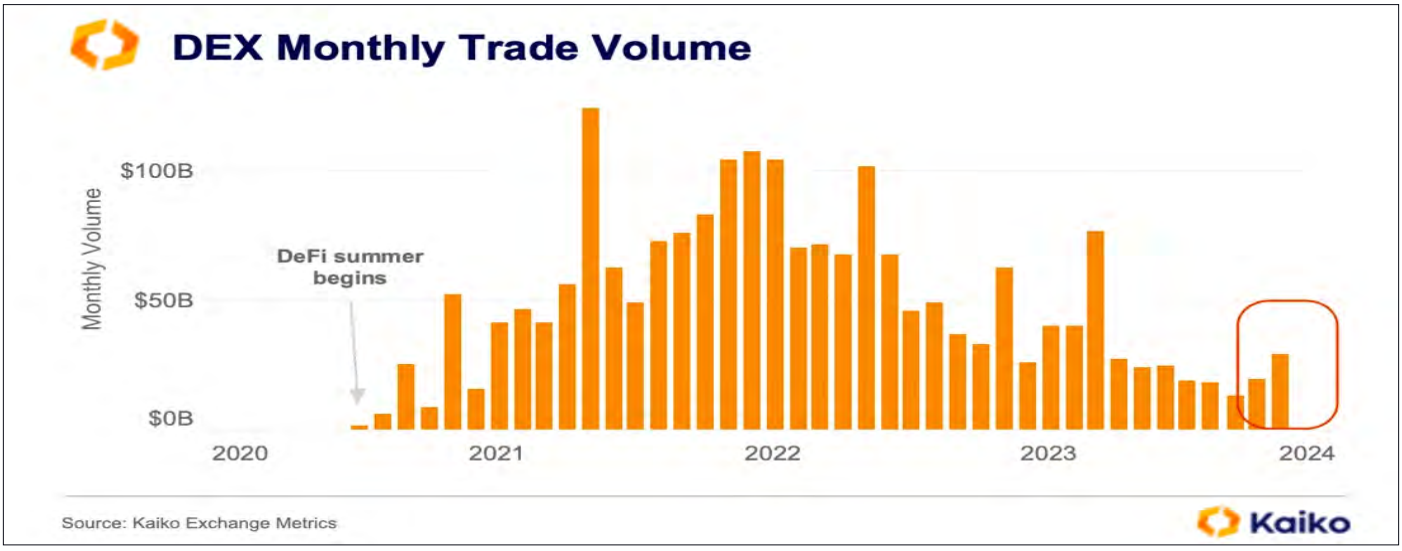


The good news is that volumes are picking up again. November 2023 experienced the largest trading volumes in over 2 years. Note the strict relation between these volumes and the price of Bitcoin.

The outlook for perpetual and perps trading protocols seems to be shining. After the fallout of FTX the issue of transparency and counterparty risk in crypto has played a pivotal role in starting moving volumes from CEXs to DEXs. The perps' evolution has contributed to bringing down trading fees and now users can have a CEX-like experience but on-chain (look at DyDx v4). Protocols like d8x claim to have fees as low as a CEX.

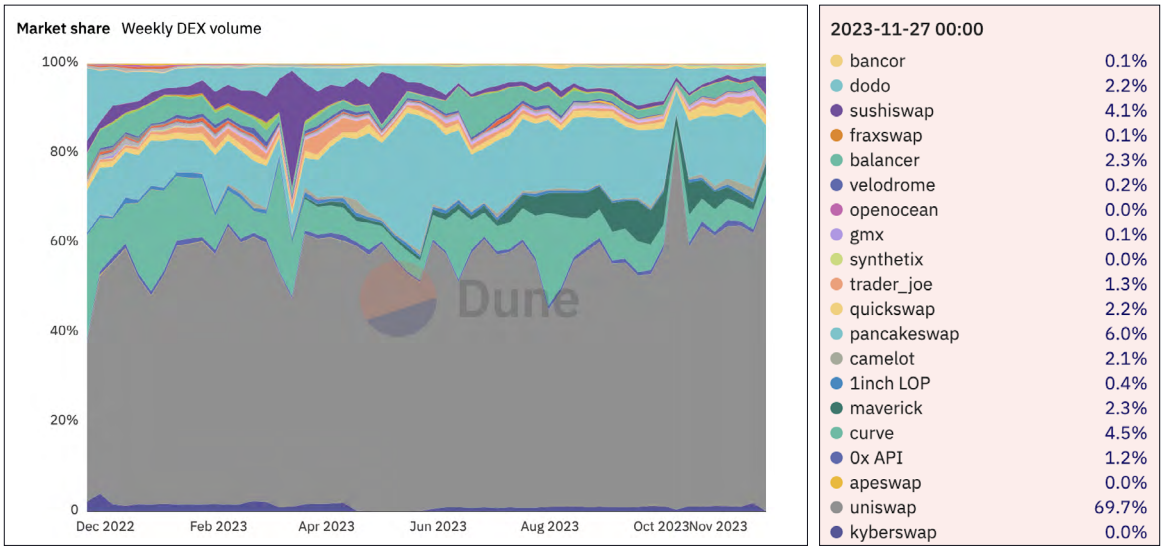
The revive of the DEXes

DEXes did suffer much more during the last bear market. The plot below shows the trading volumes for DEXes over the last 3 years.
The effect of the 'DeFi summer' is striking. So is the impact of the bear market.
Trading volumes have however stabilised and started to increase from October 2023.
DEX monthly trade volume rose to $29b in November, recovering from a multi-year low hit in September.
This could be a sign that the bear market is also over for DEXes.



Source: https://research.kaiko.com/insights/diving-into-dexs

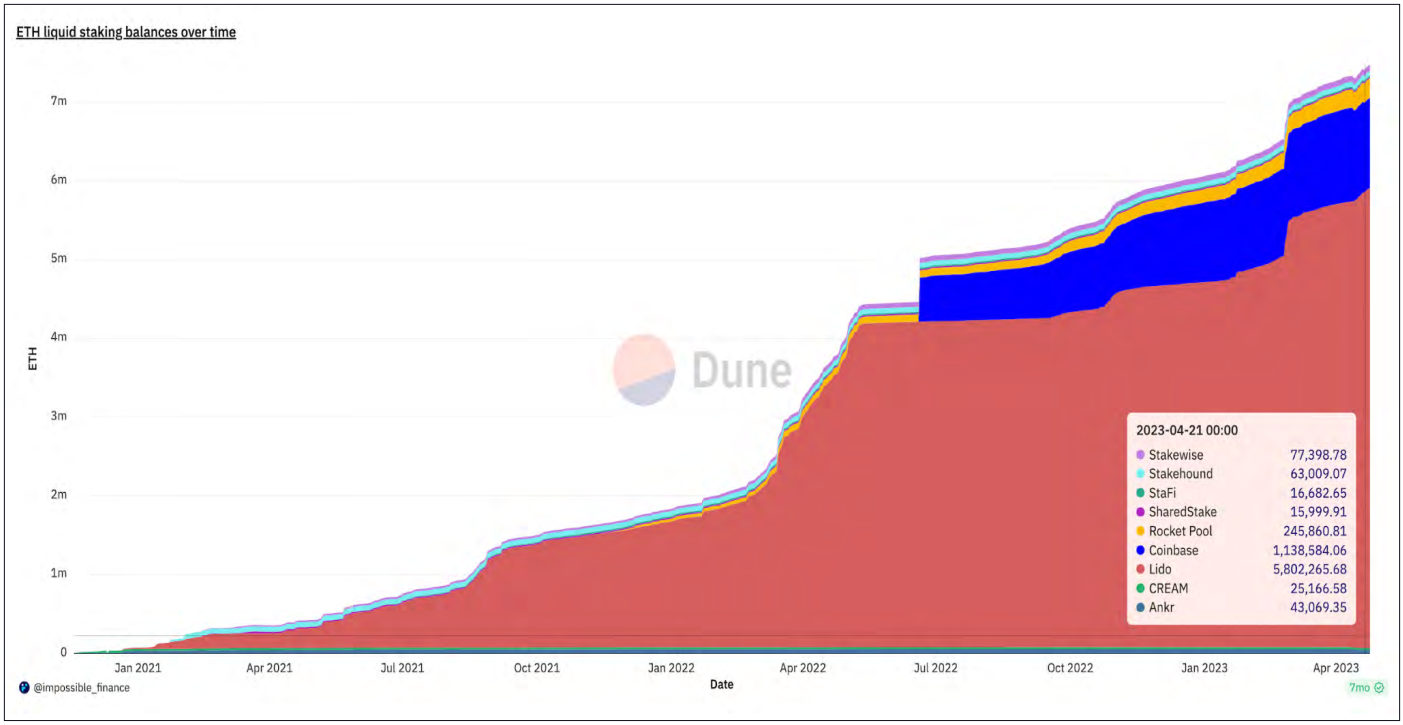
Regarding DEX dominance, the prime actor remains Uniswap (grey area) with over 50% of the overall volumes. The second is PancakeSwap (turquoise area). Curve, TraderJoe, and Balancer follow.



Source: https://dune.com/hagaetc/dex-metrics

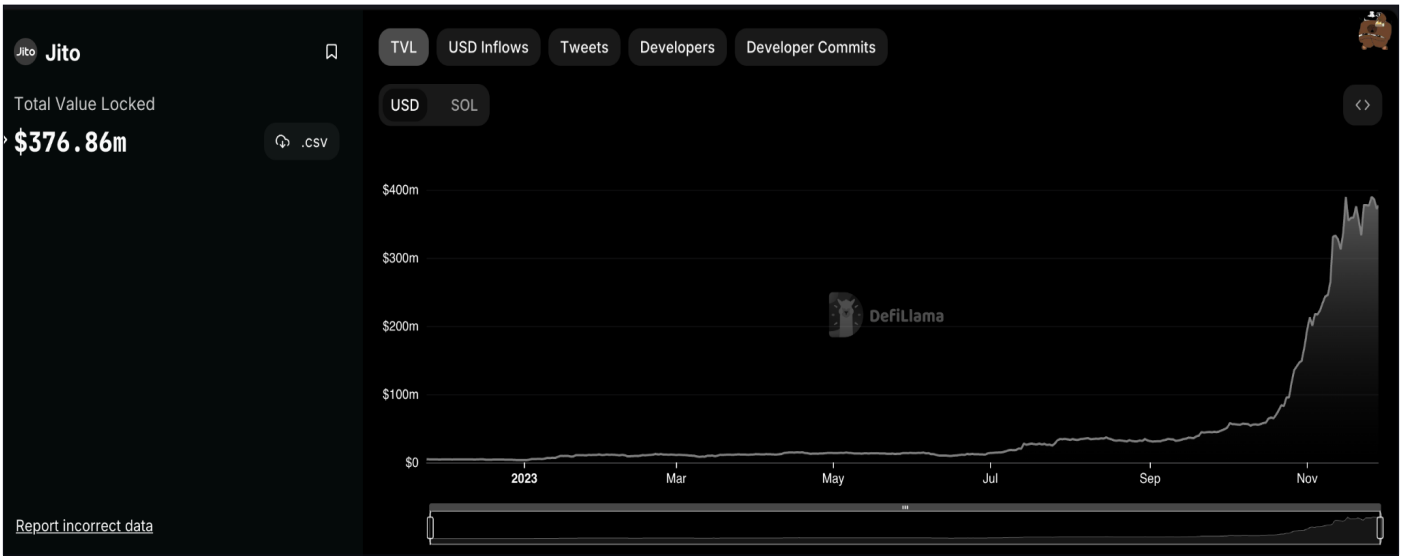
Liquid Staked TokensDerivatives (LSTDs)

The plot below shows the evolution of staked ETH per liquid staking protocol.
Lido clearly emerges as the market leader. But Coinbase and, to a lesser extent, Rocket Pool are catching up. Also, the increasing trend of ETH staked over time is impressive and shows how this business model has become the biggest contributor to the total DeFi TVL, with 50% of dollar locked in TVL to be found in a liquid staking protocol ($26b over a total DeFi TVL of $52b, source: DefiLlama)



Source: https://dune.com/impossiblefinance/liquid-staking-derivatives

A special mention is reserved for Jito Networks. This protocol is the new kid on the block when it comes to SOL staking. Its TVL increase is impressive and has reached around $400m TVL or 6.7m SOL staked.

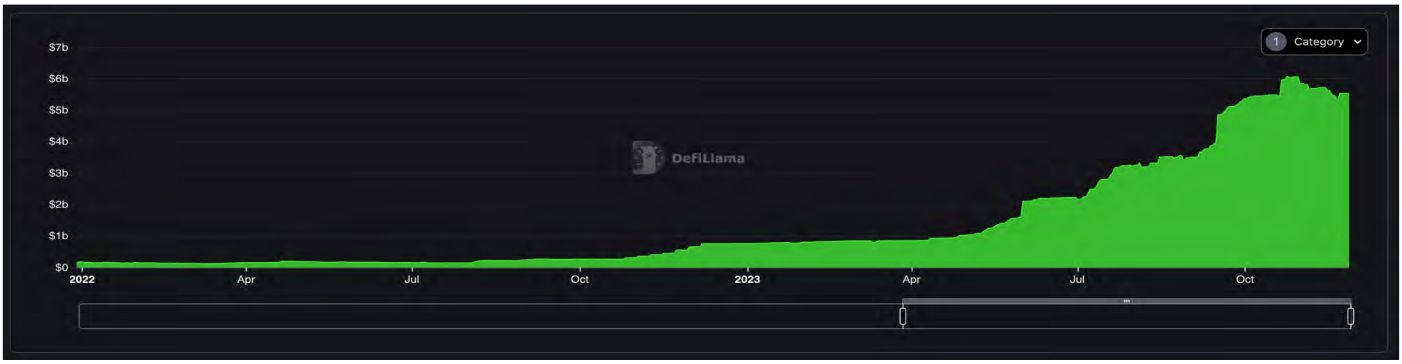


The traction over LST is not about to end with a new primitive entering the game called restaking, powered by EigenLayer. EigenLayer is creating a market of decentralised trust where projects can re-use the Ethereum network's security by asking users to allocate their restaked tokens to them.

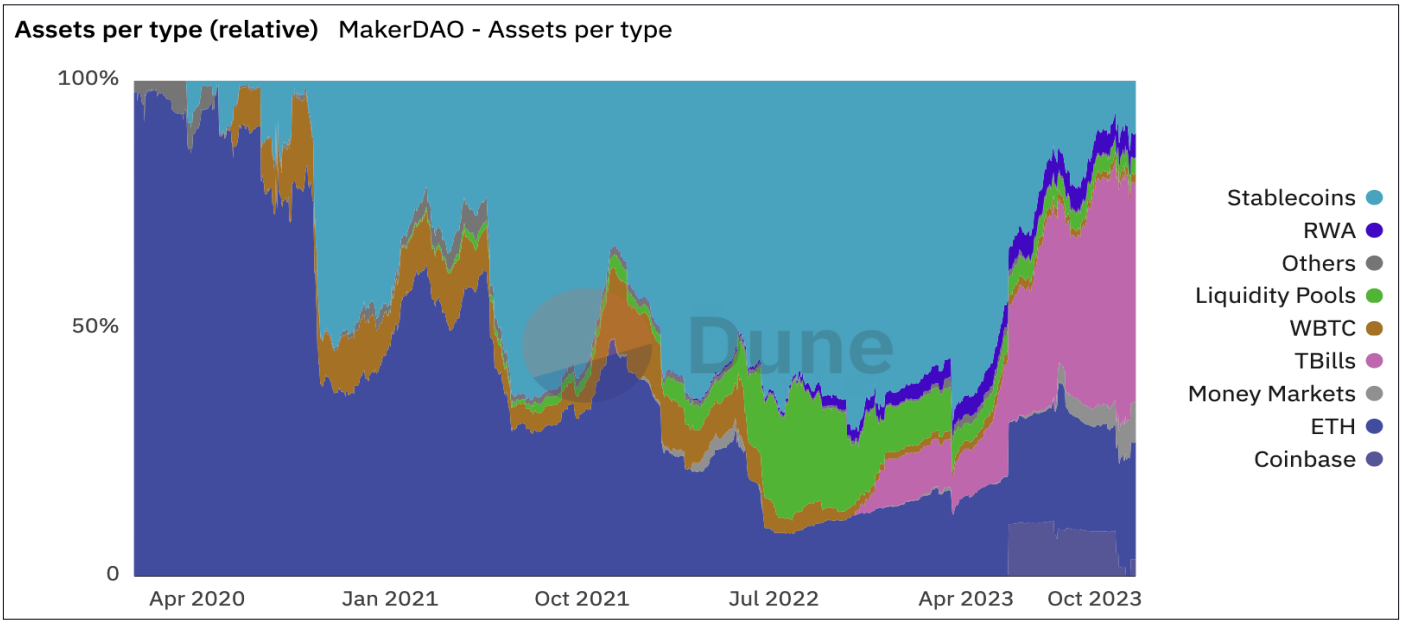
Real World Assets (RWA)

Real World Assets (RWA) in crypto refers to the tokenisation of tangible assets that exist in the physical world that are brought on-chain. They also include the growing issuance of capital market products on-chain, where digital securities are tokenised and offered to retail customers.

The plot below displays the TVL increase associated with the RWA protocol.
The two largest contributors to today's $5b TVL are MakerDao ($2.8b) and stUSDT ($2.3b).
MakerDao is the famous creator of the DAI stablecoin.
stUSDT is less well known and offers a unique blend of traditional finance and crypto by allowing users to stake USDT stablecoins for a yield. These tokens provide a way to diversify crypto portfolios by tapping into real-world assets like US government bonds. Because of this, stUSDT is facing scrutiny over its governance and unclear affiliations, notably with Tether and Justin Sun.
Most of the stUSDT TVL can be found in the Tron blockchain.



Last, we display the breakdown of MakerDAO assets per type. While in the past, most if its assets were held in the form of ETH first and stablecoins after, today, it's mostly T-bills. This again shows the turn toward RWA assets taken by MakerDAO (please note that the RWA asset in legend is misleading as it is related to loans, etc...).



Source: https://dune.com/sportreally/makerdao-dashboard

We see new protocols coming on this segment with the arrival of private credit on-chain.
Private credit has been there for a while in traditional finance. We think the DeFi infrastructure can help simplify the relationship between the lender and the borrowers.
We can see that Goldfinch and Maple are already active in the sector and driving close to 100M TVL each.

The question is also whether these products linked to TBills will be sustained in a low-yield environment.

DeFi Challenges

Centralization

There's a risk of misrepresentation known as "decentralisation theatre."
What's commonly labelled as DeFi can frequently exhibit significant centralisation. Often, DeFi protocols rely on centralised data feeds and can be steered or impacted by individuals holding "admin keys" or a heavily concentrated governance token allocation (offering voting rights). While partial centralisation might not be inherently negative, it's crucial to clearly distinguish between authentic decentralisation and entities that tout themselves as DeFi while essentially offering centralised infrastructure.
Other issues related to a more centralised DeFi are related to

- Governance Concentration: Concentrated token holdings or governance structures controlled by a few entities can lead to centralisation of decision-making within DeFi protocols.
- Regulatory Pressures: Regulatory concerns and compliance requirements might push some DeFi projects to adopt more centralised structures to navigate legal landscapes, deviating from the original decentralised ethos.
- User Experience Trade-offs: Some DeFi platforms prioritise user experience and scalability, often at the cost of decentralisation. Centralised solutions might be chosen to ensure smoother user interactions and faster transactions.

Some protocols like Morpho with Morpho Blue take an approach that goes in the motto of decentralisation where they focus on being trustless, efficient and permissionless (more information here).

Hack

A total of 57 crypto hacks have been identified in 2023 (at the time of writing, the year is not over yet) by rekt.com. Excluding HTX (Huobi, a CEX) and Midas Finance (likely a rug pull), all other 54 cases of hacks are related to crypt protocol. The total monetary loss amounts to around $4.5b (although, in some cases, the hacker has sent the funds back, e.g. in Euler's hack). This is nonetheless a huge amount of money lost in DeFi.

Whether related to smart contracts vulnerabilities or economic exploits, this space seems not to learn from its past mistakes.

Code vs reality

This brings us to the last DeFi topic, code vs reality.
"Code vs Reality" in DeFi refers to the conceptual design or intentions of a decentralised finance protocol (as reflected in its code or whitepaper) versus how it operates in practice, often highlighting discrepancies between the idealised vision and the actual implementation or outcomes.

To be more specific:
Code Refers to the technical specifications, smart contracts, and algorithms written to define how a DeFi protocol should function. It embodies the intended principles of decentralisation, transparency, automation, and trustlessness.
Reality: This encompasses how the DeFi protocol functions in practice, considering factors such as actual user behaviour, the impact of external influences (regulatory, market conditions), vulnerabilities, bugs, exploits, centralisation tendencies, or unintended consequences that may not align with the original design or objectives.

The "Code vs Reality" concept often sheds light on:

- Centralization Issues: The extent to which a protocol is genuinely decentralised or relies on centralised elements.
- Security and Bugs: Discrepancies between the intended security measures and actual vulnerabilities or exploits found in the code.
- User Experience: Variances between the expected user experience and the practical usability or challenges faced by users interacting with the protocol
- Compliance and Regulation: Differences between the intended regulatory compliance and the actual legal landscape a protocol operates within.

DeFi is not perfect. But DeFi is here to stay. It just needs to learn from its mistakes.

Outlook for 2024

There is still a long way to go before institutional investors adopt DeFi. Nonetheless, we expect some trends for 2024, which should at least aim to improve the challenges outlined in the previous section:

Maturation and Evolution: By 2024, DeFi protocols should mature, offering more sophisticated financial services beyond lending and decentralised exchanges. New use cases might emerge, integrating traditional finance with decentralised systems. Many existing protocols are also transitioning into not being a product but rather a foundational brick of this new financial infrastructure called DeFi; the play is not in being used by retail but built on top by others and capturing their value.
The vision of Fintech being DeFi-backed is probably something we will see in the distant future.

Regulatory Evolution: Regulatory frameworks around DeFi might become clearer or more defined. Compliance measures could shape how DeFi protocols operate, potentially leading to a convergence of decentralised and traditional financial systems.

Interoperability and Cross-Chain Solutions: Efforts to achieve interoperability between different blockchains and DeFi ecosystems will intensify. Cross-chain solutions could become more prevalent, allowing assets to move seamlessly between various blockchain networks; we already have solutions like LayerZero and Chainlink CCIP emerging, so it's only a matter of time before we don't hear about them while they power entire ecosystems.

Scalability Improvements: Solutions addressing scalability issues in DeFi, such as Layer 2 solutions or alternative consensus mechanisms, might become more widespread, reducing congestion and transaction costs. The Solana ecosystem could be very well positioned when it comes to improving DeFi scalability. App chains might emerge from this, and this is perhaps where we could see the value of true innovation at the blockchain level to power the best product possible.

Enhanced Risk Measurement: It is time for DeFi to learn from TradFi. 'Code is the law' has been proven to be an abstract idea that does not really work in practice. Risk limits, improved oversight, and mitigating actions would help detect malicious actors (e.g., hackers and exploiters) and increase a protocol's chances of survival.

Institutional Involvement: Increased participation from institutional investors and traditional financial institutions in DeFi could occur, driven by improved regulations, enhanced infrastructure, and greater market maturity.

We are confident that DeFi could be powering the backbone of finance in the next few years, but for now, there is still a long road ahead to achieve that.

8. DeSci: Can it be the future of science?

Florent Koenig Twitter: @tradetheflow_

Decentralised science (DeSci) has emerged as an interesting movement within the realm of science. DeSci attempts to revolutionise the traditional way of doing science by leveraging blockchain technology and the Web3 ethos of data ownership, community-centricity, and open-source methodologies.

However, to better understand the implications and potential opportunities of DeSci, it is important first to understand the current challenges faced by traditional science (TradSci).

Traditional Science: Overview and Obstacles

The Encyclopedia Britannica defines science as a system of knowledge involving unbiased observations and systematic experimentation to uncover general truths or fundamental laws.

The scientific method follows a sequence of steps to get as close as possible to those general truths. Mainly, it begins with forming a hypothesis and making a prediction. Then, the next step is to secure funding before conducting an experiment and drawing any conclusions. After that, scientists usually try to share their findings in different publications. Finally, other researchers conduct independent experiments to ensure integrity and robustness, engaging in peer review and replication to validate or challenge initial findings. This last step is essential to ensure the quality of the scientific method.

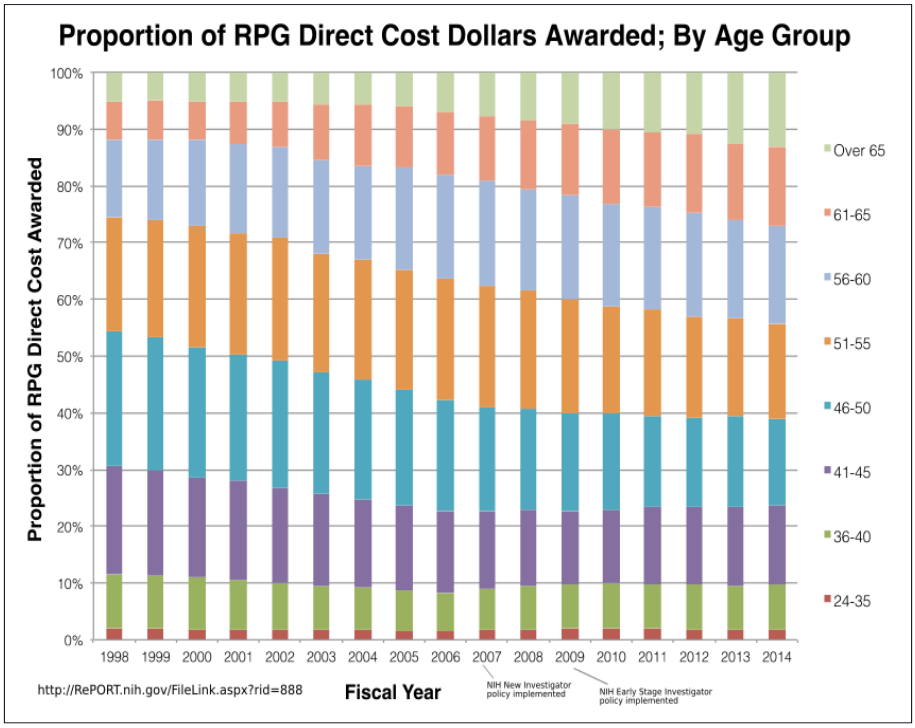
Only once this process is complete can newly acquired knowledge extend beyond theoretical realms and find practical application. Sometimes, derivatives of this knowledge yield practical improvements in our lives.

However, there are many problems and drawbacks in how science is currently conducted:

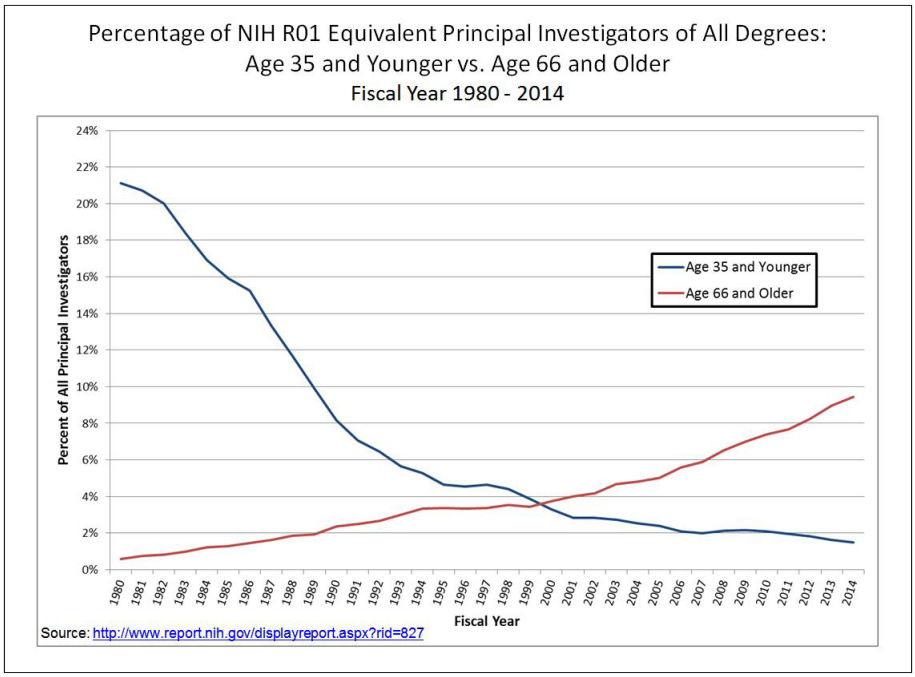
Unequal Access to Funding

Science is made possible by distributing funds to people who propose to complete a specific study. However, gaining access to such funding takes time and effort for scientists. First, access to grants is unequal. In 2017, the National Institute of Health (NIH) gave 11% of its total budget to 1% of grantees.

Another concern is that early-career scientists are usually at a disadvantage. We observe that individuals between 24 and 35 get access to only 2% of the total grant.



This trend is not getting better as the percentage of funding going to old scientists keeps increasing while the share of funding to younger ones decreases. This is arguably one of the biggest examples of institutional conservatism.



Furthermore, the funding process is slow, bureaucratic, and quite opaque. It is estimated that about half of scientists' time is spent writing grant proposals, diverting time away from doing true science. Even when the proposal is complete, the selection process is subjective, and only 17% of grants are approved in the US.

This incentivizes predictable, safe, and consensual research. Ultimately, this weakens the long term credibility of the scientific method which aims to challenge the current status quo.

Intellectual Property Ownership Issues

Intellectual property (IP) is a category of ownership that includes the intangible creation of human knowledge. Even though this factor incentivizes creativity and innovation, it faces numerous obstacles in traditional science.

First, the registration of an IP is an archaic process that necessitates a good understanding of the law. As the legal framework of IP varies widely between countries, this can create uncertainty for creators and owners of intellectual property assets.

Additionally, IP can also be hard to value and managing IP assets can be a challenge, especially for smaller organisations that don't have the proper resources to enforce their rights.

Last but not least, it can result in an unfair distribution of resources as a small number of scientists will get all the credit while the broader scientific community, who contributed nonetheless to the research progress, receives little or no benefits.

Lack of Collaboration

Science is a global public good; yet, much of it is trapped behind private and expensive databases. Despite the fact that the biggest contributors to science are often the government, and as a corollary, taxpayers like us, access to journals is expensive for both scientists wishing to publish, and individuals who want to get access to that knowledge.

To illustrate this, a single article in Science costs $30; a year-long subscription to Cell will cost $279; Elsevier publishes 2,000 journals that can cost up to $10,000 or $20,000 a year for a subscription.

This has significant implications as it discriminates science based on which scientists have the biggest budget, and hence can afford to be published. But it also creates unequal access to quality knowledge. This is unfair and is indicative of the lack of collaboration in traditional science.

Reproducibility Crisis

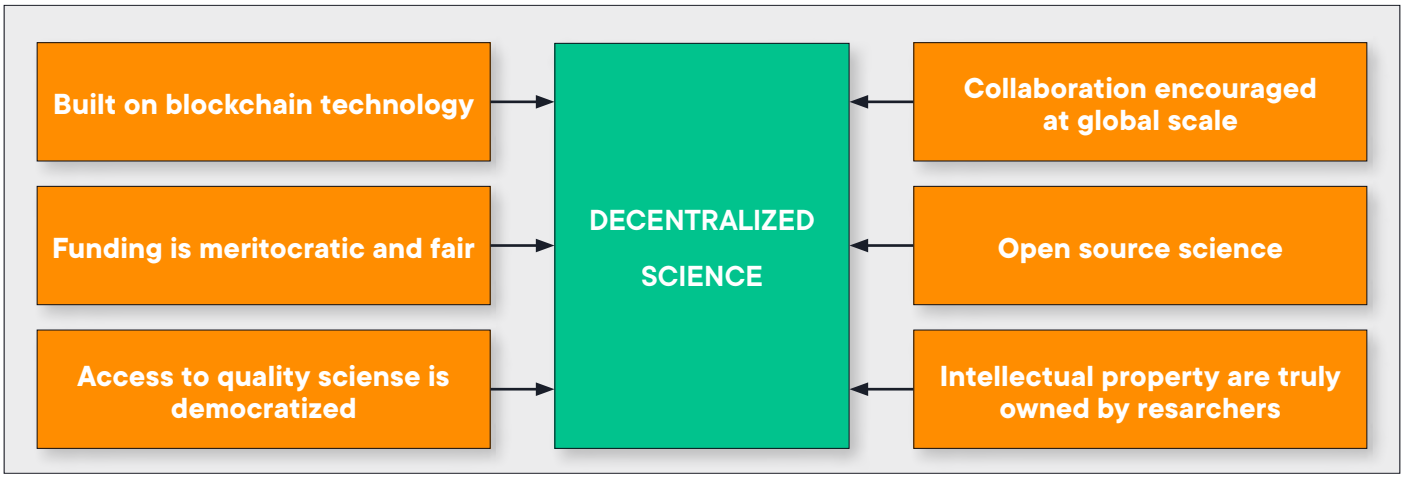
Replication is a foundational concept of quality science. It is the process by which researchers take an older study that they want to test and then try to reproduce it to see if the findings hold up.

However the incentives for that aren't well aligned. Funding initiatives prefer to support researchers that find new information instead of confirming old results. Further, even when they wish to replicate the study, they often can't do it because the methods are too opaque, the studies have too few participants and are badly designed, or access to proper data is too expensive.

The Case for DeSci

DeSci is, in a lot of ways, similar to what the DeFi (Decentralised Finance) movement was back in 2020: early and nascent, but with a lot of promises to leverage blockchain technology to challenge the status quo.

While the first blockchain-based science initiative dates back to 2015, it didn't lead to a true movement until 2021. Currently DeSci is more of a concept than a reality, and its application ranges from pure theoretical ideas to small scale enterprises and different experiments with established players. Nonetheless DeSci is offers a great alternative to the broken traditional science model.



Better Funding

DeSci can disrupt the current funding model with different incentive models developed by DAOs and Web3 more broadly. Different future possibilities include:

- Retroactive and Quadratic funding,
- Projects selling their public good tokens
- DAO Governance,
- Leveraging different DeFi tools

As mentioned above, all of this is still in a conceptual phase within the DeSci landscape.
Nonetheless, this represents an exciting alternative to incentivise more equal access to funding.

More Transparent and Aligned Ownership

Ownership of digital assets (such as scientific data or articles) is something Web3 excels at.
In the DeSci movement, various aspects of science, from peer review intellectual property and reputation systems, can be governed by separate decentralised communities, and scientific knowledge can be tokenised through non-fungible tokens (NFTs). In this paradigm, those communities become the new owners of scientific knowledge.
Furthermore, NFT could help better access scientific reputation and hence the ability to secure funding. For instance, researchers could earn NFTs by performing different tasks such as peer reviewing, mentoring, training, or sharing data openly. Then, those NFTs could be used as a "proof of reputation" for funding, which goes beyond just the number of articles published and the seniority level of the researcher.

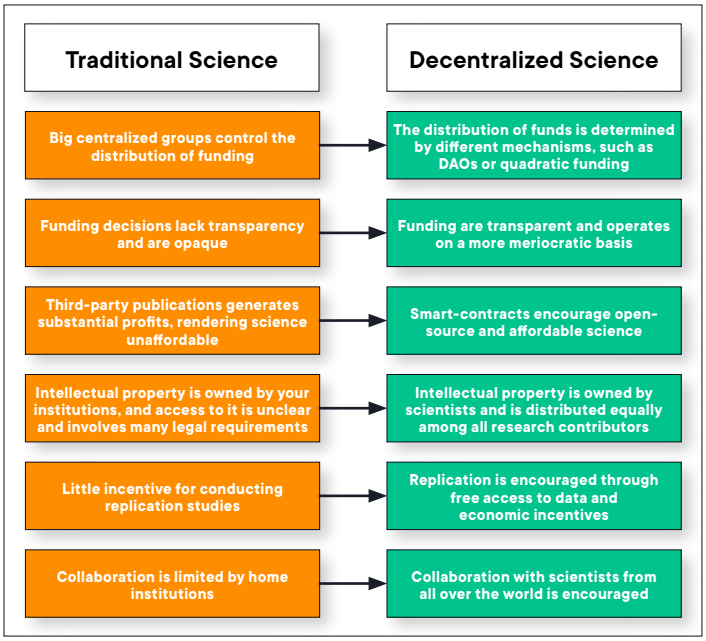
Towards Open Source Science

In an ideal world, the starting point for science should be free access to data. DeSci can make that happen by creating a system where scientists holding the proper credentials can get access to any form of data stored in a decentralised storage space. Different NFT and DAO initiatives can also be used to make collaboration at the heart of science and encourage communities to work together for the common good of science.
Moreover, DeSci can make science more accessible. Today, the total fees to publish an individual paper are often five figures, and the cost for readers is also non-negligible. All of this goes directly to publishers instead of scientists doing most of the work. Smart contracts can drastically reduce the need for big third-party publishers who extract the majority of profits, and they are one of the reasons why access to science is so expensive.

Incentive for Replication

Web3 tools also encourage scientists to participate in replication studies, as the latter can be economically advantageous while strengthening the integrity of the scientific method. As said previously, data can be free if one has enough scientist credentials (through different NFTs), and by doing replication studies, one can earn attestation of the work to help secure more funding later, but also to get a share of what publications earn in the form of tokens or NFTs.

This movement has the potential to radically alter how a generation of scientists is incentivised to tackle many of the great problems of our world. At the same time, it can make science publications and associated data accessible while enabling a perpetual community review process that reinforces the quality of the work.



Current Challenges

Technological & Design

For now, DeSci is far from being ready to challenge conventional methods of doing science. It is still nascent and currently in the "experimental" phase. The different mechanisms in place to better align incentives within the scientific community are promising but haven't found product-market fit.

Further, interacting with other blockchains is not a seamless process yet. Understanding how to interact with different blockchains is time-consuming, and many scientists are not interested in that. Therefore, efforts still need to be made on that front for true breakthroughs in DeSci to happen.

Last but not least, DeSci lacks a proper legal framework. There remains an ongoing risk that regulators might quickly change laws, which could impact DeSci projects. As scientific research is often sensitive and can have an important and lasting impact on the direction of society, clearer regulation around DeSci is essential for it to challenge the current way of doing science.

Scientific Quality

DeSci communities are still narrow and do not yet represent the global scientific community. Today, it is much more common to see crypto people going into science rather than the opposite. This means there is an ongoing debate around the quality and the reach of the science produced.

Further, renowned scientists who benefit from the conventional system are not incentivised to make the shift to DeSci as they have more to gain than to lose. This slows progress and can influence the general public's view, who will likely listen to the voices of the most famous scientists.

Also, DAO governance is often unclear, and how DAOs will collaborate to address the unknown and unexpected issues arising as the movement gains traction remains to be seen.

Conclusion

DeSci has emerged as a promising shift from traditional science. By leveraging blockchain technology and Web3 principles, it can solve different problems such as funding inequality, issues around intellectual property or a lack of collaboration within the scientific community. The bull case for DeSci is that it has become the new GitHub of science.

Yet, DeSci remains an evolving experiment. The ecosystem is nascent and has yet to reach its inflexion point. Challenges persist, from technological complexities, limited adoption, credibility issues, and legal uncertainties.

Despite the obstacles, DeSci holds significant promise. It will likely transition toward greater mainstream adoption in the coming years with real and new use cases for improving science.

Some Projects to Follow

Here is a list of potential interesting projects and initiatives to get up to speed within this emerging narrative:

ResearchCoin: https://www.researchhub.com/
Molecule: https://www.molecule.xyz/
VitaDao: https://www.vitadao.com/
GenomeDAO: https://genomes.io/
Bio.xyz: https://www.bio.xyz/launchpad
Rejuve.AI: https://www.rejuve.ai/
ValleyDAO: https://www.valleydao.bio/grow

Sources:

- https://ethereum.org/en/desci/
- https://balajis.com/p/the-purpose-of-technology
- https://newscience.org/nih/#fn:36
- https://medium.com/paradigm-research/decentralized-science-desci-web3-mediated-future-of-science-2547f9a88c40
- https://www.molecule.xyz/blog/exploring-decentralized-science-with-balaji-srinivasan-on-the-desci-podcast
- https://a16zcrypto.com/posts/article/what-is-decentralized-science-aka-desci/
- https://link.springer.com/chapter/10.1007/978-3-319-51277-8_4
- https://www.gitcoin.co/blog/desci-the-case-for-decentralised-science
- https://www.vox.com/2016/7/14/12016710/science-challeges-research-funding-peer-review-process#1

9. Unlocking the Potential of Restaking: A Deep Dive into EigenLayer

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What is restaking?

In the ever-evolving landscape of cryptocurrency, the concept of staking has emerged as a powerful mechanism for securing networks. Staking involves participants, known as validators, playing a crucial role in storing data, processing transactions, and contributing to the addition of new blocks to a blockchain. One platform that is revolutionising the staking game is EigenLayer, offering users the opportunity not only to stake their assets but also to restake them, creating a ripple effect of benefits across various networks.

Staking on Ethereum grants users the chance to become validators, requiring a stake of 32 ETH or the deposit of ETH into Liquid Staking Tokens (LSTs) like stETH, rETH, frxETH, and more. The number of ETH staked directly correlates with the security of the network, making it harder for any single entity to control the majority of validators.

EigenLayer takes staking to the next level by introducing the concept of restaking. This innovative feature allows users to continuously secure the Ethereum network while also contributing to other protocols. EigenLayer addresses the challenge of the high costs associated with bootstrapping new networks, providing a solution that allows projects to tap into the vast pool of Ethereum stakers.

What are the benefits of restaking?

Restaking, facilitated by EigenLayer, brings forth several advantages, including

- **enhanced security for new networks:** EigenLayer allows protocols to leverage Ethereum's security layer by providing incentives for ETH stakers to stake on their protocol. This strategy enables the protocol to access a broader range of validators, significantly enhancing its initial security measures.
- **improved tokenomics:** Protocols will be able to retain sovereignty and have the freedom to customise their architecture, including the type of consensus mechanism, slashing conditions etc...
- **increased ETH yields:** This is the result of the increased capital efficiency brought in by the restaking since stakers can now earn rewards from multiple protocols with the same capital.

With EigenLayer, tokens no longer need separate incentives for security, streamlining the overall efficiency of the ecosystem.

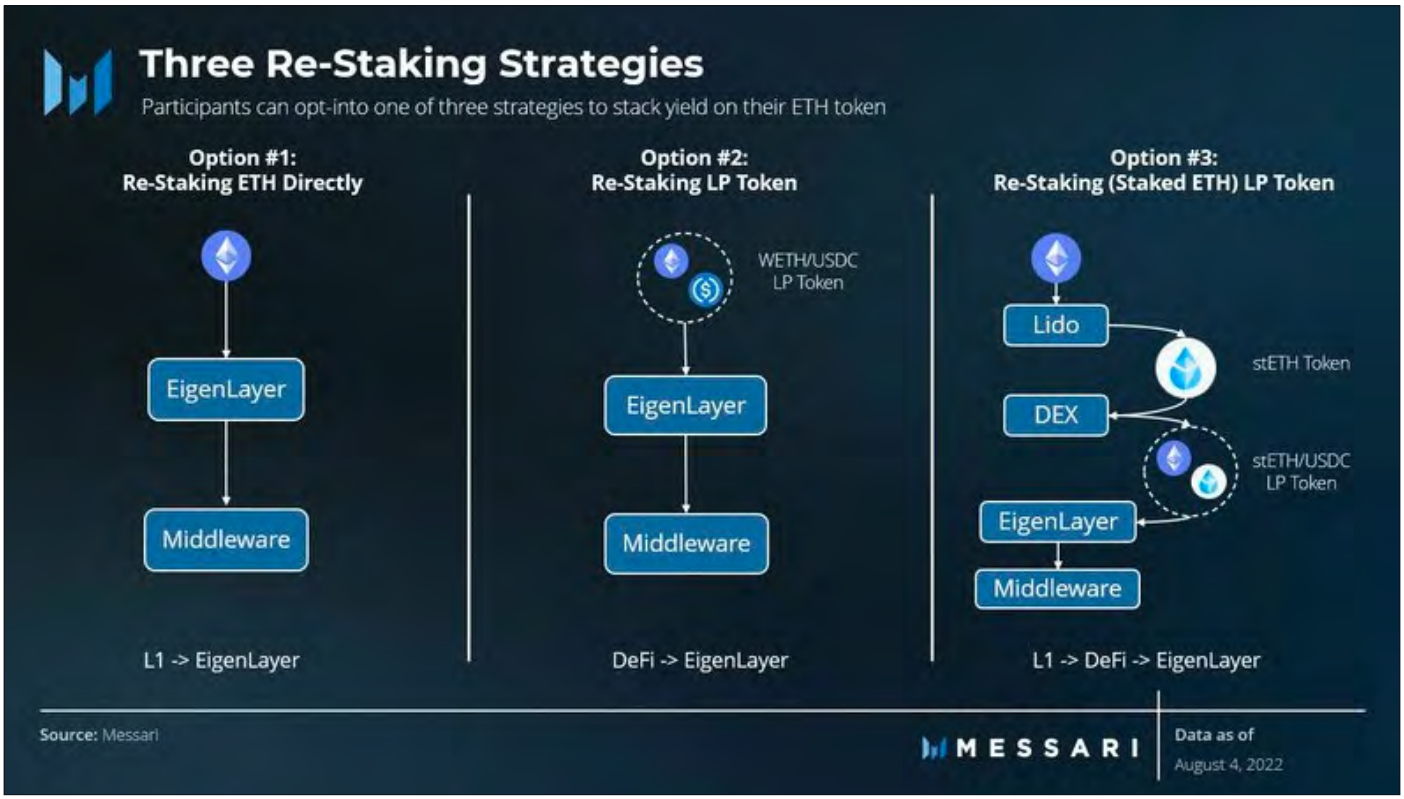
How restaking works

EigenLayer operates through nodes run by operators, where users can delegate their funds for restaking. These operators validate Actively Validated Services (AVS), which encompass systems requiring distributed validation, such as side-chains or Layer 2 solutions. EigenLayer's inaugural service, EigenDA, serves as a data availability layer, enabling rollups to benefit from increased throughput, reduced costs, and the security provided by restakers.

There are multiple restaking methods offered by EigenLayer (risk level from least to most):

1. Native Restaking: Validators restake their staked ETH
2. ETH LP Restaking: Validators restake the LP token of a pair which includes ETH.
3. ETH LSD:
    a. LSD Restaking: Validators restake assets that are already staked via liquid staking providers, including Lido, Rocket Pool etc.
    b. LSD LP Restaking: Validators restake the LP token of a pair which includes a liquid staking ETH token

The diagram below shows how the various options work in practice:



EigenLayer sets the stage for future projects, such as Rollups-as-a-Service, leveraging restaking to fortify their infrastructure. Notable migrations, like Celo[1]'s shift from Layer 1 to Layer 2, choose EigenDA for its multifaceted benefits. Projects like Espresso[2] are also exploring the potential of EigenLayer to secure decentralised sequencers for rollups, showcasing the platform's versatility and relevance in the evolving crypto landscape.

[1] https://celo.org
[2] https://www.espressosys.com/

The risks:

Slashing

The main risk of restaking is related to slashing for the staked (e.g. Lido) and restaked tokens.
During the restaking process, ETH tokens are subject to slashing conditions imposed by the other protocol, as well as Ethereum. Suppose a validator participating in the restaking process **loses their staked ETH tokens due to the slashing conditions of another protocol. In that case, they will still lose their staked ETH tokens** even if they have followed the Ethereum network's rules. Currently, suppose a validator on the Ethereum network engages in malicious behaviour. In that case, they may lose up to half of their staked 32 ETH tokens, up to a maximum of 16 ETH through slashing. EigenLayer allows slashing the remaining half.
Participating in restaking doubles the risk, but it also increases the total reward by receiving not only Ethereum block rewards but also rewards from other protocols.

Centralization

Restaking introduces greater centralisation risks as ETH stakers redirect their withdrawal credentials to EigenLayer, which creates a systemic risk for the Ethereum network.

Security

Restaking raises concerns regarding security incidents. If a significant portion of the ETH staked in the Ethereum network is employed for restaking in other protocols, a security breach in one of those protocols could potentially result in a significant amount of ETH being slashed. This could then cause a decrease in the security of the Ethereum network. To prevent this, a conservative approach should be taken when investigating and selecting protocols for restaking. Moreover, the conditions for slashing should not be overly severe. Preventive measures could include insurance pools or the implementation of delays before slashing and going through governance processes.

Conclusions:

In conclusion, EigenLayer emerges as a transformative force in the world of staking, offering users a secure avenue for token staking and a strategic mechanism for optimising and diversifying their crypto portfolios.

Sources:

- https://ethereum.org/en/staking/
- https://eigenlayer.xyz
- https://docs.eigenlayer.xyz/
- https://www.blog.eigenlayer.xyz/twelve-early-projects-building-on-eigenlayer/

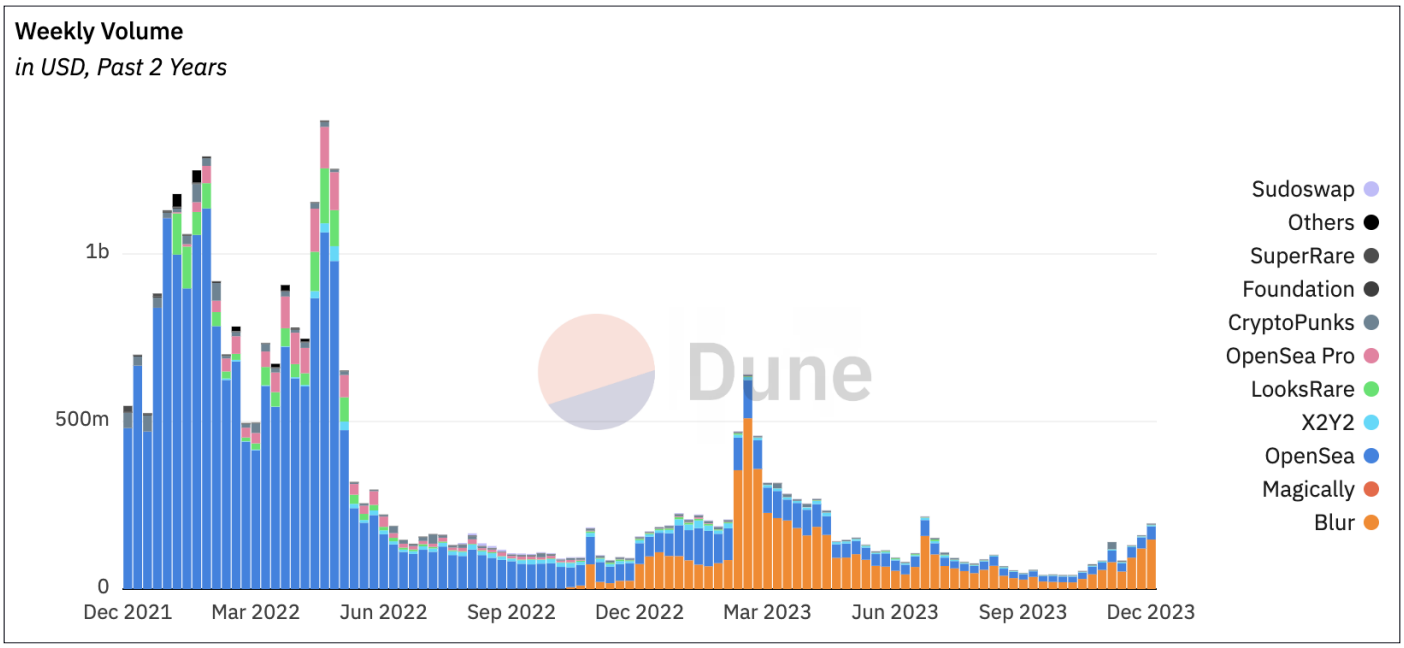
10. The NFT Market

Matteo Bonato Twitter: @mattiboySB

What happened in 2023?

Let's be honest. No, 2023 has not been the best year for the NFT market. The crypto winter of late 2021 has dragged itself well into 2023. Only now can one see signs of relief.
The NFT market was not immune to this risk-off mood.

The plots below show the weekly volumes of the NFT trading activity. Except for an early 2023 blimp (e.g., when Bitcoin doubled in price between early January and April), the trend goes in only one direction: down.



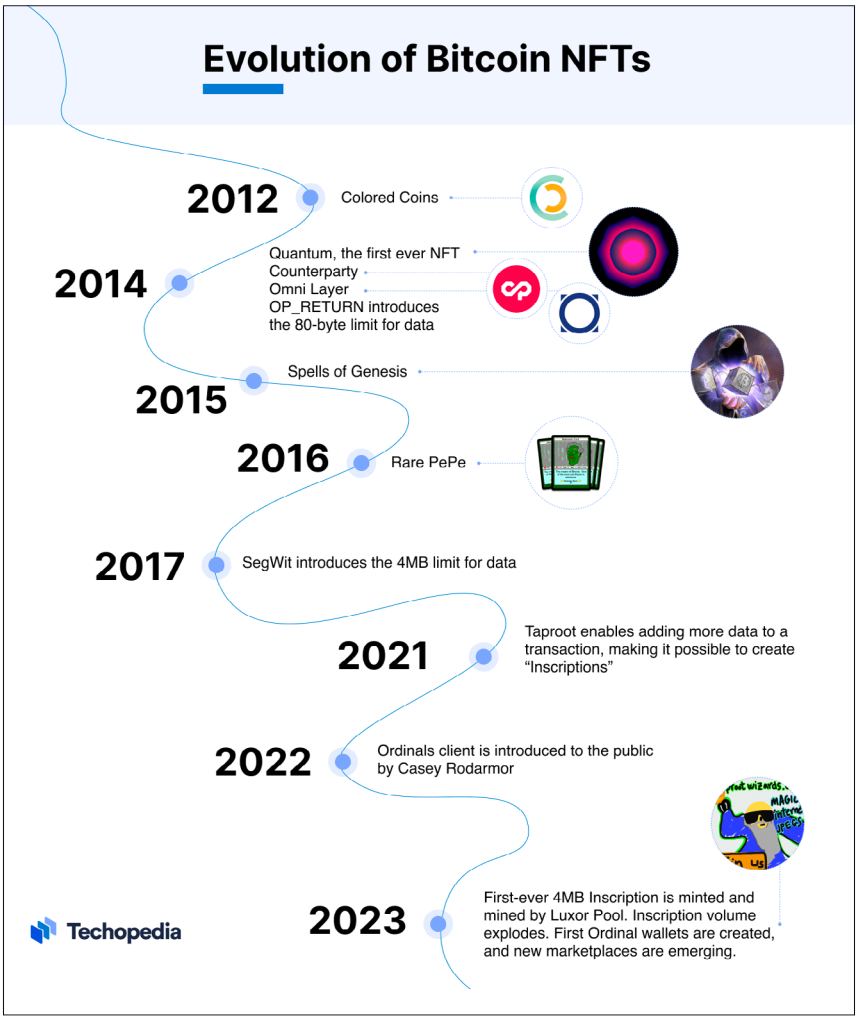
Source: https://dune.com/hildobby/NFTs

There is light at the end of the tunnel, though. Since October, the NFT market has been gaining traction again.
Indeed, NFTs have surged back into the spotlight with a remarkable 32% spike in sales this October 2023. This revitalisation heralds a new chapter for the digital art market, backed by a robust $40 billion valuation that continues to escalate.

Bitcoin Ordinals: the latest crypto play

Tokens based on the Bitcoin blockchain are emerging as the newest play among market participants after weeks of a bitcoin (BTC) price frenzy driven by expectations of an exchange-traded fund approval.

Bitcoin Ordinals is a protocol that allows individual satoshis (SATs) in a Bitcoin blockchain to be assigned a unique identifier and transacted with extra data attached.
The protocol, credited to Casey Rodarmor, effectively allows non-fungible tokens (NFTs) to be created directly on the Bitcoin blockchain.
Bitcoin NFTs may also be referred to as Ordinal NFTs.



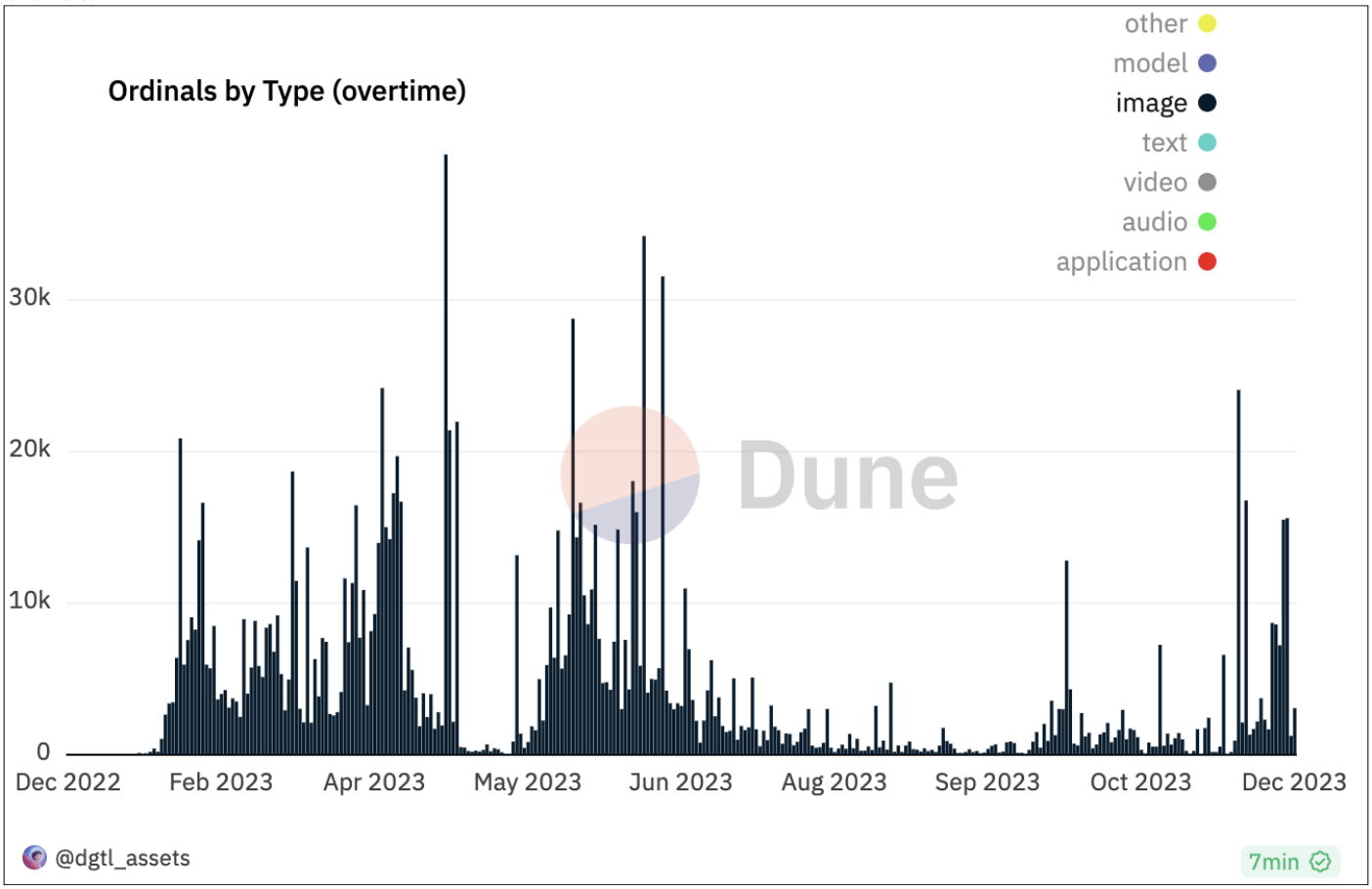
The rapid evolution of BTC ordinals (image) NFT can be seen in the plot below.
This trend emerged in early 2023 and is picking up again.

The Ordinals Protocol allows users to embed data into the Bitcoin Blockchain. This includes

- Ordinals, a numbering mechanism for Sats (the smallest Bitcoin denomination)
- Inscriptions, blobs of arbitrary data and associated metadata, the latter of which tells a Bitcoin node how to render said data (is it an image? Text? Something else?).

This is essentially creating Bitcoin-native Digital Artefacts. Note that not only images but also text, video, etc., can be created via

Ordinals.



Some of the currently popular Bitcoin Ordinals NFT projects are based on their Ethereum counterparts. For example, Ordinal Punks and Bitcoin Punks are based on CryptoPunks.
Top NFT brands are also participating in Ordinals. Yuga Labs, the company behind the top Ethereum-based NFT collection Bored Ape Yacht Club, recently launched its first Bitcoin Ordinals NFT collection called TwelveFold. It consists of 300 unique generative art pieces inscribed onto satoshis on the Bitcoin blockchain.

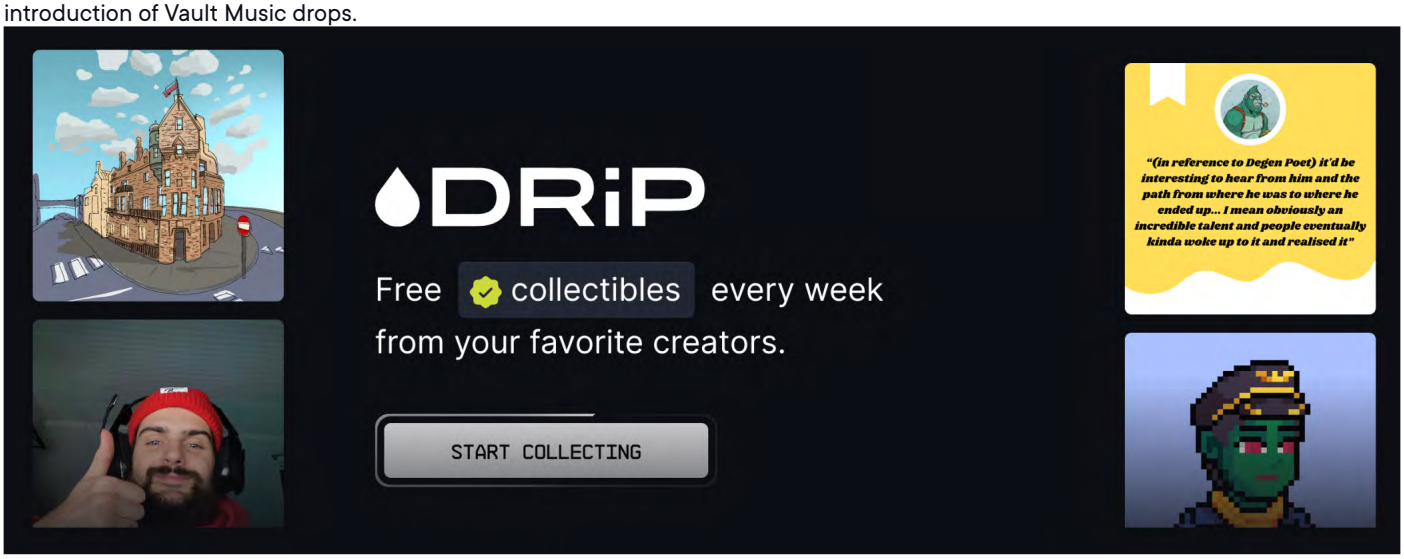
Bitcoin Ordinals have ignited significant debate among the crypto community. Critics express concerns about block space utilisation by inscribed satoshi Ordinals, predicting potential rises in Bitcoin transaction fees and slower transaction speeds for network users. On the other hand, supporters of Bitcoin Ordinals are exploring avenues to leverage the protocol for broadening Bitcoin's applications beyond just financial transactions.

Exploring Drip Haus: Delving into the Future of NFTs through Inclusive Innovation

DRiP Haus is an NFT release platform operating within the Solana ecosystem. It acts as a central space where users can obtain digital collectables from their preferred creators without charges. As of October 2023, DRiP Haus has minted over 36 million collectables, seen 1.2 million sales in secondary markets, and has cultivated a community of 500,000 collectors.
DRiP Haus operates in a simple manner. Users can explore the latest releases on the platform, ranging from legendary to ultimate collectables.

Beyond offering collecting opportunities, the platform highlights featured creators such as Bork The Viking Pug, Degen Poet, and Vault Music. By supporting these creators, collectors can curate their distinctive collections and become part of a thriving community. Moreover, DRiP's creative endeavours encompass not only visual art but also extend into the music realm with the

introduction of Vault Music drops.



Last but not least, DRiP Haus has been pioneering the concept of NFT compression.
NFT compression involves reducing the size of Non-Fungible Token (NFT) files while preserving their essential features and visual integrity.
This process utilises data compression techniques employing algorithms designed for efficient data encoding. The primary objective is to improve blockchain network storage, transmission, and processing capabilities. This optimisation plays a crucial role in blockchain platforms where efficiency and scalability are vital.

DRiP Haus offers high-quality digital assets that consume fewer resources through this approach. This strategy ensures that the digital assets maintain quality while enhancing scalability and cost-effectiveness on the blockchain.
Essentially, DRiPHaus's technology enables the same immersive digital experience but with a reduced impact on the blockchain, thus making it more accessible to a wider audience.

Royalties are back

The multi-chain NFT marketplace Magic Eden revealed in early November its plans to introduce a new Ethereum platform by year-end, partnering with Yuga Labs, the creator of Bored Ape Yacht Club. Their collaboration comes with a steadfast pledge to uphold creator royalties on all NFT sales.

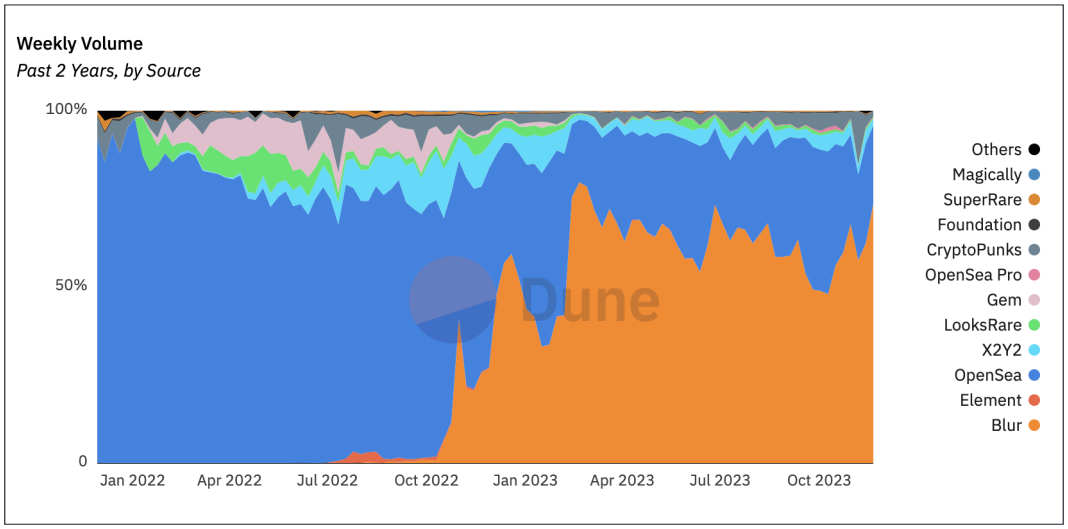
A creator royalty refers to a nominal charge designated by the artist or organisation linked to a project, deducted from the subsequent sale price of an NFT. Typically ranging from 2.5% to 10% of the sale amount, these fees were initially widely respected by platforms as a means for creators to gain long-term benefits from their tokenised creations.
However, during the latter half of 2022, as NFT sales waned, some emerging marketplaces found ways to bypass these creator fees or reduce their payments. Subsequently, leading platforms followed suit to stay competitive, and the prominent marketplace OpenSea eventually opted to cease enforcing royalties in August, despite its prior commitment to honouring such fees.

The revamped Ethereum marketplace by Magic Eden will implement new smart contracts, housing the code fueling decentralised apps (dapps). Through technical advancements, the platform asserts that these contracts will guarantee the disbursement of royalties during secondary market sales.

Ethereum and Polygon: Driving Forces Behind a Flourishing Market

The resurgence of NFTs owes much to Ethereum, Solana, and Polygon. Their accessible and economical platforms have played a pivotal role in fostering the NFT boom.
Ethereum's recent upgrades have elevated its appeal among artists and investors, while Polygon's scalable solutions present an

efficient alternative for enthusiasts.
Finally, the resurgence of Solana unquestionably exerts a positive influence on the NFT space, exemplified clearly by the MadLads explosion (detailed in the following section).
The landscape of NFT market participants is evolving.
The graphic below delineates the market shares among various NFT trading platforms.
The traditional dominance of OpenSea in NFT trading is now under scrutiny. In 2023, Blur emerged as the preferred destination for NFT trading.

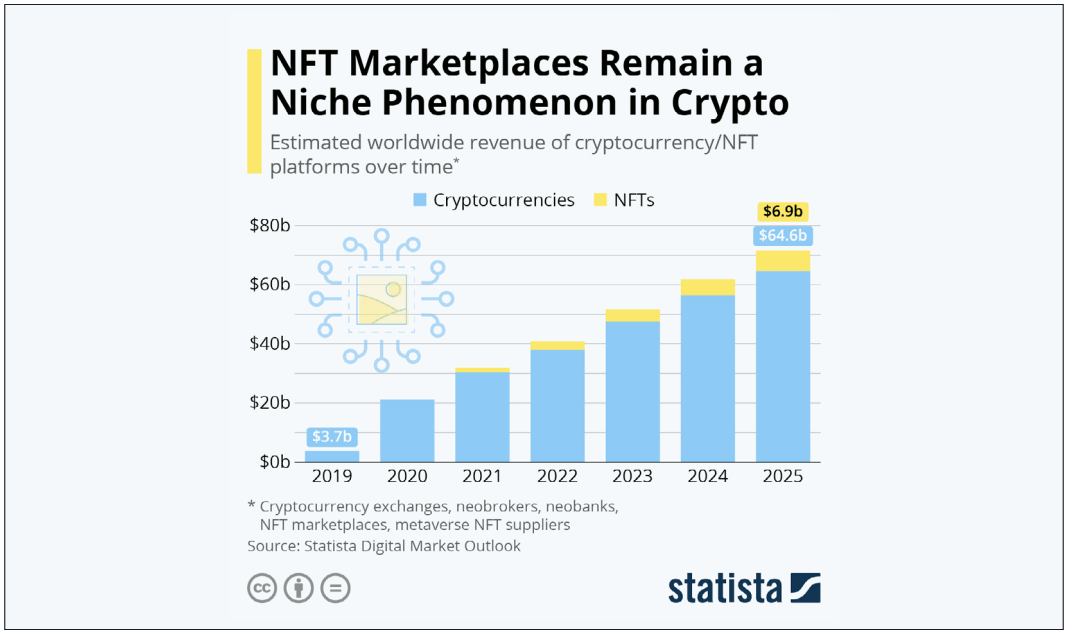


Source: https://dune.com/hildobby/NFTs

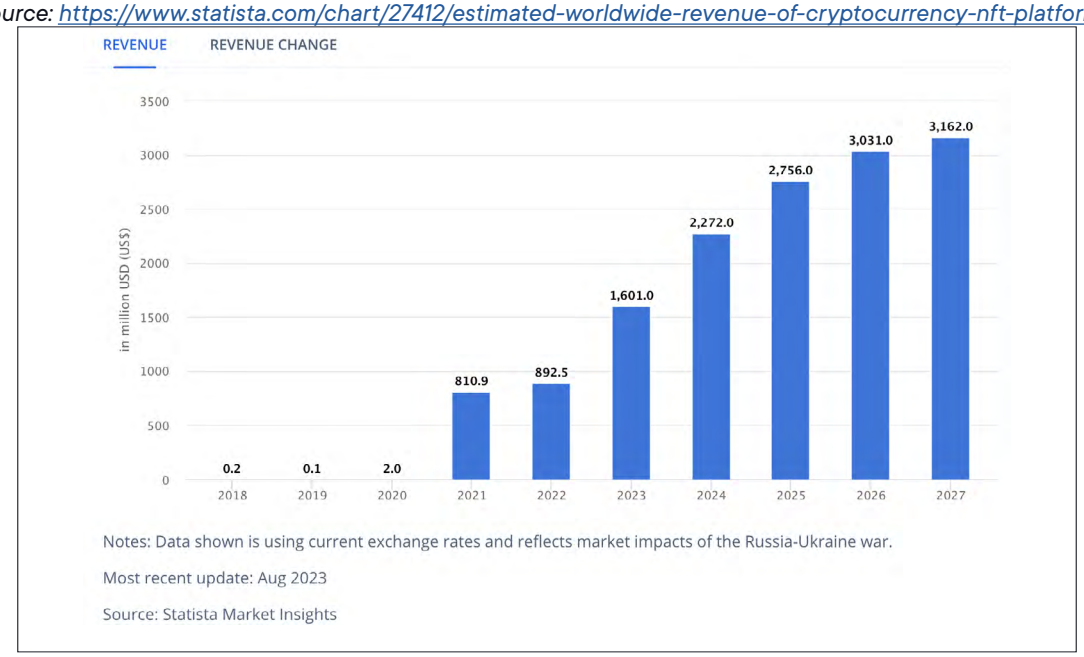
Outlook for 2024

Despite the optimistic prospects, the NFT industry encounters hurdles, notably concerning environmental sustainability and the balance of costs versus profits for creators. Nevertheless, the community's proactive initiatives and the emergence of platforms like BNice.io or Magic Eden's new platform indicate a promising future for NFTs.

The chart below illustrates the projected revenues for NFT platforms compared to crypto protocols over time. While NFTs currently occupy a niche within the crypto space, their revenues are anticipated to significantly increase, reaching an estimated $7 billion by 2025.



Source: https://www.statista.com/chart/27412/estimated-worldwide-revenue-of-cryptocurrency-nft-platforms/



https://www.statista.com/outlook/fmo/digital-assets/nft/worldwide#revenue

Mad Lads take over DeGods?

It is now official: Mad Lads have taken over DeGods. At least, this is what the current floor prices tell us.

How we got there

It was Christmas day 2023 when the decision was announced to the public: DeLabs founder Rohun "Frank" Vora revealed his latest "social experiment," announcing that DeGods NFTs would be "bridged" over to Ethereum—the largest and most valuable NFT market, with projects like the Bored Ape Yacht Club and CryptoPunks.
When DeGods launched in late 2021, Solana was experiencing a near-all-time high price, and its NFT ecosystem rapidly expanded. However, the following summer, despite becoming Solana's most valuable large-scale NFT project, SOL had significantly decreased in value, coinciding with a broader decline in the NFT market.
By late December, shortly after announcing a potential move, SOL had plummeted by approximately 97% from its peak price. Frank had previously hinted at a possible migration during Solana's struggles later in the year, which led to substantial backlash from the community. Nevertheless, rumours persisted as DeLabs continued outlining its plans.
Ultimately, they did proceed with the move to Ethereum.

Mad Lads

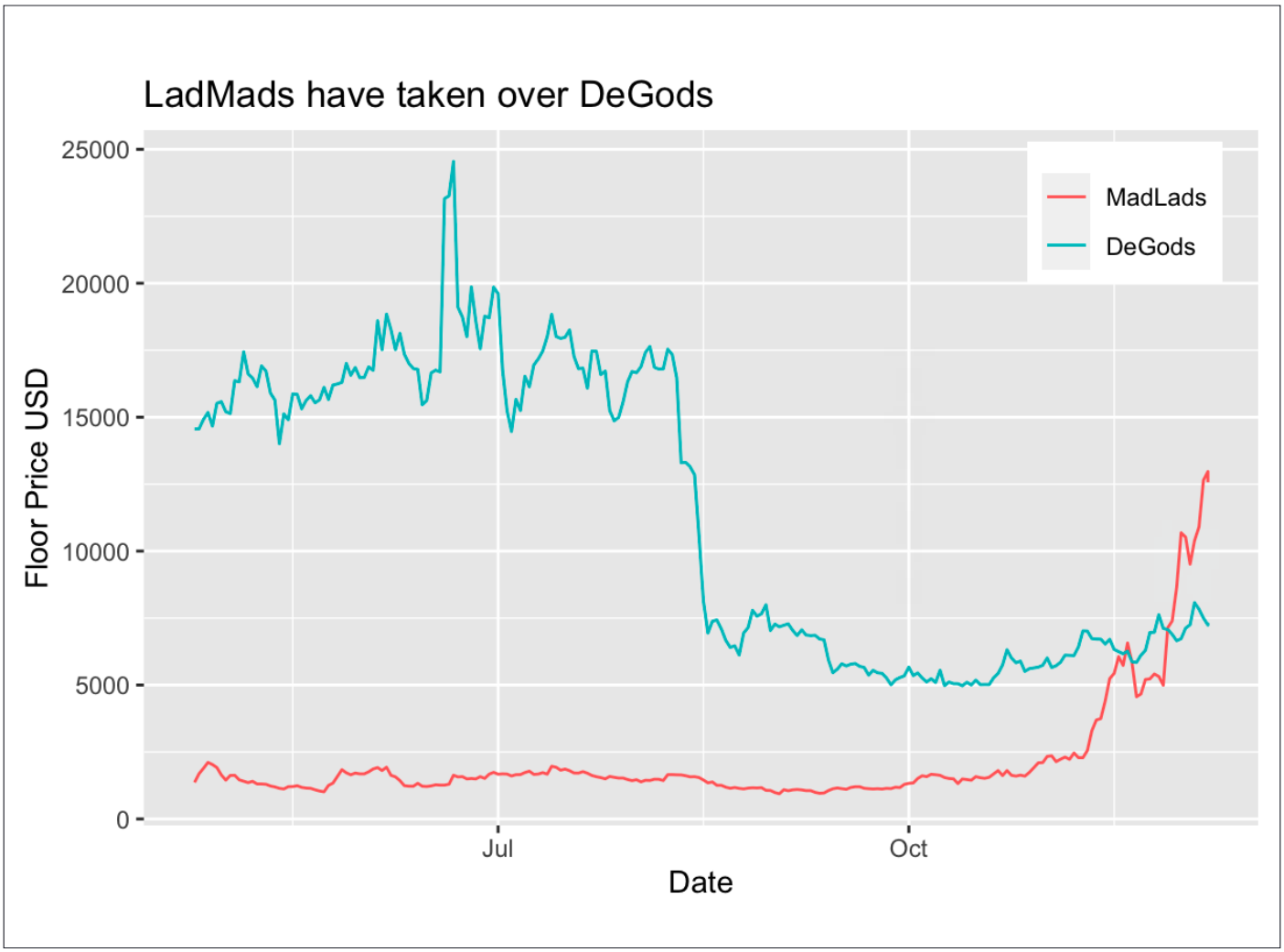
At the time, this decision appeared astute; however, the resurgence of Solana in late 2023 has propelled a new luminary in the realm of NFTs: Mad Lads.

The NFT project originated from the Solana framework development company Coral, led by Armani Ferrante and Tristan Yver. When Mad Lads debuted, it did so exclusively through Backpack, an all-in-one wallet app developed by Coral. Although having exclusive access via a wallet might seem unconventional for a new collection aiming to establish itself, in hindsight, it appears to have been an inventive approach for Backpack to establish its presence on Solana.

This strategy aligns with Backpack's focus on "executable NFTs" (xNFTs), often touted as the next evolution in NFT technology. xNFTs are essentially non-fungible tokens integrated with executable programming scripts, effectively turning xNFTs themselves into platforms for functional Web3 applications.

Following the manoeuvre executed by Backpack, news swiftly circulated, sparking an upsurge in secondary sales. This surge caused floor prices to soar, igniting fervent participation across the NFT community and generating an astounding $8+ million within a mere 24-hour period. Notably, Anatoly Yakovenko and Raj Gokal, Co-Founders of the Solana blockchain, also joined as collectors, showing their support for the emerging project by updating their Twitter profile pictures.

That marked just the outset. During the 2023 Solana Breakpoint event held in Amsterdam in early November, Armani Ferrante revealed the introduction of a new decentralised exchange (DEX) called 'X'. Additionally, holders of Mad Lads were slated to receive an allocation from Pyth's airdrop, a Solana oracle service. Concurrently, the value of SOL surged more than threefold between October and December 2023.
The USD floor price of Mad Lads, initially around $2,000, gained substantial traction and, as of the present moment, is nearing $10,000. Notably, as depicted in the chart below, Mad Lads has surpassed DeGod in its impact and market standing.



Data Source: https://www.coingecko.com/en/nft

11. The fallout of CEXes and SwissBorg MEX revolution

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In CEXs' hands

When on April 1 2002, SwissAir ceased its operations and declared bankruptcy, people continued to take planes. It was a corporate failure. Sure, equity and bondholders lost money. But this event did not have much to do with the safety of a flight.

Crypto is different.

When FTX spectacularly crashed, dragging retail investors, investment funds, and anyone with their crypto deposits on the exchange to the ground, the blame was quickly attributed
to the cryptocurrencies themselves.
A story of greed, stupidity and gross mismanagement (one of many in Corporate America; think about Enron, etc.) was turned into an anti-crypto manifesto. And not without reason.

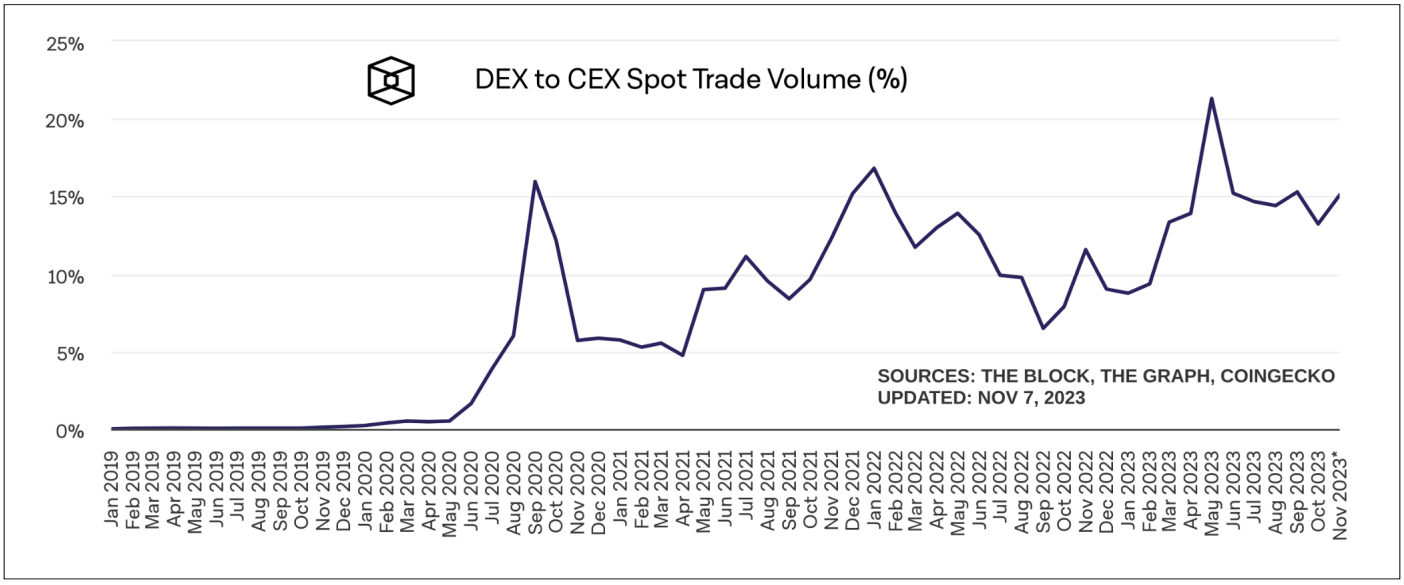
The lack of transparency around FTX operations paved the way for its collapse.
The overlooked conflict of interests between Alameda and FTX, coupled with insufficient oversight, regulatory scrutiny, and the centralised holding of tokens, were the underlying causes of FTX's failure. Unfortunately, these issues seem more prevalent in the crypto market than in Corporate America.

However, every cloud has a silver lining.
The FTX episode was a wake-up call for all those involved with the crypto industry. While some critics strongly called for more regulation, and crypto maximalists promoted a paradigm change to full decentralisation (i.e. decentralised exchanges), it was clear that one action had to be immediately taken to calm crypto users and investors: **enhance transparency.**

Concepts like **'proof of reserves'** and **'proof of liability'** were introduced to enhance clarity around CEXs operations and regain users' confidence. Notably, such steps were also undertaken by crypto marketplaces such as SwissBorg.

So, what's the future of CEXs?
The picture below (source: intotheblock.co) gives a rough idea of the relationship between DEX to CEX trading volume.
The trend is clear: most crypto transactions happen in CEXs (with Binance accounting for 44% of all transaction volumes as of September 2023). But DEXs are catching up, and fast.

Will we be in DEX's hands soon? - Only time will tell.



SwissBorg MEX revolution

The discussion from above on the CEXs and DEXs brings us to the following topic: which one should an investor use? Well, the answer is simple: the one giving the best price!
While the answer is simple and, to some extent, obvious, its implementation is far from straightforward.

The good news is that SwissBorg did it, thus becoming the first true meta-exchange to aggregate liquidity and trading pairs across various platforms. This includes FX, centralised exchanges (CEXs), and decentralised exchanges (DEXs). This approach guarantees optimal execution in terms of price and slippage, making it a dream come true for traders and investors.

SwissBorg is breaking new ground in the cryptosphere by integrating with the high-velocity Solana blockchain, creating a seamless, bi-directional passage between conventional fiat and the decentralised world. This groundbreaking initiative into the realm of DeFi does not just establish a conduit for smooth and transparent financial interactions across both platforms but introduces a gateway to an infinite, secure, and user-centric financial adventure.

SwissBorg's Smart Engine Graduates to DeFi

With the integration of Solana's leading decentralised exchange, Orca, and the newly launched Phoenix, SwissBorg has become the first platform to combine the power of Centralised (CEX), Decentralised (DEX), and Foreign Exchange (FX) platforms, all in one easy to use interface - the SwissBorg Smart Engine.

Merging the off-chain and on-chain worlds has always been a mission we take seriously at SwissBorg. We strongly believe in breaking down the barriers between the traditional and decentralised finance worlds to create something practical and hopeful for the future of finance. We are proud to be the first to truly offer this experience to our community of SwissBorg app users.

Advantages of SwissBorg's Smart Engine

Enhanced Liquidity: SwissBorg's Smart Engine gives users access to all exchanges' combined liquidity, endorsing smoother and more efficient trading, even for substantial transaction volumes.

Optimal Pricing: It scans multiple exchanges in real-time, ensuring users receive the best price for their trades by tapping into diverse liquidity pools and trading pairs.

Diversified Trading Opportunities: Users can access a vast array of assets and pairs across different exchanges without creating accounts on multiple platforms.

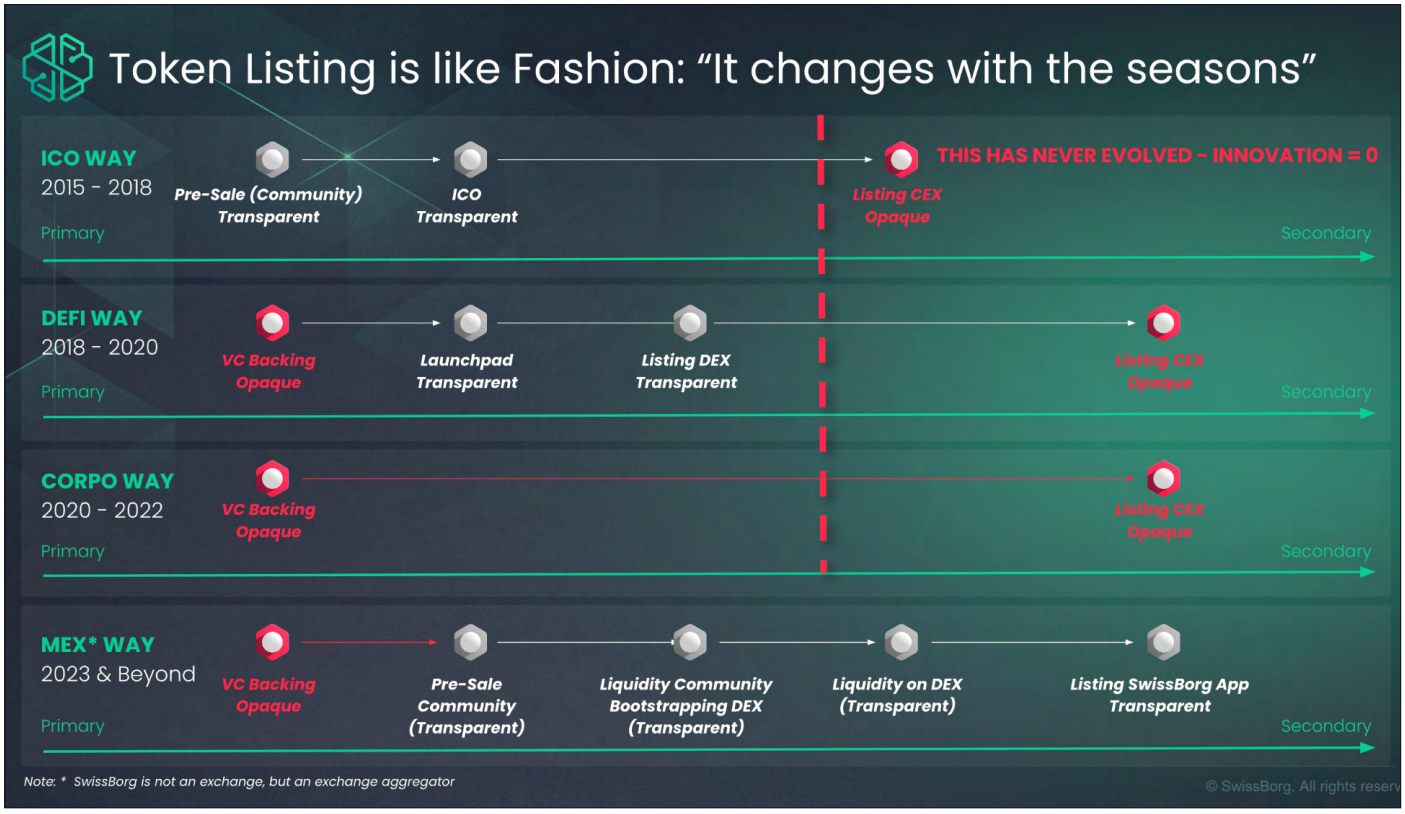
Reduced Slippage: The amplified liquidity and broad access to various exchanges assist in diminishing slippage during trading.

Streamlined User Experience: Users can administer their investments and trades across multiple platforms in milliseconds from a single SwissBorg interface.

Better than CEX!

Technical innovations are meaningless if they do not directly benefit users. Ultimately, the MEX is a revolution because it allows us to disrupt the monopolistic position of CEX & TradeFi over freshly created tokens and provides us with all the tools to offer our users the opportunity, from the very early days, to purchase a token and get involved into its journey all the way to full maturity:

- Alpha: Seed/Serie A/Pre-Listing Investment
- Alpha: Liquidity Deals Before Token Listing
- Marketplace (i.e. exchange): Listings Token in SB Marketplace, before they get listed in Binance or other CEX



The MEX is the concretisation of the desire to bring communities and projects closer together, enabling them to compound their efforts to reach maximum potential without having to rely exclusively on VC or market makers. This approach avoids excluding users from accessing the best investment opportunities.

Mark our words: In 2024, SwissBorg is the most open of the centralised apps, and it's only the beginning.

VIII. Guest Contributions

2024: The Year of All Hope

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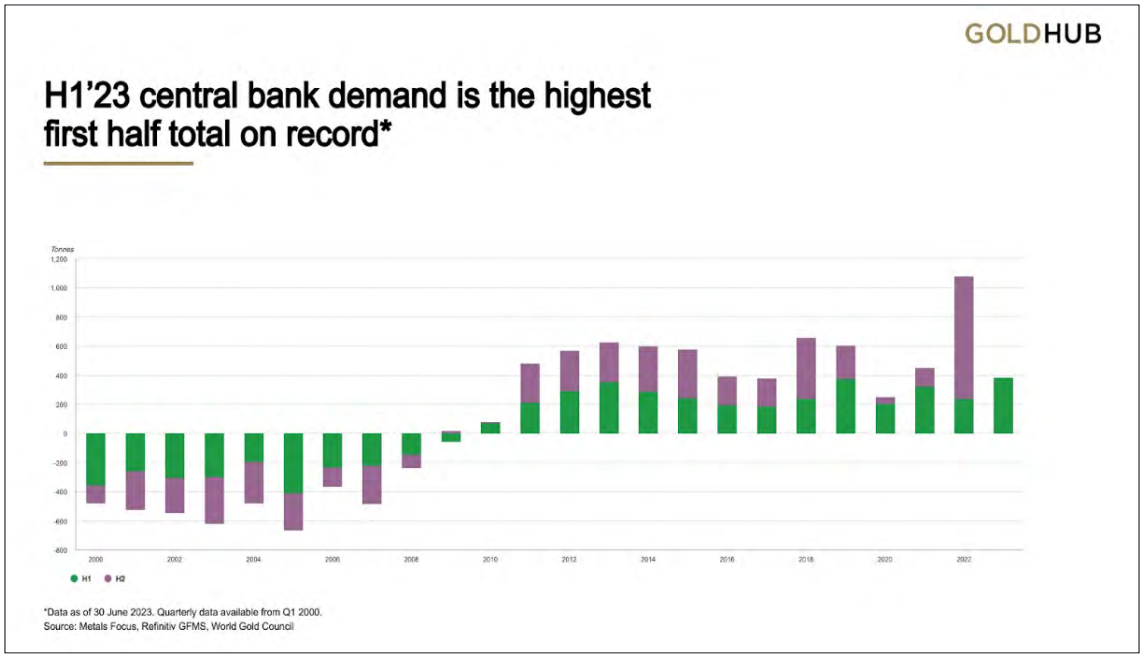
Author of "Gold and silver: Complete guide to learn and invest" and "Creation and destruction: The mystery behind economic and human cycle"

2024 will be the year of the halving for Bitcoin and this is often considered a good omen by the crypto community. But 2024 will also be the year of the end of monetary tightening, the end of rising interest rates, and the deceleration of inflation. Hence, it seems next year has all the qualities to gather the best hopes for crypto. But can the same be said for the stock markets, real estate, or gold?

Gold does not disappoint

In 2023, gold has performed over 10%, flirting with its historical highs. A very reasonable performance that leaves room for even greater bullish hopes ahead. Between 2016 and 2020, this precious yellow metal has had a strong performance of over 60%. But since 2020, rising interest rates and stock market crash led to a stagnation in the price of gold.

How then can we explain a potential bullish reversal trend in the price of gold? Over the last two years, central banks bought gold at an unprecedented rate in history! In 2022, central banks added close to 1,100 tons of gold to their stocks, representing an increase of more than 3% in the global gold reserves of central banks. This demand for gold was driven by increasing inflation in many countries which significantly reduced foreign exchange reserves (global foreign exchange reserves decreased by over $1 trillion in 2022, an unprecedented drop). In other words, inflation destroys the value of currencies, forcing central banks to go back to monetary fundamentals, meaning a return to gold.



Another, less official reason, is the rise in international geopolitical tensions. There is an increasing number of countries that prefer to hold their reserves in gold rather than in US dollar due to a lack of confidence in the United States or the West more generally. But in the end, global demand for "monetary" gold has significantly asserted itself in the global gold market over the past years (central banks represent around 25% of all gold demanded).

Producing gold is getting more and more expensive

Mining gold requires huge financial and human resources. In 2000, producing one ounce of gold cost less than $300. Today, producing this same amount costs on average up to $1,400. This information becomes even more key when we consider that nearly 75% of physical gold supply is provided by mining companies.

The "fundamental price" of gold, assuming sufficient and regular demand, is close to its production cost. By the end of 2023, we can estimate that, at a price of between $2,000 to $2,100 per ounce, gold is fundamentally well-evaluated. In 2024, the production cost borne by mining companies is expected to remain stable or increase slightly. However, these past two years have seen a significant increase in production costs that has not been accompanied by a rise in gold price.

This asymmetry can lead some to assert that gold has all its chance to initiate a bullish breakout. In that way, the end of 2023 is encouraging. It is, therefore, still relevant to remind the reader that a rise in gold prices to over $2,500 per ounce is statistically and fundamentally possible.

In any case, gold proves its stability in terms of purchasing power. Last century, one "Louis d'or" was enough to buy a bicycle, and that is still the case today! Gold is an intriguing metal where, beyond its economic qualities, holds evident historical and symbolic value for most civilizations. It is the jewel of humanity...

A new economical context

The past two years have been marked by the return of inflation and rising interest rates. Some anticipated a sharp decrease in gold price as rising interest rates are generally detrimental to gold. However, the decline in gold price did not materialize because the "fundamentals" remained strong (a constant price under $1,700 would have been catastrophic to mining companies).

Now, banks should stabilize interest rates as inflation decelerates. This environment is generally bullish for gold. Conversely, the risk of a recession should remain moderate in 2024, but it could increase in 2025 or 2026. It is worthy to note that bull markets on stocks often benefit the price of gold by a mechanical effect. Therefore, any bullish dynamics in the price of gold could be accompanied by the long-term economic context.

Finally, gold also benefits from geopolitical instability, which can prompt economic agents to protect themselves by exposing themselves to gold. The recent bank failures also contributed to the gold price in recent months.

Gold and crypto: The eternal debate

2024 is the year of the halving. This event is welcomed by the crypto communities, and so far, bitcoin has followed a 3.5 to 4 years cycle related to this key event. At this stage, Bitcoin should be halfway through its upward phase. In this context, some might be tempted to reaffirm the qualities of different cryptocurrencies, aiming to render gold obsolete and outdated. But things are more complex.

Indeed the structure of the two markets is very different. We have never mined as much gold as today. In contrast, we have never mined as little Bitcoin as today. Furthermore, the significance of mining supply is not as evident on Bitcoin, whose value depends more broadly on the number of users. Gold and Bitcoin are very different assets, and their utilities are also very different.

Half of the gold purchased worldwide is intended for the jewelry industry, while the other half is distributed among central banks, investors, and technology. Gold is not an asset used for transaction; at most, it is a "monetary" asset intended to preserve value. In contrast, Bitcoin is designed and conceived to serve as a means of payment and ideally to preserve value over the long term. Bitcoin is not used to make jewelry, secure monetary relations between countries (not yet, according to the most fervent supporters), or transmit historical value. Bitcoin is also significantly riskier than gold (gold has one of the best risk-return ratios), but it can also deliver substantially higher returns. Hence, these two assets: Gold and Bitcoin, are different yet complementary.

Nevertheless, gold and Bitcoin have no nationality; they are not subject to the whims of time and politics, and those assets are most importantly absolutely scarce. In any case, 2024 carries legitimate hope across several markets. But as the Italian proverb goes, "who lives of hope dies of desires"...

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